Hochfrequenztechnik 1 — english version

The module „Hochfrequenztechnik 1“ is considered for students in 5th semester of the B.Sc. program „Elektrotechnik und Informationstechnik“ (i.e., electrical engineering) at University of Stuttgart. The module comprises weekly lectures (90 minutes) and weekly exercises (90 minutes) over approximately 14 weeks time span, followed by a written exam (120 minutes). The weight of the module is 6 credit points (ECTS).

Courses in the M.Sc. program „Electrical Engineering“ at University of Stuttgart cover (among others) the field of microwave engineering in more width and depth. These courses require a basic understanding of microwave concepts and techniques, such as those covered by the module „Hochfrequenztechnik 1“ as part of prior B.Sc. studies.

In order to help those students who enter the M.Sc. courses in microwave engineering, but who attended insufficiently or differently focussed lectures in their undergrad studies, this translated scriptum of the module „Hochfrequenztechnik 1“ is considered a valuable resource. The scriptum comprises most lectures slides and a few problems from exercises.

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Suggested literature:

The art of microwave engineering is about 100 years old. Fundamentals do not change. Thus, the content of this lecture is covered in most, if not all, textbooks about the topic. However, the depth of this lecture is limited, as is the time for studying, therefore most content is discussed in more depth in the textbooks. On the other hand, this lecture is more focussed and limited in the variety of examples. In addition, this scriptum is based on presentation slides with little text (for practical reasons, primarily, to allow simpler updating, extensions and other changes, over the years), while textbooks typically discuss things in a more prosaic way.
For students interested in referring to a textbook, the following two books are suggested:

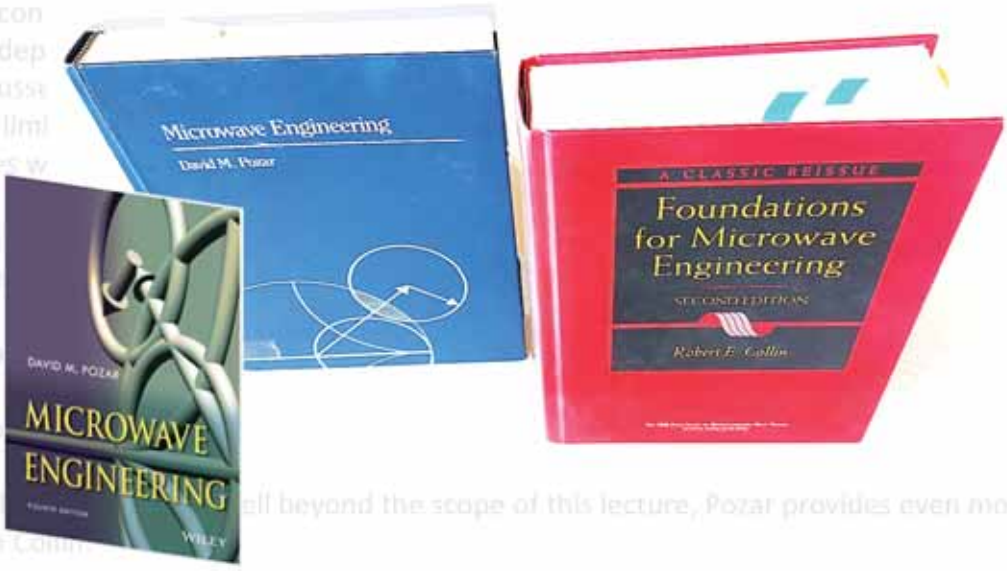
- David M. Pozar, Microwave Engineering, Wiley

- Robert E. Collin, Foundations for Microwave Engineering, IEEE Press

While both books go well beyond the scope of this lecture, Pozar provides even more depth than Collin.

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Suggested literature:



Hochfrequenztechnik 1 — english version

Content:

Repeating Maxwell‘s equations (Faraday‘s law, Ampère‘s law, permittivity)

Plane waves (polarization, skin effect, reflection, diffraction, transmission)

Waveguides (parallel-plates waveguides, rectangular waveguide)

Lumped components (inductor, capacitor, resistor)

Waves on transmission lines (guided wave, co-axial line, microstrip line, equivalent circuit, impedance transformation, Smith chart, lossy transmission line)

Wave matrices (S-parameters, conjugate-complex match, T-matrix, examples)

Transmission line circuits (Wilkinson divider, even-mode-odd-mode-analysis, 90-degrees hybrid, 180-degrees hybrid, transmission line couplers)

Impedance transformers and filters (impedance transformers, distributed components, Q-factor, filter synthesis, Richards transformation, Kuroda identities, examples)

Motivation

Microwave Engineering today covers electronic devices, circuits and concepts operating at frequencies between, roughly, 100 MHz and 100 GHz. In a wider context, some concepts also apply to higher frequencies up to (and including) optics (infrared and visible-light).

One main application of microwave engineering is communication: the physical infrastructure of wireless and fiber-optic communication. Other applications of today are in radar and sensing, localization, medicine and in industrial applications such as heating. Thus, microwave engineering applications are ubiquituous, and the related commercial impact is gigantic and growing.

If I have seen further than others, it is by standing upon the shoulders of giants.

Isaac Newton

Motivation

From early mobile telephony and radar …









… within less than 100 years …

[https://www.youtube.com/watch?v=U5V1sxAKu5I] Lucent Technologies

Motivation

[Facebook]

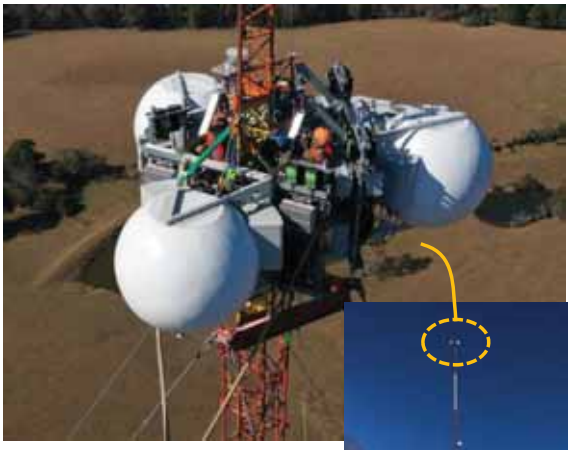
... covering all earth and outer space:







[JPL, NRAO Alma]







https://submarine-cable-map-2021.telegeography.com/

Content:

Repeating Maxwell's equations

Plane waves

Waveguides

Lumped components

Waves on transmission lines

Wave matrices

Transmission line circuits

Impedance transformers and filters

some repetition: Maxwell´s Equations

Lorentz force law: if a charge is moving across electric or magnetic field, then a force affects to this charge:

Hendrik Aanton Lorentz, 1853 – 1928, received PhD at age of 22!

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$$ … only E interacts with a non-moving charge, not H

first consider „instantaneous" quantities, e.g. $\vec{e} = \vec{e}(x, y, z, t)$

relation between electrical field strength *e* and electric displacement *d* :

$$\vec{d} = \varepsilon \vec{e}$$ [d]=As/m²

relation between magnetic field strength *h* and magnetic flux density *b* :

$$\vec{b} = \mu \vec{h}$$ [b]=Vs/m²

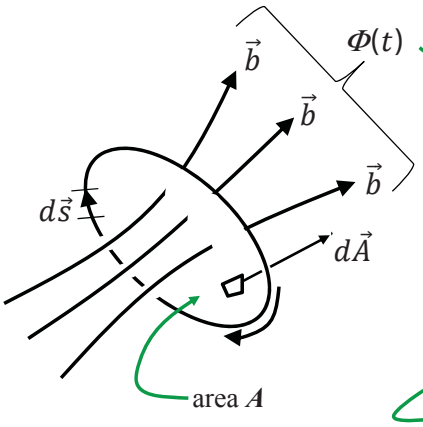
small letters: time-dependent, instantaneous quantities

Maxwell: time-varying electric and magnetic fields are interconnected. that is, e.g., conduction current causes a magnetic field, but displacement current does it as well !!

James Clerk Maxwell, 1831 – 1879, found his equations when he still was a student

Faraday's law of induction (Faraday 1831):

Michael Faraday, 1791 – 1867, physicist and chemist



magnetic flux Φ as integral of the magnetic flux density b(t) :

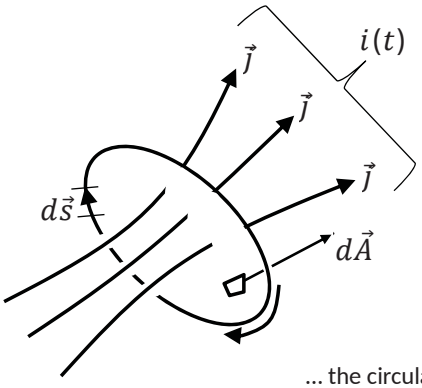
$$\Phi = \int_A \vec{b} d\vec{A}$$

$$\oint_C \vec{e} d\vec{s} = -\frac{\partial \Phi(t)}{\partial t} = -\frac{\partial}{\partial t} \int_A \vec{b} d\vec{A}$$

induced voltage (induced electromotive force) in the loop

Ampère-Maxwell law:

André-Marie Ampère, 1775 – 1836



$$\oint_C \vec{h} d\vec{s} = i_{ges}(t) = \int_A \overrightarrow{J_{ges}} d\vec{A}$$

$$\overrightarrow{J_{ges}} = \sigma\vec{e} + \frac{\partial \vec{d}}{\partial t} + \cdots$$

... the circulation of a magnetic field around a closed loop relates to the electric current (conduction current plus displacement current) going through this loop

from vector algebra – Stokes´ law:

George Gabriel Stokes, 1819 – 1903

$$\oint_C \vec{x} \cdot d\vec{s} = \int_A \nabla \times \vec{x} \cdot \overrightarrow{dA}$$

$$\text{curl}\,\vec{x} = \nabla \times \vec{x}$$

$$\text{rot}\,\vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

→ differential forms:

$$\text{curl}\,\vec{e} = \nabla \times \vec{e} = -\frac{\partial \vec{b}}{\partial t} = -\mu\frac{\partial \vec{h}}{\partial t}$$ … „Faraday plus Stokes“

… in German notation: „rot“

$$\text{curl}\,\vec{h} = \nabla \times \vec{h} = \sigma\vec{e} + \varepsilon\frac{\partial \vec{e}}{\partial t}$$ … „Ampère-Maxwell plus Stokes“

… displacement current

… conduction current

complex-valued quantities in case of harmonic time dependency:

$$\vec{e}(x,y,z,t)|_\omega = Re\{\underline{\vec{E}}(x,y,z)e^{j\omega t}\}$$

$$\frac{\partial \vec{e}}{dt} = Re\{j\omega\underline{\vec{E}}e^{j\omega t}\}$$

→ differential form for time-harmonic quantities: $\frac{\partial}{\partial t} \cong j\omega$

$$\text{rot}\,\underline{\vec{E}} = -j\omega\mu\underline{\vec{H}}$$ … Faraday's law

$$\text{rot}\,\underline{\vec{H}} = \sigma\underline{\vec{E}} + j\omega\varepsilon\underline{\vec{E}}$$ … Ampère-Maxwell's law

→ integral form for time-harmonic quantities:

$$\oint_C \underline{\vec{E}}\,d\vec{s} = -\int_A j\omega\mu\underline{\vec{H}}\,d\vec{A}$$ … Faraday's law

$$\oint_C \underline{\vec{H}}\,d\vec{s} = \int_A (\sigma + j\omega\varepsilon)\underline{\vec{E}}\,d\vec{A}$$ … Ampère-Maxwell's law

Gauss' law for time-harmonic quantities:

$$\text{div}\,\underline{\vec{D}} = \nabla\cdot\underline{\vec{D}} = \rho$$ *sources: electric charges*

… the net outflow of electric field through a closed surface around an electric charge is proportional to the amount of enclosed charge

$$\text{div}\,\underline{\vec{B}} = \nabla\cdot\underline{\vec{B}} = 0$$ *there are no „magnetic charges" in reality*

polarisation:

in vacuum: $\underline{\vec{D}} = \varepsilon_0\underline{\vec{E}}$ and $\underline{\vec{B}} = \mu_0\underline{\vec{H}}$

in material: $\underline{\vec{D}} = \varepsilon_0\underline{\vec{E}} + \underline{\vec{P}} = \varepsilon_0(1+\chi_e)\underline{\vec{E}}$ → $\varepsilon_r = 1+\chi_e$

$\underline{\vec{B}} = \mu_0\underline{\vec{H}} + \underline{\vec{M}} = \mu_0(1+\chi_m)\underline{\vec{H}}$ → $\mu_r = 1+\chi_m$

… vacuum
… effect of material („polarisation")
… susceptibility: χ

generally, susceptibilities (therefore also permittivity and permeability) are complex values (i.e., the imaginary part of permittivity accounts for loss):

$$\underline{\varepsilon} = \varepsilon' - j\varepsilon'' = \varepsilon_0(\varepsilon_r' - j\varepsilon_r'')$$

negative sign chosen!

in addition, the loss effect due to conductivity σ can be included into the complex permittivity:

$$\vec{J} = \sigma\underline{\vec{E}} + j\omega(\varepsilon' - j\varepsilon'')\underline{\vec{E}} = j\omega\varepsilon_{ges}\underline{\vec{E}} = j\omega\left[\varepsilon' - j\left(\varepsilon'' + \frac{\sigma}{\omega}\right)\right]\underline{\vec{E}}$$

$$\varepsilon_{ges} = \varepsilon_0\varepsilon_{r,ges} = \varepsilon_0\left[\varepsilon_r' - j\left(\varepsilon_r'' + \frac{\sigma}{\omega\varepsilon_0}\right)\right]$$

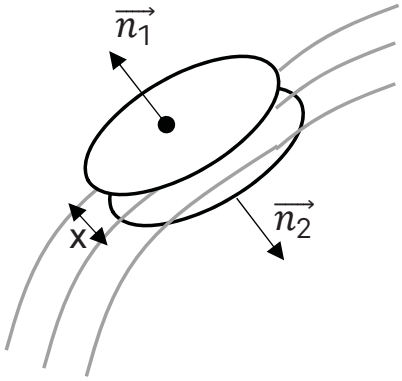
boundary conditions

$$q = \iiint \rho\, dV$$

$$\oiint \underline{\vec{D}}\, d\vec{A} = \iiint \rho\, dV$$

@ ρ = 0, x → 0 :

$$\vec{D}_1 \cdot \vec{n}_1 + \vec{D}_2 \cdot \vec{n}_2 = 0$$



$$\oiint \underline{\vec{B}}\, d\vec{A} = 0$$

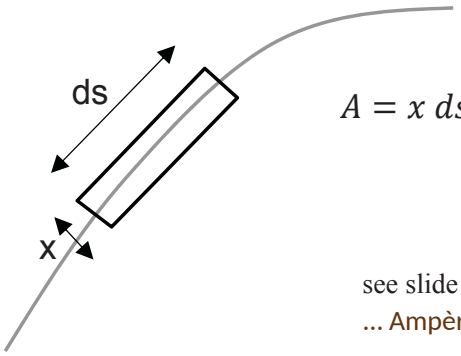
@ x → 0 :

$$\vec{B}_1 \cdot \vec{n}_1 + \vec{B}_2 \cdot \vec{n}_2 = 0$$

… continuity of normal components of D and B

see slide 1.5 →
... Faraday's law

$$\oint_C \vec{E} d\vec{s} = -\int_A j\omega\mu\vec{H} d\vec{A}$$



$$A = x\, ds$$

@ x → 0 :

$$\vec{E}_{\tan,1} - \vec{E}_{\tan,2} = 0$$

see slide 1.5 →
... Ampère's law

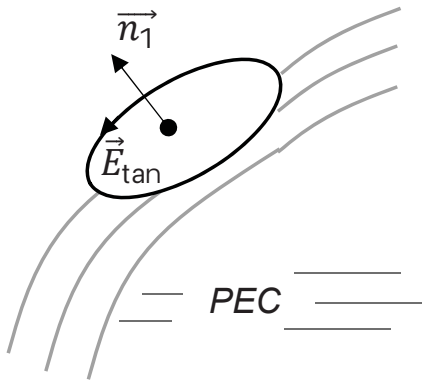
$$\oint_C \vec{H} d\vec{s} = \int_A (\sigma + j\omega\varepsilon)\vec{E} d\vec{A}$$

@ x → 0 :

$$\vec{H}_{\tan,1} - \vec{H}_{\tan,2} = 0$$

… continuity of tangential components of E and H

boundary to the perfect electrical conductor (PEC): $\vec{E}_{\tan} = 0$



↓

$$\oint_C \vec{E} d\vec{s} = 0 = -\int_A j\omega\mu\vec{H_n} d\vec{A}$$

↓

$$\vec{H}_n = 0$$

… at the PEC-boundary:
- tangential-only H-field
- normal-only E-field

… current along the PEC boundary according to:

$$\vec{J} = \vec{n}_1 \times \vec{H}_{\tan}$$

→ non-PEC (finite σ): skin effect

Content:

Repeating Maxwell‘s equations

Plane waves

Waveguides

Lumped components

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Wave matrices

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Impedance transformers and filters

Plane Waves

Ampère-Maxwell‘s law plus Faraday‘s law:

$$\text{rot}\,\vec{H} = \sigma\vec{E} + \varepsilon\frac{\partial\vec{E}}{\partial t} \qquad \text{rot}\,\vec{E} = -\mu\frac{\partial\vec{H}}{\partial t}$$

$$\text{rot rot}\,\vec{H} = -\sigma\mu\frac{\partial\vec{H}}{\partial t} - \varepsilon\mu\frac{\partial^2\vec{H}}{\partial t^2} \qquad \text{rot rot}\,\vec{E} = -\sigma\mu\frac{\partial\vec{E}}{\partial t} - \varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\text{rot rot}\,\underline{\vec{A}} = \text{grad div}\,\underline{\vec{A}} - \Delta\,\underline{\vec{A}}$$

… assuming there are no sources

$$\Delta\,\vec{H} = \sigma\mu\frac{\partial\vec{H}}{\partial t} + \varepsilon\mu\frac{\partial^2\vec{H}}{\partial t^2} \qquad \Delta\,\vec{E} = \sigma\mu\frac{\partial\vec{E}}{\partial t} + \varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2}$$

… these are vectorial wave equations

→ **differential operators in cartesian coordinates**:

$$\text{grad } f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\text{div } \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\text{curl } \vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

- **loss-less medium** $\sigma = 0$: $\Delta \vec{E} = \varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$, $\Delta \vec{H} = \varepsilon\mu \frac{\partial^2 \vec{H}}{\partial t^2}$

- **time-harmonic process** $\frac{\partial}{\partial t} \cong j\omega$: $\Delta \underline{\vec{E}} = j\omega\mu(\sigma + j\omega\varepsilon)\,\underline{\vec{E}}$

$$\Delta \underline{\vec{H}} = j\omega\mu(\sigma + j\omega\varepsilon)\,\underline{\vec{H}}$$

- **definitions**:
  - wave number k $\quad k^2 = -j\omega\mu(\sigma + j\omega\varepsilon)$

$$\Delta \underline{\vec{E}} = -k^2\,\underline{\vec{E}}$$

$$\Delta \underline{\vec{H}} = -k^2\,\underline{\vec{H}}$$

  - phase constant β [rad/m]
  - attenuation constant α [Np/m] $\quad k = \beta - j\alpha$

… α and β are real numbers

- **a simple solution of the wave equation:**

  - cartesian coordinates
  - time-harmonic process → factor: $\cdot e^{j\omega t}$
  - loss-less
  - *E*-field with only component $\underline{E}_x(z)$

$$\left.\begin{matrix} E_y = 0 \\ E_z = 0 \end{matrix}\right\} \rightarrow \operatorname{curl}\vec{\underline{E}} = -j\omega\mu\vec{\underline{H}} \rightarrow \left.\begin{matrix} H_x = 0 \\ H_z = 0 \end{matrix}\right\}$$

$$\left.\begin{matrix} \operatorname{curl}\vec{\underline{H}} = \sigma\vec{\underline{E}} + j\omega\varepsilon\vec{\underline{E}} \rightarrow \dfrac{\partial \underline{H}_y}{\partial x} = 0 \\ \text{source-free}: \quad \operatorname{div}\vec{\underline{H}} = 0 \rightarrow \dfrac{\partial \underline{H}_y}{\partial y} = 0 \end{matrix}\right\} \rightarrow \underline{H}_y(z)$$

$$\rightarrow \frac{d^2\underline{E}_x}{dz^2} + k^2\underline{E}_x = 0$$

… this is a differential equation of 2nd order with constant coefficients → can be solved with „Ansatz“ method (meaning: guess the solution and calculate the coefficients)

- „Ansatz“: $\underline{E}_x = \underline{E}_{x0}\, e^{\gamma z}$

- propagation constant γ: $\gamma^2 = -k^2 = -\beta^2$

} omitted: $\cdot e^{j\omega t}$

- general solution: $\underline{E}_x(z) = \underline{E}_{x0,h}\, e^{-j\beta z} + \underline{E}_{x0,r}\, e^{+j\beta z}$

„forward wave“ (in **+z** -direction) | „backward wave“ (in **–z** -direction)

- complex magnitude: $\underline{E}_{x0,hr} = E_{xo,hr}\, e^{j\varphi_{hr}}$

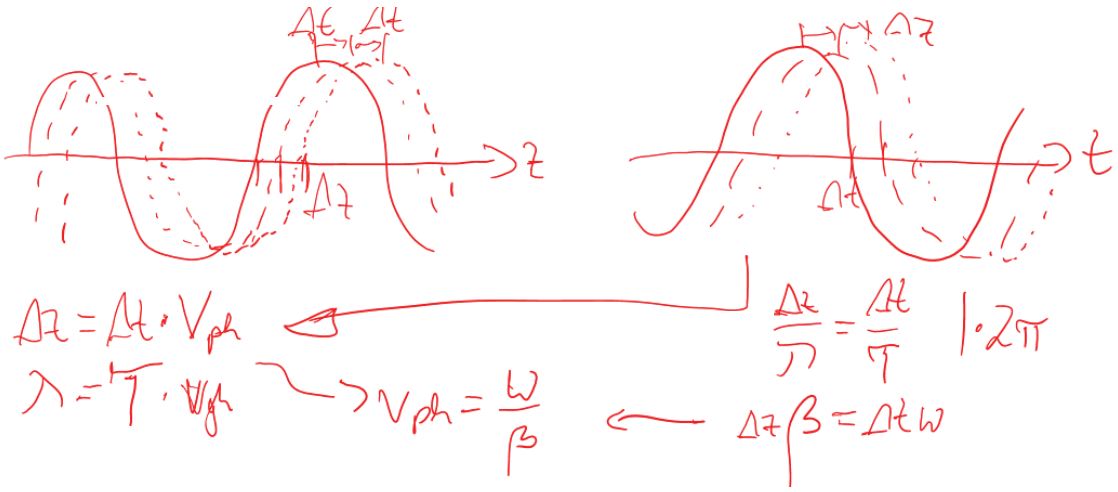
- instantaneous value:

$$e_x(z,t) = \left|E_{x0,h}\right| \cos(\omega t - \beta z + \varphi_h) + \left|E_{x0,r}\right| \cos(\omega t + \beta z + \varphi_r)$$

$$e_x(z,t) = \left|E_{x0,h}\right| \cos(\omega t - \beta z + \varphi_h) + \left|E_{x0,r}\right| \cos(\omega t + \beta z + \varphi_r)$$

... periodicity in space (z):

... periodicity in time (t):



- wave characteristic: the fields are reproducing identically
  - in time after a period T
  - in space after a wavelength λ

$$T = \frac{1}{f} = \frac{2\pi}{\omega}$$

$$\beta = \frac{2\pi}{\lambda}$$

- velocity of propagation of a wave: $v = \lambda f$

- in loss-less medium: $\beta = \omega\sqrt{\varepsilon\mu}$, $v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\varepsilon\mu}}$

- wave impedance:

$$\underline{E}_x(z) = \underline{E}_{x0,h}\, e^{-j\beta z} + \underline{E}_{x0,r}\, e^{+j\beta z}$$

$$\underline{H}_y(z) = \underline{H}_{y0,h}\, e^{-j\beta z} + \underline{H}_{y0,r}\, e^{+j\beta z}$$

, $\frac{d\underline{E}_x}{dz} = -j\omega\mu\underline{H}_y$

$$\frac{\underline{E}_{x0,h}}{\underline{H}_{y0,h}} = \frac{\omega\mu}{\beta} = Z_w \quad , \quad \frac{\underline{E}_{x0,r}}{\underline{H}_{y0,r}} = -\frac{\omega\mu}{\beta} = -Z_w$$

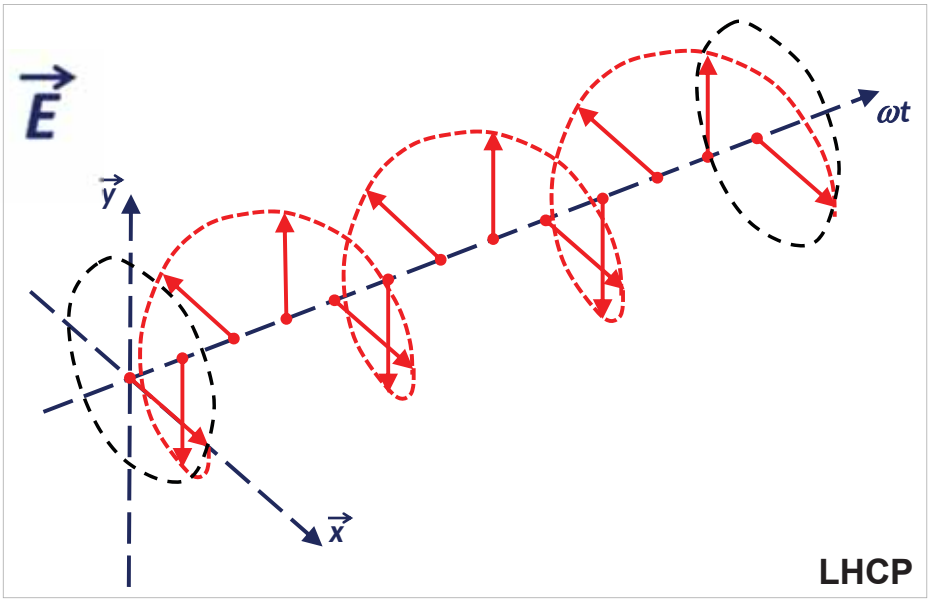
$$Z_w = \sqrt{\frac{\mu}{\varepsilon}} = Z_{w0}\sqrt{\frac{\mu_r}{\varepsilon_r}} \qquad , \qquad Z_{w0} \approx 120\pi\ \Omega \approx 377\ \Omega$$

- **general properties of plane waves:**
  - *E* and *H* are in-phase in a loss-less medium
  - infinite extent → requires infinite distance from a source of infinite extent (and, infinite power)

… obviously impossible, but in many cases, the „plane wave“ is a good (reasonably precise, and simple) approximation

- **polarisation**:
  - defined by direction of vector of *E*-field (in example above: x-direction, i.e., „linearly polarised“)
  - circular / elliptical polarisation, rotation direction defined by: view in direction of propagation, referring to the rotation direction of the *E*-field vector over progressing time (i.e., at a fixed location); notations: „RHP“, „RHCP“, „LHCP“, „LHP“

… example/remark: sound does not have polarisation, because pressure is a scalar (while electromagnetic wave is based on vector-quantities, E and H)



$$e_x(z,t) = |E_{x0}|\cos(\omega t - \beta z) \qquad , \qquad e_y(z,t) = |E_{y0}|\cos\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

… propagation direction: +z

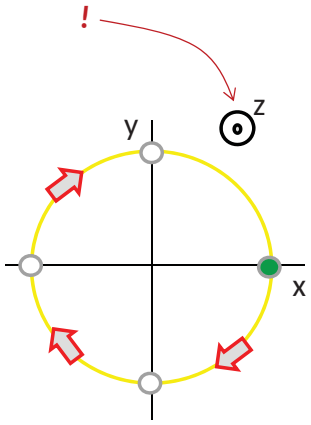
- considering a fixed location, e.g., z=0
- considering (progressing) flow of time:

$\omega t = 0$: $E_x = +1$, $E_y = 0$ ●

$\omega t = \pi/2$: $E_x = 0$, $E_y = -1$ ○

$\omega t = \pi$: $E_x = -1$, $E_y = 0$ ○

$\omega t = 3\pi/2$: $E_x = 0$, $E_y = +1$ ○



- looking in direction of propagation (transmit antenna, [IEEE definition])

--> sense of rotation if looking in direction +z : CCW

--> sense of rotation and direction of propagation form a left-handed-system → LHCP

+z / Δφ = +π/2 → LHCP / CCW
-z / Δφ = +π/2 → RHCP / CW
+z / Δφ = -π/2 → RHCP / CW
-z / Δφ = -π/2 → LHCP / CCW

--> Δφ = +π/2 corresponds to LHCP / CCW if direction of propagation is +z

- **wave propagation in lossy media:**

$$\Delta \underline{\vec{E}} = -k^2 \underline{\vec{E}}$$
$$\Delta \underline{\vec{H}} = -k^2 \underline{\vec{H}}$$
$$k^2 = -j\omega\mu(\sigma + j\omega\varepsilon)$$
$$k = \beta - j\alpha$$

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}} \sqrt{\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1} \geq 0 \quad \text{[Np/m]}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2}} \sqrt{\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1} > 0 \quad \text{[rad/m]}$$

… in a lossy medium, v_{ph} and λ both vary as a function of σ !

$$\underline{Z}_w = \frac{\omega\mu}{k} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \left|\underline{Z}_w\right| e^{j\theta_{Z_w}}$$

… in a lossy medium, the wave impedance is complex, that is, *E* and *H* of a plane wave are not in phase !

- wave propagation in lossy media:

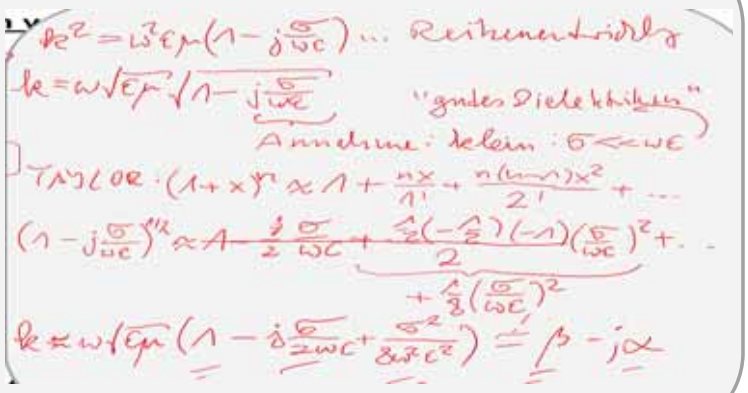
$$\Delta \underline{\vec{E}} = -k^2 \, \underline{\vec{E}}$$

$$\Delta \underline{\vec{H}} = -k^2 \, \underline{\vec{H}}$$

$$k^2 = -j\omega\mu(\sigma + j\omega\varepsilon)$$

$$k = \beta - j\alpha$$

… series expansion:



… in a lossy medium, v_{ph} and λ both vary as a function of σ !

$$\underline{Z}_w = \frac{\omega\mu}{k} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \left|\underline{Z}_w\right| e^{j\theta_{Z_w}}$$

… in a lossy medium, the wave impedance is complex, that is, *E* and *H* of a plane wave are not in phase !

special case #1: … good dielectric / small loss : $\sigma \ll \omega\varepsilon$

$$tan\delta = \frac{\sigma}{\omega\varepsilon}$$ … „loss tangent“ (small for a good dielectric)

… approximations from series expansion:

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon}\left(1 + \frac{\sigma^2}{8\omega^2\varepsilon^2}\right)$$

special case #2: … good conductor : $\sigma \gg \omega\varepsilon$

… approximations from series expansion:

$$\alpha \approx \beta \approx \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\underline{Z}_w \approx \sqrt{\frac{\pi f\mu}{\sigma}}\,(1+j)$$

… inside a good conductor, $\underline{Z}_w$ is complex with a phase of 45° :

- *H* follows *E* with 45°,
- *Z* is very small → *H* is very large,
- β is very large → λ is very small.

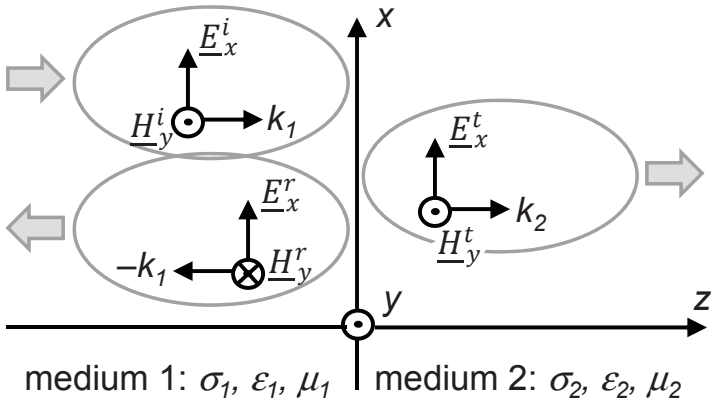
… skin effect :

- the magnitude of a wave in a „good“ conductor is decreasing with $e^{-\alpha z}$. After a distance of $z = \delta$ the magnitude has fallen to *1/e* .

- skin depth: $\delta \approx \sqrt{\frac{1}{\pi f \mu \sigma}}$

- e.g.: Cu @ 1 MHz → δ = 67 µm,
  Cu @ 10 GHz → δ = 0.67 µm

- **reflection and transmission — normal incidence**



total fields:

$$\underline{E}_{x1} = \underline{E}_x^i + \underline{E}_x^r \qquad \underline{E}_{x2} = \underline{E}_x^t$$

$$\underline{H}_{y1} = \underline{H}_y^i + \underline{H}_y^r \qquad \underline{H}_{y2} = \underline{H}_y^t$$

- at the boundary: $\underline{E}_x^i + \underline{E}_x^r = \underline{E}_x^t$

$$\frac{\underline{E}_x^i}{Z_{w1}} - \frac{\underline{E}_x^r}{Z_{w1}} = \frac{\underline{E}_x^t}{Z_{w2}}$$

- reflection coefficient: $\Gamma = \dfrac{\underline{E}_x^r}{\underline{E}_x^i} = \dfrac{Z_{w2} - Z_{w1}}{Z_{w2} + Z_{w1}}$

- transmission coefficient: $T = \dfrac{\underline{E}_x^t}{\underline{E}_x^i} = \dfrac{2\,Z_{w2}}{Z_{w2} + Z_{w1}}$

$$T = 1 + \Gamma$$

- total fields (loss-less media):

$$\underline{E}_{x1} = \underline{E}_x^i\, e^{-j\beta_1 z}\left(1 + \Gamma e^{+2j\beta_1 z}\right)$$

$$\underline{H}_{y1} = \frac{\underline{E}_x^i}{Z_{w1}}\, e^{-j\beta_1 z}\left(1 - \Gamma e^{+2j\beta_1 z}\right)$$

- „open“ : $Z_{w2} \to \infty$ $\quad \underline{E}_{x1} = 2\underline{E}_x^i \quad \underline{H}_{y1} = 0 \quad \Gamma = \dfrac{\underline{E}_x^r}{\underline{E}_x^i} = +1$

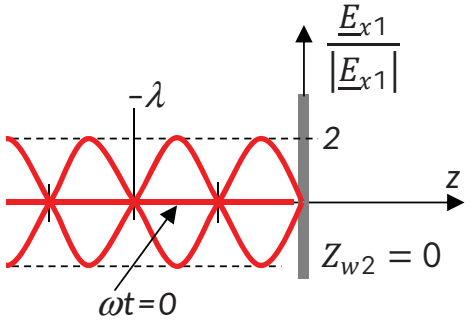
- „short“ : $Z_{w2} = 0$ $\quad \underline{E}_{x1} = 0 \quad \underline{H}_{y1} = 2\dfrac{\underline{E}_x^i}{Z_{w1}} \quad \Gamma = \dfrac{\underline{E}_x^r}{\underline{E}_x^i} = -1$

$$e^{jx} - e^{-jx} = 2j\sin x$$

$$\underline{E}_{x1} = -2j\,\underline{E}_x^i \sin(\beta_1 z)$$

$$e^{jx} + e^{-jx} = 2\cos x$$

$$\underline{H}_{y1} = 2\frac{\underline{E}_x^i}{Z_{w1}}\cos(\beta_1 z)$$



Discussíon: This is a „standing wave“.

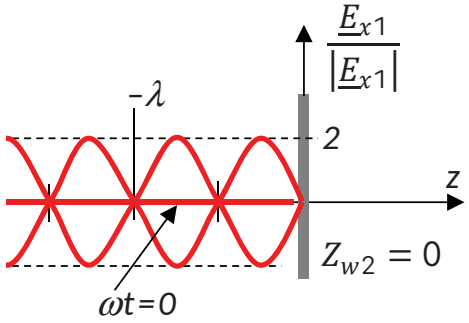
- It is caused by the boundary condition at z=0, here Z_{w2}=0, causing tangential E_{tan}=0.
- It oscillates over time, the zero-line corresponds to ωt=π/2. But there is an envelope.
- Its max. magnitude is 2.
- Its zeros are periodic.
- H-field (H_y) is displaced by π/4, w.r.t. E-field.

$$e^{jx} - e^{-jx} = 2j \sin x$$

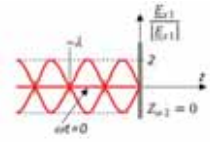
$$\underline{E}_{x1} = -2j\, \underline{E}^i_x \sin(\beta_1 z)$$

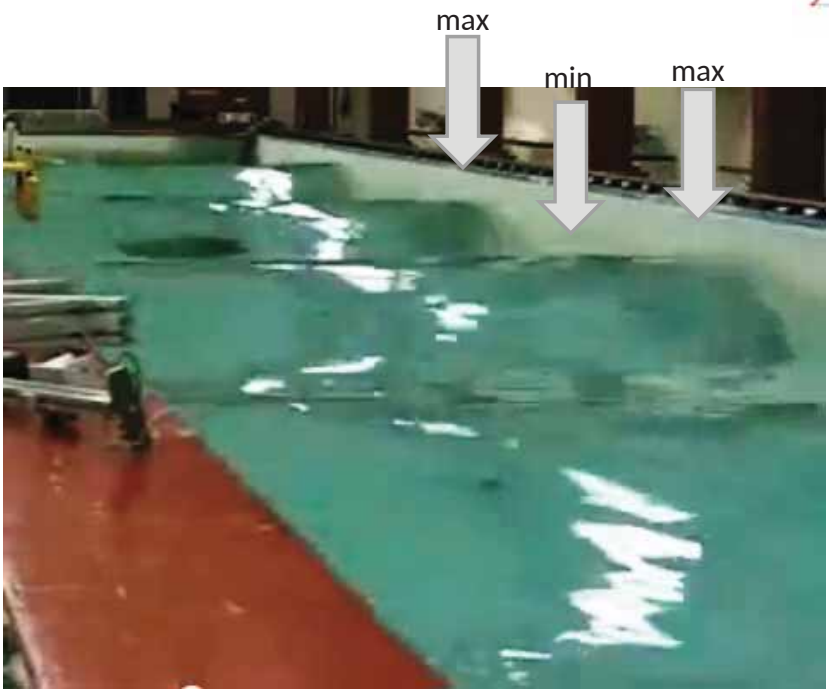
$$e^{jx} + e^{-jx} = 2 \cos x$$

$$\underline{H}_{y1} = 2 \frac{\underline{E}^i_x}{Z_{w1}} \cos(\beta_1 z)$$

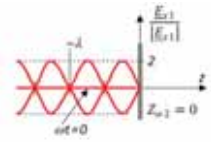


http://www.youtube.com/watch?v=NpEevfOU4Z8

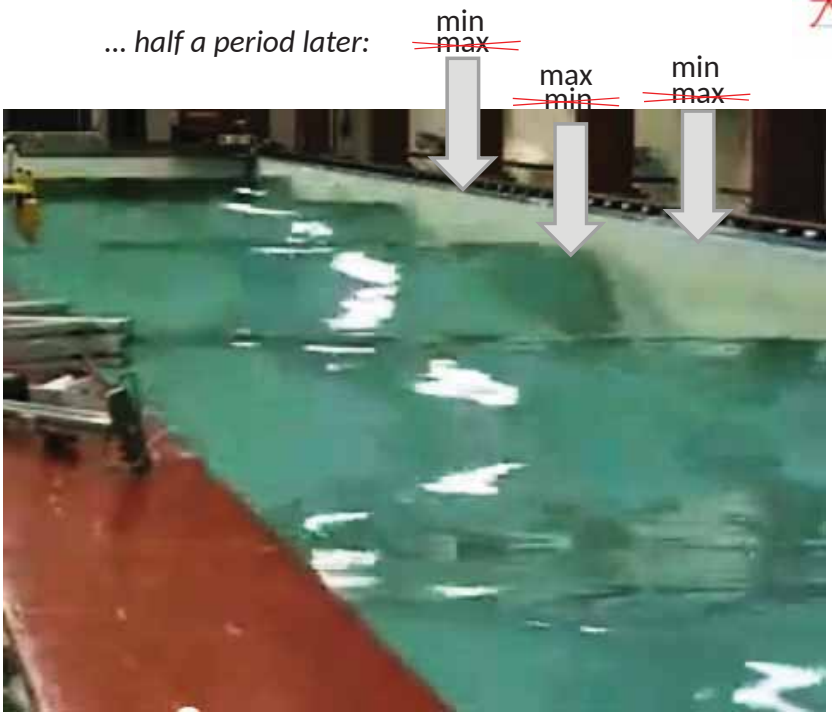




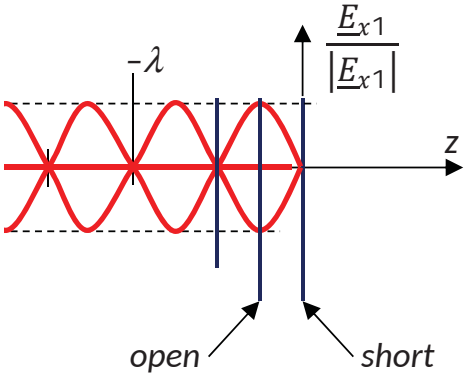
http://www.youtube.com/watch?v=NpEevfOU4Z8



... half a period later:



- layered dielectrics → **impedance transformation** :



... the same standing wave (for z<<0) caused by short, will be caused by open if the location of reflection is moved by $\lambda/4$!

→ reflection coefficient Γ is periodic with (multiples of) $\lambda/2$:

$$\Gamma_{z=-l} = \Gamma_{z=0}\, e^{-2j\beta_1 l}$$

...This is important !

... the ratio between E and H is an impedance, which varies (and is periodic) along the standing wave:

$$Z_{z=-l} = \left.\frac{\underline{E}_{x1}}{\underline{H}_{y1}}\right|_{z=-l} = Z_{w1}\frac{1+\Gamma_{z=0}\,e^{-2j\beta_1 l}}{1-\Gamma_{z=0}\,e^{-2j\beta_1 l}}$$

... see next slide for this calculation

$$Z_{z=-l} = Z_{w1}\frac{Z_{w2}+j\,Z_{w1}\tan(\beta_1 l)}{Z_{w1}+j\,Z_{w2}\tan(\beta_1 l)}$$

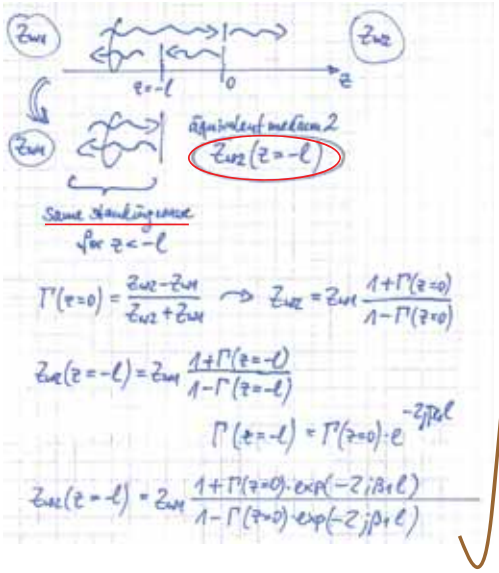
... i.e., the standing wave creates an impedance transformation !

...This is important !

... e.g. $l=\lambda/4$: $Z_{z=-\lambda/4} = \dfrac{Z_{w1}^2}{Z_{w2}}$

... the ratio between E and H is an impedance, which varies (and is periodic) along the standing wave:

$$Z_{z=-l} = \left.\frac{\underline{E}_{x1}}{\underline{H}_{y1}}\right|_{z=-l} = Z_{w1}\frac{1+\Gamma_{z=0}\,e^{-2j\beta_1 l}}{1-\Gamma_{z=0}\,e^{-2j\beta_1 l}} \quad , \quad \Gamma_{z=-l} = \Gamma_{z=0}\,e^{-2j\beta_1 l} \quad , \quad \Gamma = \frac{\underline{E}_x^r}{\underline{E}_x^i} = \frac{Z_{w2}-Z_{w1}}{Z_{w2}+Z_{w1}}$$



$$\Gamma(z=0) = \frac{1-Z_{w2}/Z_{w1}}{1+Z_{w2}/Z_{w1}}$$

$$Z_{w2}(z=-l) = Z_{w1}\frac{1+\frac{Z_{w2}}{Z_{w1}}+\left(1-\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(-2j\beta l)}{1+\frac{Z_{w2}}{Z_{w1}}-\left(1-\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(-2j\beta l)}$$

$$Z_{w2}(z=-l) = Z_{w1}\frac{\left(1+\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(+j\beta l)+\left(1-\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(-j\beta l)}{\left(1+\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(+j\beta l)-\left(1-\frac{Z_{w2}}{Z_{w1}}\right)\cdot\exp(-j\beta l)}$$

$$\exp(+j\beta l)+\exp(-j\beta l) = 2\cos(\beta l)$$

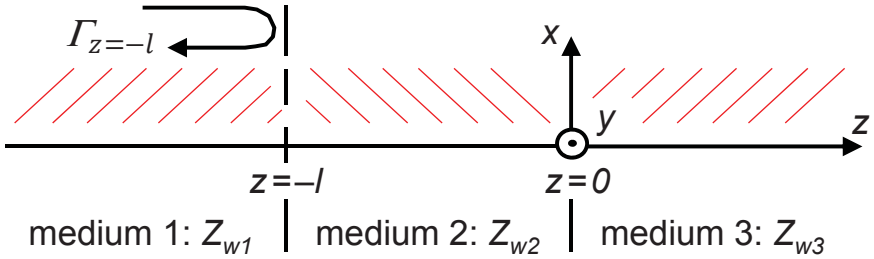
$$\exp(+j\beta l)-\exp(-j\beta l) = 2j\sin(\beta l)$$

$$Z_{w2}(z=-l) = Z_{w1}\frac{2\cos\beta l + \frac{Z_{w2}}{Z_{w1}}\cdot 2j\sin\beta l}{2j\sin\beta l + \frac{Z_{w2}}{Z_{w1}}\cdot 2\cos\beta l} \quad \left|\cdot\frac{Z_{w2}}{2\cos\beta l}\right.$$

$$\cdot\frac{Z_{w2}}{2\cos\beta l}$$

$$Z_{z=-l} = Z_{w1}\frac{Z_{w2}+j\,Z_{w1}\tan(\beta_1 l)}{Z_{w1}+j\,Z_{w2}\tan(\beta_1 l)}$$

... yet another boundary at $z=-l$ (previous boundary remains at $z=0$), and some re-numbering of media (no. 1,2,3 from left to right) gives:



$$\Gamma_{z=-l} = \frac{\Gamma_{12} + \Gamma_{23}\, e^{-2j\beta_2 l}}{1 + \Gamma_{12}\,\Gamma_{23}\, e^{-2j\beta_2 l}}$$

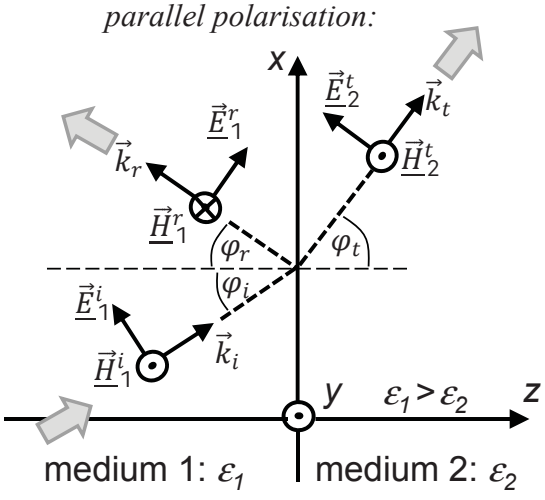
$$\Gamma_{z=-l} = \frac{Z_{w2}\left[Z_{w3} + Z_{w2} + (Z_{w3} - Z_{w2})\, e^{-2j\beta_2 l}\right] - Z_{w1}\left[Z_{w3} + Z_{w2} - (Z_{w3} - Z_{w2})\, e^{-2j\beta_2 l}\right]}{Z_{w2}\left[Z_{w3} + Z_{w2} + (Z_{w3} - Z_{w2})\, e^{-2j\beta_2 l}\right] + Z_{w1}\left[Z_{w3} + Z_{w2} - (Z_{w3} - Z_{w2})\, e^{-2j\beta_2 l}\right]}$$

- reflection and transmission — tilted incidence

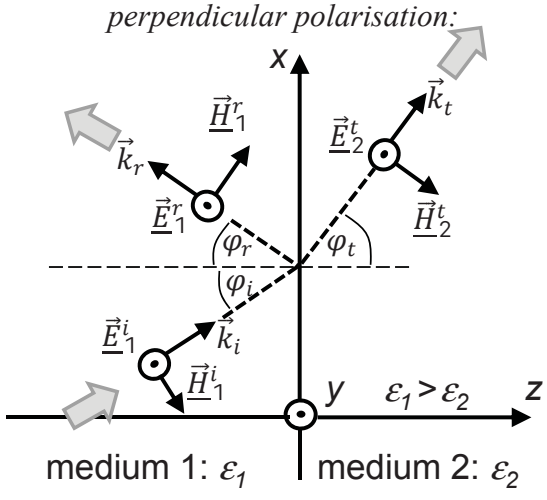
- isotropic media: any wave is a combination of parallel and perpendicularly polarized waves
- where polarisation is defined by *E* field
- loss-less media
- suitably aligned coordinate system:

... Note: „parallel polarised“ meaning, *E* is parallel to the drawing plane (and not: parallel to the boundary – this is denoted as „perpendicularly polarized“)

parallel polarisation:



perpendicular polarisation:



- for any polarisation: $\varphi_r = \varphi_i$, $\frac{\sin\varphi_i}{\sin\varphi_t} = \frac{k_2}{k_1} = \sqrt{\frac{\mu_2\varepsilon_2}{\mu_1\varepsilon_1}}$ Snell's law

- total reflection: $\varphi_t = 90°$ … requires $\varepsilon_1 > \varepsilon_2$

- parallel polarisation:

$$\Gamma_{/\!/} = \frac{Z_{w2}\cos\varphi_t - Z_{w1}\cos\varphi_i}{Z_{w2}\cos\varphi_t + Z_{w1}\cos\varphi_i} \qquad T_{/\!/} = \frac{2\,Z_{w2}\cos\varphi_i}{Z_{w2}\cos\varphi_t + Z_{w1}\cos\varphi_i}$$

$$1 + \Gamma_{/\!/} = T_{/\!/}\,\frac{\cos\varphi_t}{\cos\varphi_i}$$

Γ and T refer to field magnitudes, not to power

- Brewster angle: $\Gamma_{/\!/} = 0 \quad \rightarrow \quad \tan\varphi_i^{\text{Brewster}} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \quad \& \quad \mu_1 = \mu_2$

- perpendicular polarisation:

$$\Gamma_{\perp} = \frac{Z_{w2}\cos\varphi_i - Z_{w1}\cos\varphi_t}{Z_{w2}\cos\varphi_i + Z_{w1}\cos\varphi_t} \qquad T_{\perp} = \frac{2\,Z_{w2}\cos\varphi_i}{Z_{w2}\cos\varphi_i + Z_{w1}\cos\varphi_t}$$

$$1 + \Gamma_{\perp} = T_{\perp}$$

field magnitudes, not power

Content:

Repeating Maxwell's equations

Plane waves

Waveguides

Lumped components

Waves on transmission lines

Wave matrices

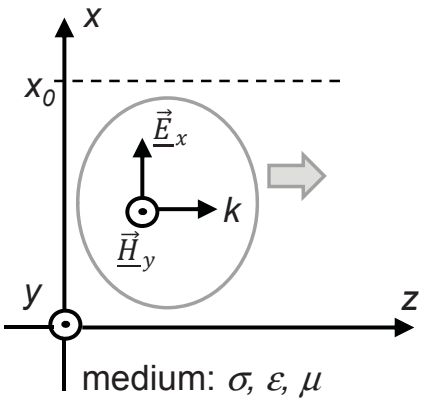
Transmission line circuits

Impedance transformers and filters

Waveguides

Parallel plate waveguide:

- consider an x-polarized wave: introduction of an infinite PEC-plane at x=0 does not affect the field! (because boundary conditions are fulfilled)
- similarly for a PEC-plane at any $x=x_0$



what happens when introducing a PEC-plane at y=0 ???

- cuts space in two halves
- reflection
- standing wave & interference

... PEC ≡ perfect electric conductor,
$\sigma \to \infty$,
Dirichlet boundary condition

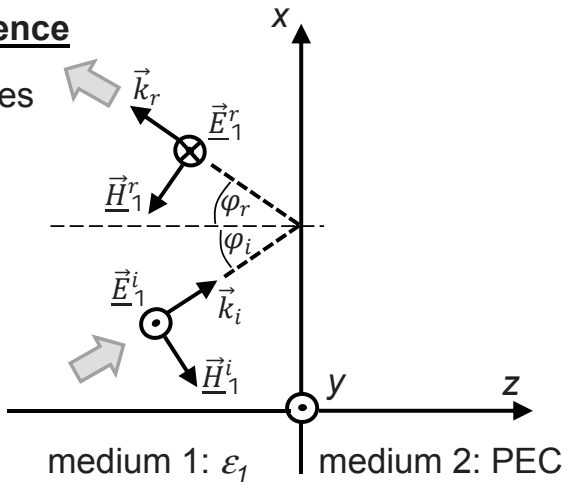
reflection at PEC — oblique incidence

- consider incident and reflected waves (projection of wave vector *k*):

$$\underline{\vec{E}}^i = \vec{e}_y \left|\underline{\vec{E}}^i\right| e^{-j k_1 (x \sin\varphi + z\cos\varphi)}$$

$$\underline{\vec{E}}^r = -\vec{e}_y \left|\underline{\vec{E}}^i\right| e^{-j k_1 (x \sin\varphi - z\cos\varphi)}$$

$$e^{jx} - e^{-jx} = 2j \sin x$$



$$\underline{\vec{E}} = \underline{\vec{E}}^i + \underline{\vec{E}}^r = -2j\, \vec{e}_y \left|\underline{\vec{E}}^i\right| e^{-j k_1 x \sin\varphi} \sin(k_1 z \cos\varphi)$$

projection of k_1 in direction *x*

projection of k_1 in direction *z*

- projection in direction ***x*** (along the PEC-plane):

  propagation with „guided“ wavelength $\lambda_g = \dfrac{\lambda}{\sin\varphi}$

  λ_g being a measure of the distance after which the „field pattern“ repeats

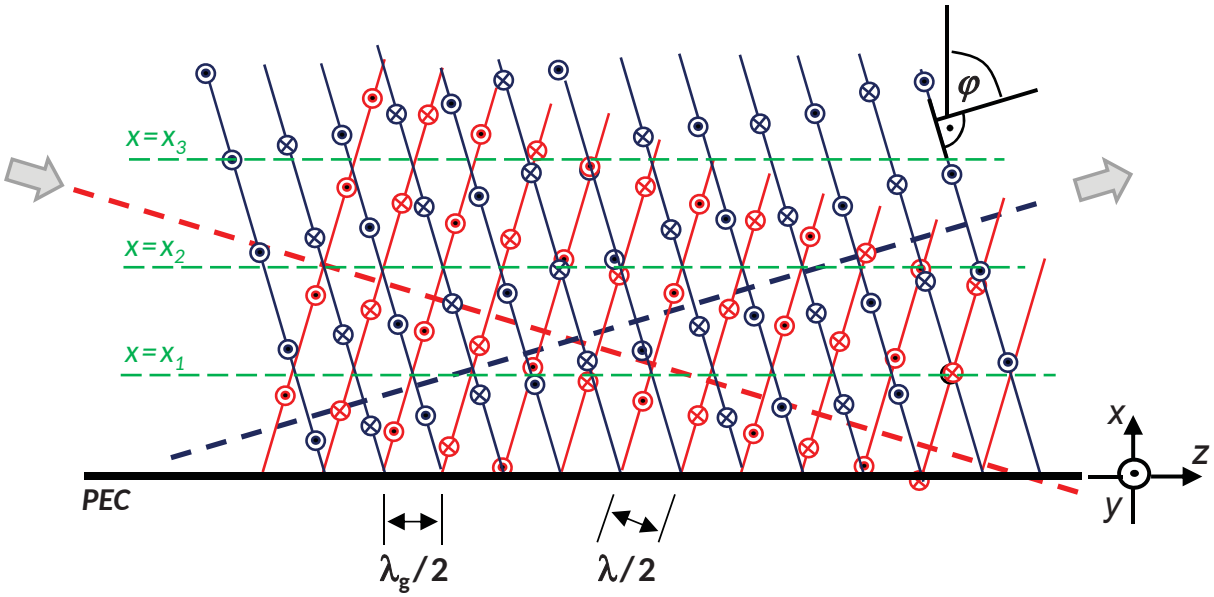
  phase velocity $v_{ph} = f\lambda_g > c$

- projection in direction ***z*** (normal to the PEC-plane):

  repetition of the „field pattern“ after Δz with $\pi = k_1 \Delta z \cos\varphi$

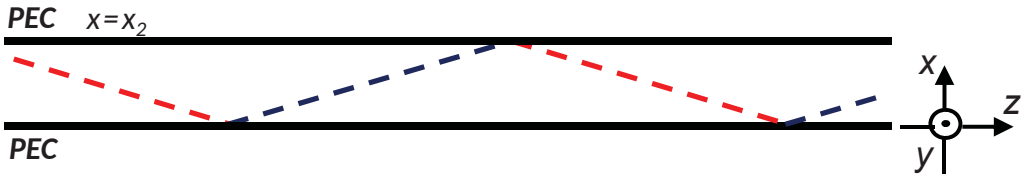
  $$\Delta z = \frac{\lambda}{2\cos\varphi}$$

  → a second PEC-plane at $z = \Delta z$ does not affect the fields!



- extinction of ***E*** at $x = 0$ (PEC boundary condition) and at $x = x_2$
- doubling of magnitude of ***E*** at $x = x_1$ and at $x = x_3$

- ***E*** is y-polarised. PEC-planes at $x=0$ and $x=x_2$:



$$x_2 = \frac{\pi}{k\cos\varphi} \quad \rightarrow \quad x_2 = \frac{m\,\pi}{k\cos\varphi}$$

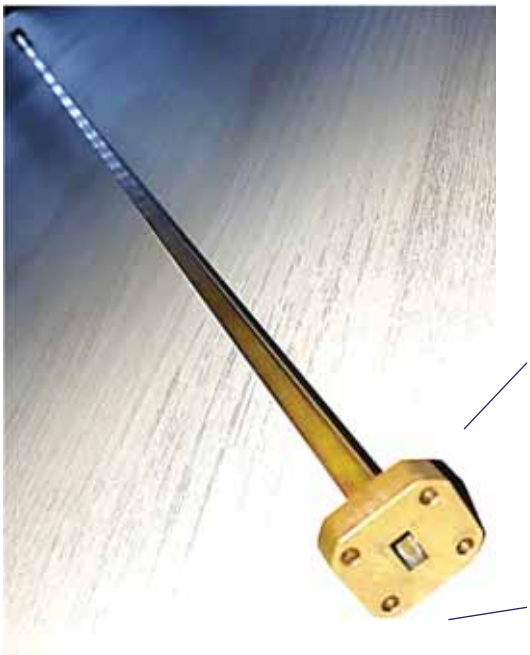
- wave vector / wave number: $k^2 = \beta^2 + k_x^2 = \omega^2\varepsilon\mu$

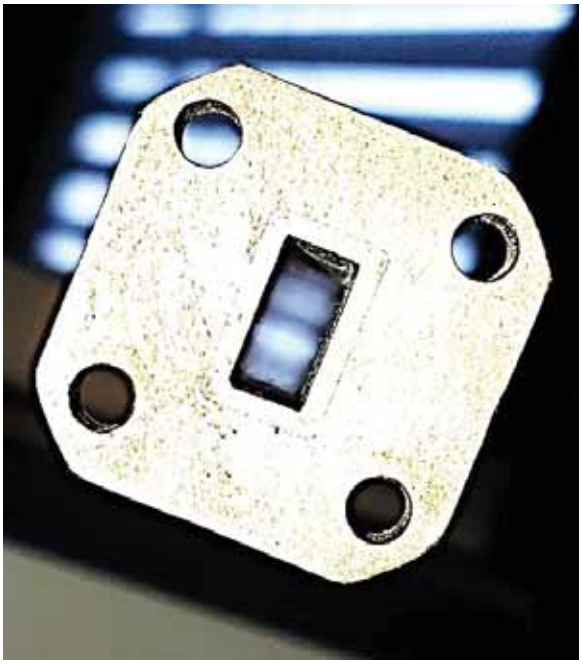
$$k_x = \frac{m\,\pi}{x_2}$$

- additional PEC-planes (#3, #4) at $y=0$ and $y=b$ (always on PEC: E is ⊥):

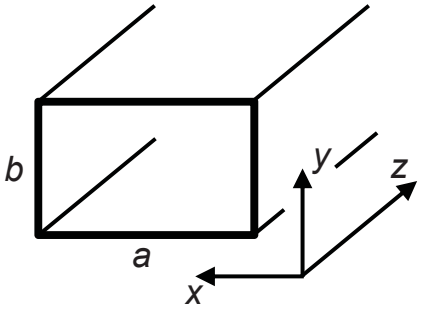
→ **rectangular waveguide (air-filled, metallic)**

rectangular waveguide





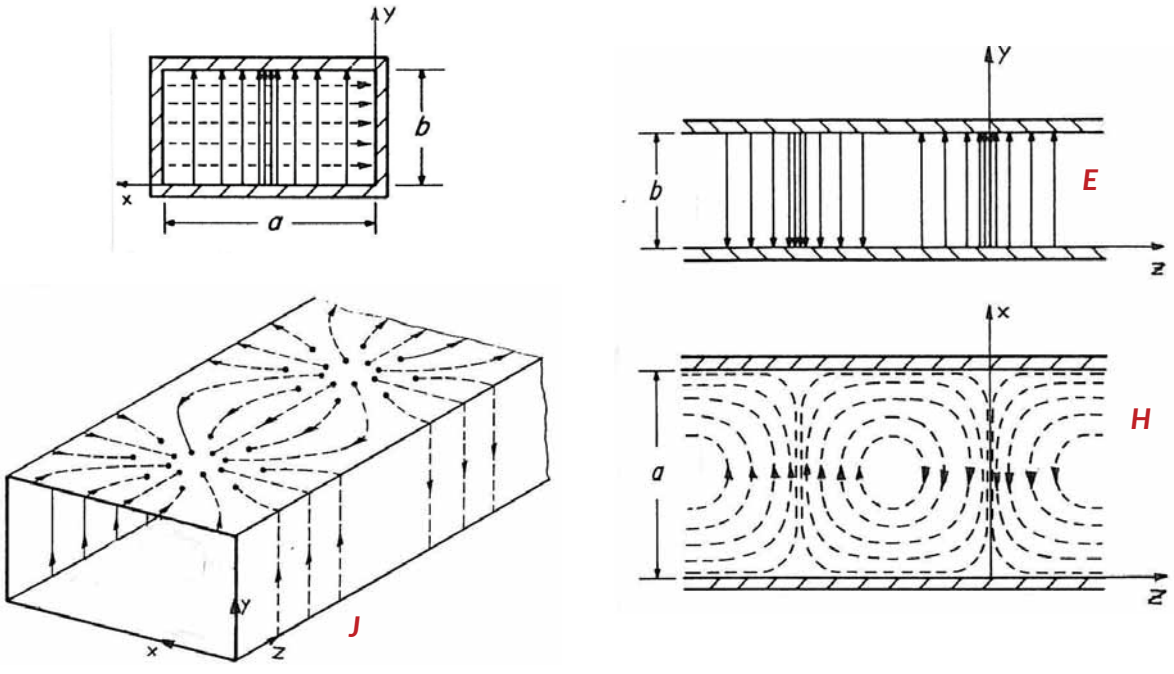
Rectangular waveguide



- cut-off frequency: for given x_2 (and a) and for positive β^2 , the operation frequency must be larger than the cut-off frequency
- cut-off frequency depends on dimension a and on modal index number m
- fundamental mode TE_{10} with $m=1$ has cut-off frequency as $f_{c,\mathrm{TE10}} = \frac{c}{2a}$ (which does not depend on b)
- below cut-off frequency: no propagating waves in the waveguide
- at sufficiently high frequency, several (orthogonal) modes can propagate at the same time

similar characteristics for all cylindrical (i.e., cross-section does not depend on z) metallic or/and dielectric (meaning, with at least two different dielectrics) structures. E.g.: optical fibre waveguide

fields and wall currents (instantaneous views) of TE_{10} mode (H_{10} mode) :



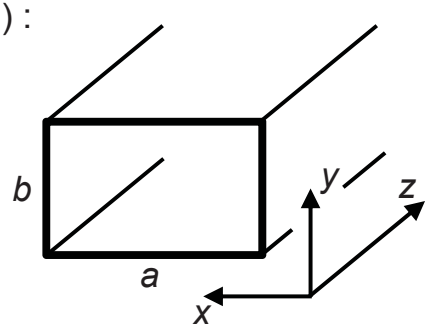
field components of the TE_{10} mode (H_{10} mode) :

$$\underline{H}_z = \underline{H}_0 \cos\frac{\pi x}{a}\, e^{-j\beta z}$$

$$\underline{E}_y = -j\omega\mu\,\frac{a}{\pi}\,\underline{H}_0 \sin\frac{\pi x}{a}\, e^{-j\beta z}$$

$$\underline{H}_x = j\beta\,\frac{a}{\pi}\,\underline{H}_0 \sin\frac{\pi x}{a}\, e^{-j\beta z}$$

$$\underline{E}_x = \underline{E}_z = \underline{H}_y = 0$$





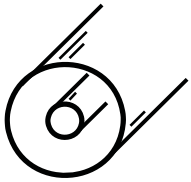
[dorado]

standardised cross-sections of rectangular waveguides:

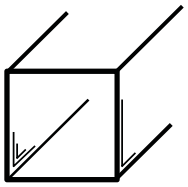
TE_{10} cut-off [GHz]	useable frequency range [GHz]	inner dimensions a x b [mm]	designation of frequency band	waveguide designation
0.909	1.14 – 1.73	165.1 x 82.55	L	WR650
1.37	1.72 – 2.61	109.22 x 54.61	LS	WR430
2.08	2.60 – 3.95	72.14 x 34.04	S	WR284
3.15	3.9 – 6.0	47.55 x 22.15	C	WR187
6.56	8.2 – 12.5	22.86 x 10.16	X	WR90
9.49	11.9 – 18.0	15.80 x 7.90	Ku	WR62
14.06	17.6 – 26.7	10.67 x 4.32	K	WR42
21.09	26.4 – 40.0	7.112 x 3.556	Ka	WR28
39.9	50 – 75	3.759 x 1.880	V	WR15
48.4	60 – 90	3.048 x 1.524	E	WR12
59.1	75 – 110	2.540 x 1.270	W	WR10
73.8	90 – 140	2.032 x 1.016	F	WR08
98.4	110 – 170	1.524 x 0.762	D	WR06
118.1	140 – 220	1.270 x 0.635	G	WR05

Other waveguides (examples):

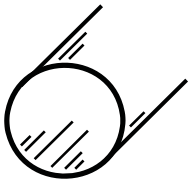
- co-axial waveguide



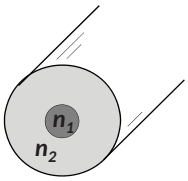
- square waveguide



- (circular) cylindrical waveguide

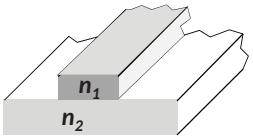


- optical fibre



... without metal!

- (dielectric) ridge waveguide



Content:

Repeating Maxwell's equations

Plane waves

Waveguides

Lumped components

... as frequency grows, components L, C, R behave differently as at low frequency!

Waves on transmission lines

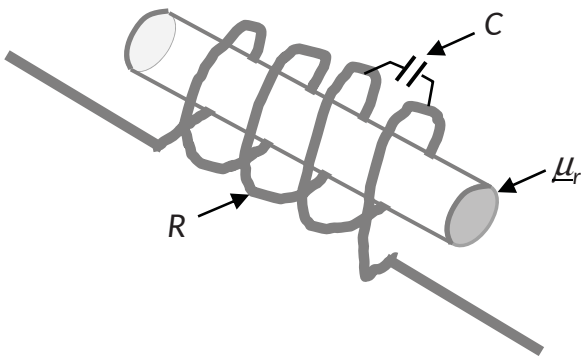
Wave matrices

Transmission line circuits

Impedance transformers and filters

Lumped components

Inductance: An inductance has, in most cases, parasitic capacitive parts, dissipative loss, and – as long as it is not *very* small compared to wavelength (or frequency *very* low) – phase/time delay effects as well as radiation loss.

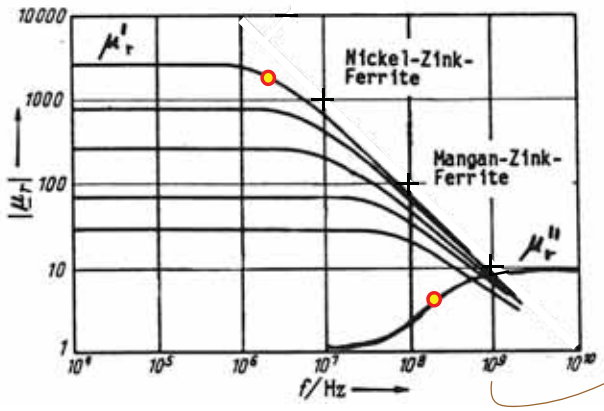


with increasing frequency, the inductance tends to lower, as:
- windings partly shorted by *C*,
- smaller μ_r '

with increasing frequency, loss increases, as:
- skin effect causes higher current density in metal,
- higher loss of permeability (μ_r'')

permeability of ferrites: these are non-conducting, sintered metal oxides

- high permeability and low loss at „HF“ frequency (< 100 MHz)

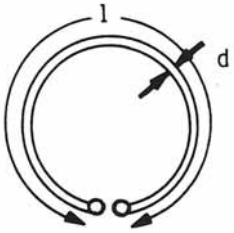


Q-factor (quality-factor) of inductance:

- series connection of *L* with loss resistance: $Q_L = \frac{\omega L}{R_S}$
- Q = 30…300 at HF…VHF frequencies

approximate formulas of coil-inductances at HF-VHF frequency:

conductor loop



Voraussetzung: $l > 50\ d$

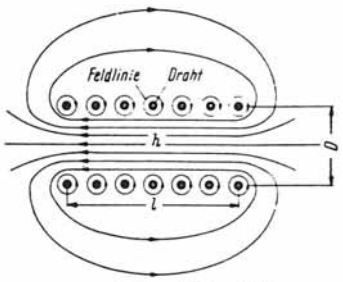
$$\frac{L}{nH} = 2\,\frac{l}{cm}\left(\ln\frac{l}{d} - K_2\right)$$

Kreis: $K_2 = 1{,}07$

Quadrat: $K_2 = 1{,}47$

gleichseit. Dreieck: $K_2 = 1{,}81$

cylinder coil in air



Luftspule mit vielen Windungen

Windungszahl: n

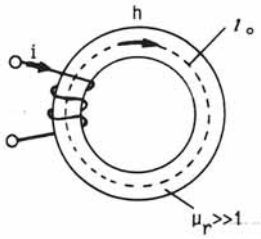
Ohne Streuung:

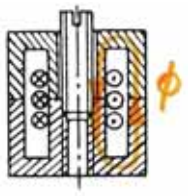
$$L = \frac{\pi}{4}\mu_0\,\frac{n^2D^2}{l}$$

Mit Streuung:

$$L = \frac{\pi}{4}\mu_0\,\frac{n^2D^2}{l + 0{,}45\,D}$$

coil on ring core



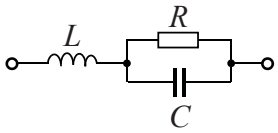


$$L = n^2 A_L$$

A_L given by manufacturer, e.g.: 1´000 nH

capacitor:

- loss mostly due to dissipation in dielectric
- parasitic series inductance due to connecting wires



ex: C = 33 nF, L = 1 nH → f_{res} = 27.7 MHz … meaning: do not chose a very large blocking C (or, place a smaller blocking C in parallel!)

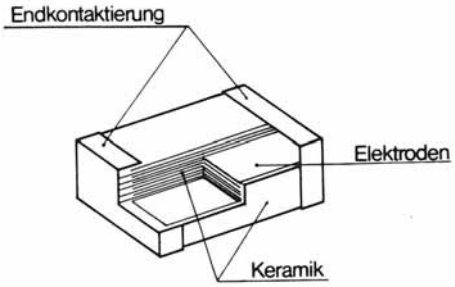
principle: parallel-plate capacitor: $C = \varepsilon_0\varepsilon_r\,\frac{A}{d}$

- materials with large ε_r mostly also have large tanδ
- tanδ grows with frequency
- small *d* contradicts high voltage strength
- some materials with high ε_r are piezo-electric (prone to vibration effects)

example – constructions of capacitors:

ceramic – multi-layer – SMT – capacitor

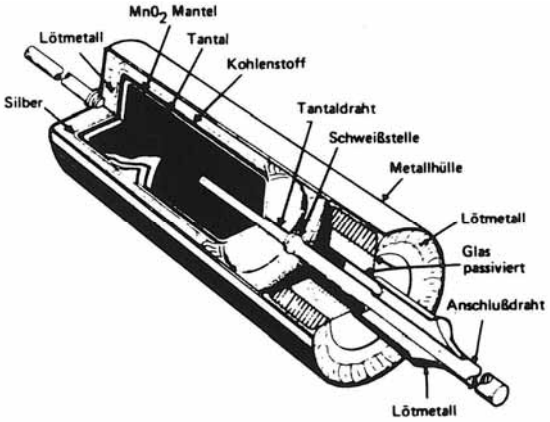
up to ca. 330 pF





Tantal – electrolyte – capacitor

up to ca. 100 µF

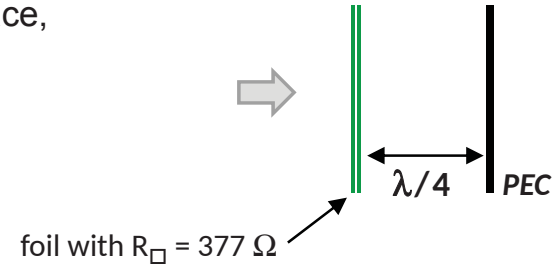


Räumlich klein. Anode: Poröser Körper aus gesintertem Tantalpulver mit großer innerer Oberfläche. Als Elektrolyt dient Mangan-Dioxid. Beim Formieren bildet sich an der Anode eine dünne Schicht von Tantalpentoxid (ε_r = 27). Polarisation!

absorption of electromagnetic waves:

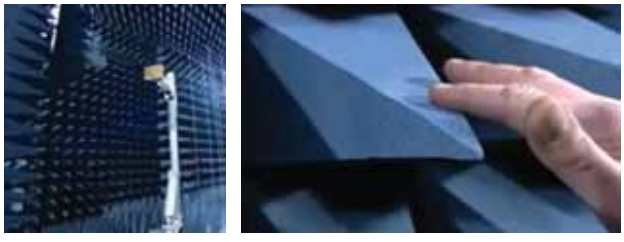
reflection-less transition “from air to loss”; two variants:

- absorbing layer with wave impedace, followed (after λ/4) by reflection (works narrow-banded and in normal incidence only)



- gradual („tapered“) transition „from air to loss“

  → pyramidal absorber



criteria for use of resistors in circuits:



- inductance of connections
- heat removal
- stability/variation over temperature
- size of component compared to wavelength



size/type	on-board footprint a x b [mm]	typ. max. power dissipation [mW]
1206	3.2 x 1.6	250
0805	2.03 x 1.27	125
0603	1.55 x 0.85	100
0402	1.0 x 0.5	63
0201	0.6 x 0.3	50
01005	0.4 x 0.2	30



Content:

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Waves on transmission lines

a typical transmission line has two, separated (isolated) metal conductors in the cross-section.
The fundamental wave traveling along such transmission line has ***E*** and ***H*** only in the cross-section → TEM-wave.

… „transversal-electric-magnetic wave“, i.e., ≤ 2 vectoral components of E-field and ≤ 2 vectoral components of H-field

in the rectangular waveguide, every mode has either E_z (→ TM wave) or H_z (→ TE wave).

… maximum of 5 vectoral components of E- and H-field, when choosing the coordinate system suitably — one z-component is zero

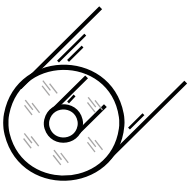
… this follows from Maxwell by some maths → to be shown in module „Radio Frequency Technology“

transmission lines / waveguides with inhomogeneous dielectric (in the cross-section) have most modes with E_z and H_z → hybrid waves / modes

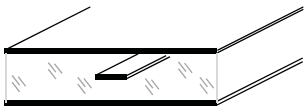
in microstrip line, the (fundamental) hybrid mode has very small E_z and H_z → quasi-TEM wave / mode

Typical TEM- and quasi-TEM- transmission lines:

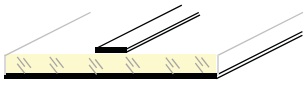
- co-axial line



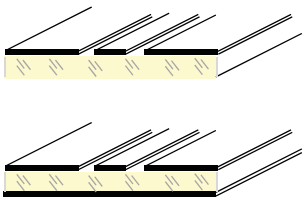
- stripline



- micro-strip line

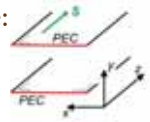


- CPW (co-planar waveguide)



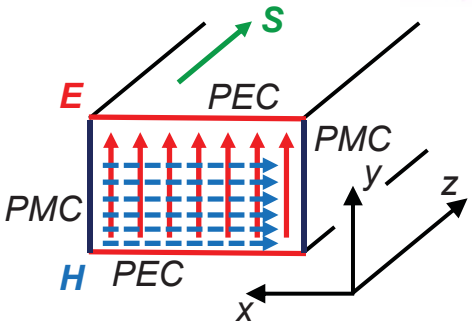
Ex: TEM transmission line, parallel-plate waveguide :

structure:



- insertion of conducting planes in the field of a linearly polarized plane wave

- conducting strips of width ***a*** in separation distance ***b*** :



$$U = \int_{y=0}^{b} E_y \, dy = E_y \, b$$

$$I = \oint H \, ds = H_x \, a$$

$$Z_{\text{Leitung}} = \frac{U}{I} = \frac{E_y \, b}{H_x \, a} = Z_{F0} \, \frac{b}{a}$$

$$\vec{J} = \vec{n} \times \vec{H}$$

… for ***b = a*** the line impedance equals the wave impedance of free space

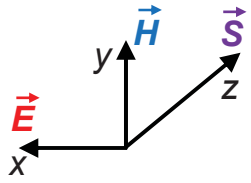
Power flow along a transmission line:

why „conjugate complex“? … see next slide

- effective power: $P = \frac{1}{2} \, Re\{U \, I^*\} = \frac{1}{2} \, Re\{E_y b \, H_x^* a\} = \frac{ab}{2} Re\{E_y \, H_x^*\}$

- power density [W/m²] : $S = \frac{P}{ab} = \frac{1}{2} Re\{E_y \, H_x^*\}$

- Poynting vector: $\vec{S} = \frac{1}{2} Re\{\vec{E} \, \vec{H}^*\} \, \vec{e_z}$



- Ex: plane wave in air (breakdown field strength E_{max} approx. 3 kV/mm):

$$S_{\text{max}} = \frac{1}{2} Re\left\{E_y \, \frac{1}{Z_{F0}} \, E_y^*\right\} = \frac{1}{2} \, \frac{E_{\text{max}}^2}{Z_{F0}} \approx 12 \, \frac{\text{GW}}{\text{m}^2}$$

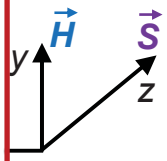
Power flo

- effective

$E_y\ H_x^*\}$

- power d

- Poynting



- Ex: plan

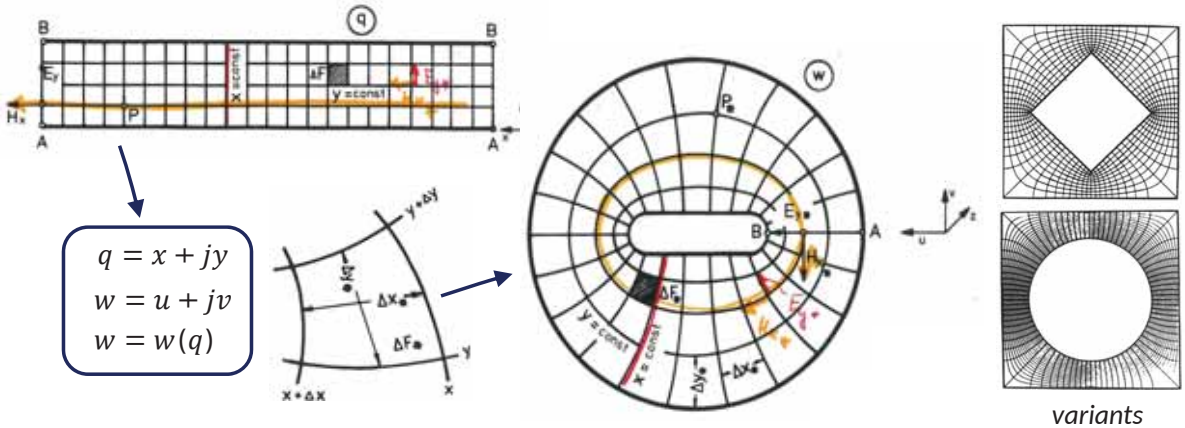
3 kV/mm):

S_{ma}

… with the "conjugate-complex" operation, the real part becomes "constant"

Co-axial transmission line:

- TEM (fundamental) wave / mode
- a conformal mapping between parallel-plate waveguide and co-axial line:

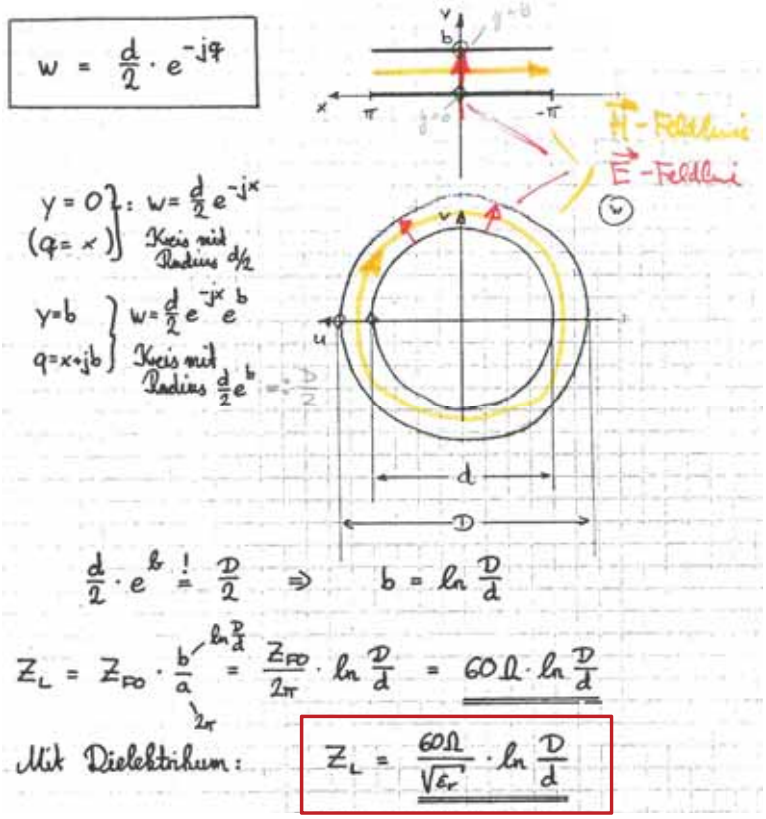


variants

- possibility for analytical calculation of impedance of co-axial line:
  … conformal: keeps right angles and boundary shapes
  … same solution of wave equation: can solve for Z, γ etc.

Co-axial line:

- impedance …
  … derived using a clever conformal mapping:



Co-axial transmission line:

- E-field strength (max. on inner conductor's surface): $E_{max} = \frac{2\,U_{max}}{d \ln D/d}$

- attenuation: $\alpha = \alpha_G + \alpha_R = \frac{\pi f \sqrt{\varepsilon_r}}{c} \tan\delta + \frac{\sqrt{\mu f}}{2\,Z_L \sqrt{\sigma \pi}} \left(\frac{1}{d} + \frac{1}{D}\right)$

  attenuation in the dielectric — attenuation in metal

- min. attenuation → D/d = 3.6 , Z_L = 77 Ω / sqrt(ε_{rel})

- max. voltage strength → D/d = 2.72 , Z_L = 60 Ω / sqrt(ε_{rel})

- max. power transfer → D/d = 1.65 , Z_L = 30 Ω / sqrt(ε_{rel})

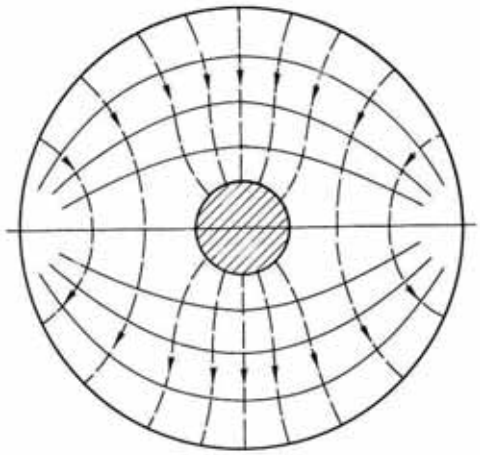
- typical RF impedance = 50 Ω

Co-axial transmission line:

- the fundamental / monomode frequency range (starts at zero frequency, DC) ends with the possible propagation (or, with the cut-off frequency) of the first higher-order mode of propagation :

$$\lambda_{c,TE11} \approx \pi \frac{D+d}{2} \sqrt{\varepsilon_r}$$

($c_0 = f_c \cdot \lambda_c$)



Cross-section view of first higher-order mode (TE11) in co-axial line. Solid lines – H-field, dashed lines – E-field.

- Ex: „type 86 cable“: D = 1.65 mm,
  d = 0.53 mm,
  ε_r = 1.86,
  λ_c = 4.67 mm,
  f_c = 64 GHz,
  → f_{max} ~ 40 GHz

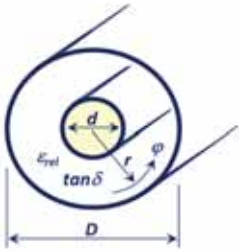


Co-axial line, attenuation / loss:

$$\alpha = \alpha_G + \alpha_R = \frac{\pi f \sqrt{\varepsilon_r}}{c} \tan\delta + \frac{\sqrt{\mu f}}{2 Z_L \sqrt{\sigma \pi}} \left(\frac{1}{d} + \frac{1}{D}\right)$$

… attenuation in the dielectric

… attenuation in metal



dielectric loss ~ f

(material property tanδ often increases with f)

conductor (metal) loss ~ $\sqrt{f}$

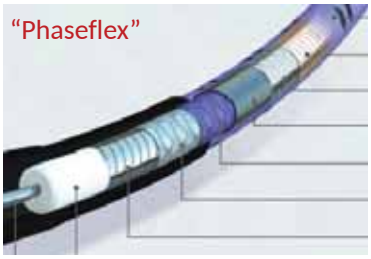
(roughness effects often decrease “effective” material property σ with f)

but: the conductor (metal) loss increases with smaller cross-section (here: d,D) !

→ structures (transmission lines, resonators) require large cross-section for low loss at high f

but: for “mono-mode” behavior, the cross-section of the transmission line must reduce with increasing f !

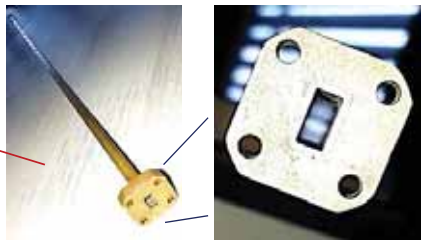
Transmission Line Loss, Generalization:

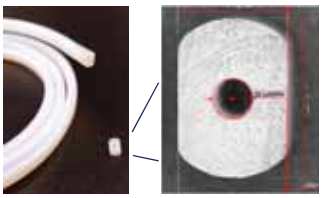


[W.L.Gore & Ass.]

"real" examples:

- coax (Ag) with foam dielectric (ε_r=1.4):
  - max f = 3 GHz: insertion loss 0.5 dB/m
  - max f = 18 GHz: insertion loss 0.8 dB/m
  - max f = 26.5 GHz: insertion loss 1.6 dB/m
    - ... brass TE_{10} waveguide 30 GHz (meas.): 1.45 dB/m
  - max f = 40 GHz: insertion loss 2.9 dB/m
    - ... PTFE fiber 60 GHz (meas.): 1.5 dB/m
  - max f = 67 GHz: insertion loss 6.4 dB/m
    - ... Ag TE_{10} waveguide 100 GHz (theory): 1.8 dB/m
  - max f = 110 GHz: insertion loss 13 dB/m





- ***much*** higher loss (and smaller cross-section) for lines on PCB and, ***much*** worse, on-chip

Co-axial transmission line:

- practical realizations. issues:
  - . frequency range / limit
  - . power handling capability
  - . attenuation
  - . mech. flexibility / stability



corrugated (2 inch diameter)



50 GHz test leads (inner diameter of outer conductor = 2.4 mm)



semi-rigid, SMA

Co-axial transmission line:

- connectors:

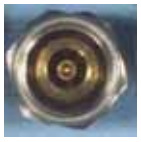
BNC  $f_{max} \approx 2$ GHz

N / QN   $f_{max} \approx 12$ / 18 GHz

APC-7  $f_{max} \approx 18$ GHz

SMA / QMA   $f_{max} \approx 18$ GHz

---

3.5mm 26 GHz  2.92mm "K" 40 GHz 

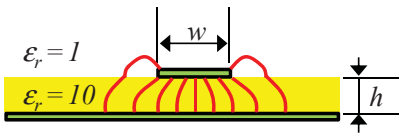
2.4mm 50 GHz  1.85mm "V" 65 GHz 

allows for mechanical connection

- and many more others !

Microstrip transmission line:

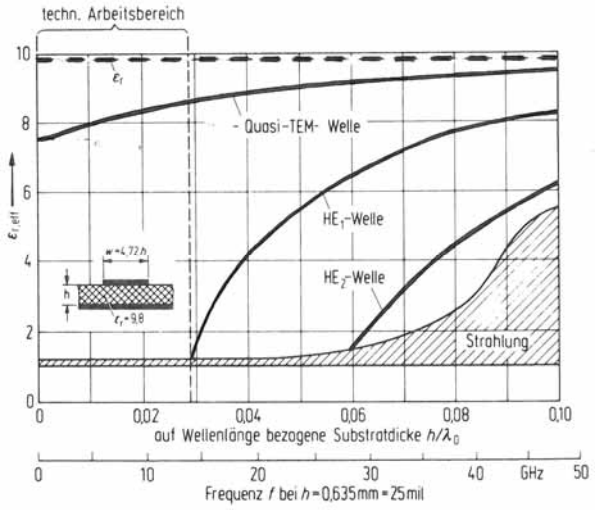
- field energy is mainly concentrated between metal strip and ground plane, the more
  . the wider the metal strip,
  . the larger substrate permittivity,
  . the thinner the substrate,
  . the higher the frequency.



→ concept of $\varepsilon_{r,eff}$

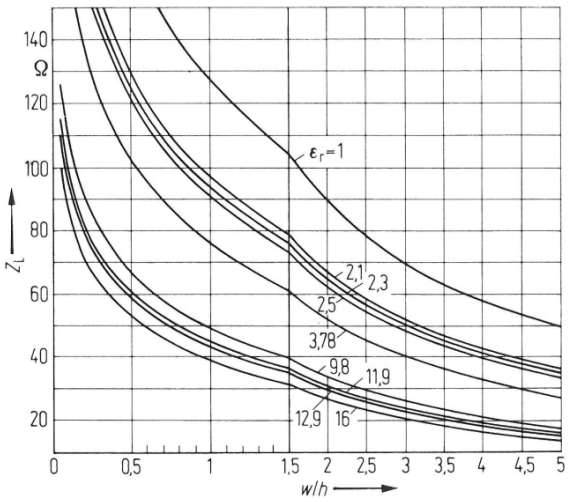
- fundamental mode („quasi-TEM“) cut-off at zero

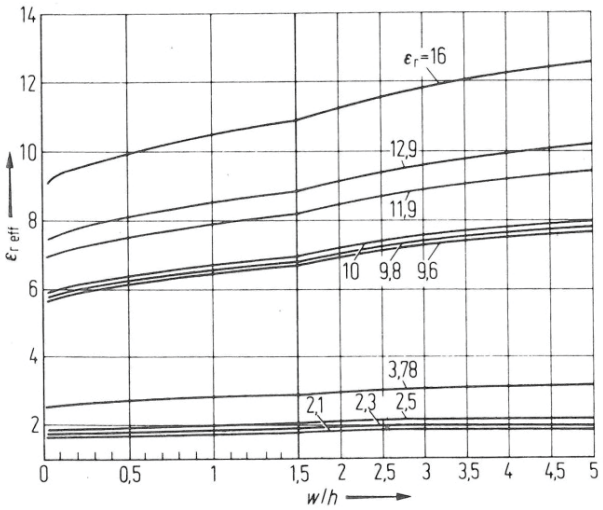
- higher-order mode limits the practically useable frequency range



Microstrip transmission line:

- Z_L and $\varepsilon_{r,eff}$ depend on line geometry:



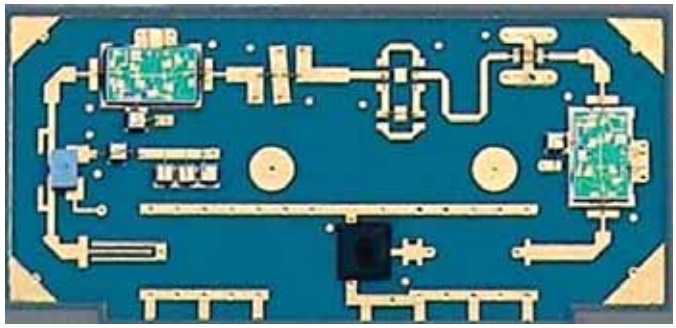


[Zinke Brunswig] , [Wheeler 1977]

Microstrip transmission line:

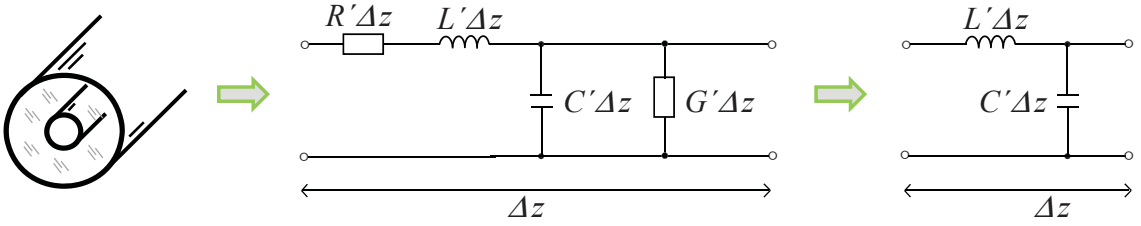
- practical issues:
  - for small line loss, use small ε_{rel} and large h
  - for small coupling (between close-by lines), use large ε_{rel} and small h
  - for small radiation leakage (or leakage into *surface waves*), use small ε_{rel} and small h
  - for large impedance, use small ε_{rel} and large h
  - for small impedance, use large ε_{rel} and small h

→ design choice is always compromise



microstrip circuits, 30 GHz, on LTCC and on GaAs MMIC

Equivalent circuit of a transmission line:



$$U(z+\Delta z,t) - U(z,t) = -L' \, \Delta z \, \frac{\partial I(z,t)}{\partial t}$$

$$I(z+\Delta z,t) - I(z,t) = -C' \, \Delta z \, \frac{\partial U(z+\Delta z,t)}{\partial t}$$

$$I(z+\Delta z,t) - I(z,t) = -C' \, \Delta z \, \frac{\partial U(z,t)}{\partial t} + L'C' \, (\Delta z)^2 \frac{\partial^2 I(z,t)}{\partial t^2}$$

$$\frac{\partial U(z,t)}{\partial z} = -L' \, \frac{\partial I(z,t)}{\partial t} \quad , \quad \frac{\partial I(z,t)}{\partial z} = -C' \, \frac{\partial U(z,t)}{\partial t}$$

$$\frac{\partial^2 U(z,t)}{\partial z^2} = L'C' \, \frac{\partial^2 U(z,t)}{\partial t^2} \quad , \quad \frac{\partial^2 I(z,t)}{\partial z^2} = L'C' \, \frac{\partial^2 I(z,t)}{\partial t^2}$$

$$\frac{\partial^2 \underline{U}(z)}{\partial z^2} = -\omega^2 \, L'C' \, \underline{U}(z) \quad , \quad \frac{\partial^2 \underline{I}(z)}{\partial z^2} = -\omega^2 \, L'C' \, \underline{I}(z)$$

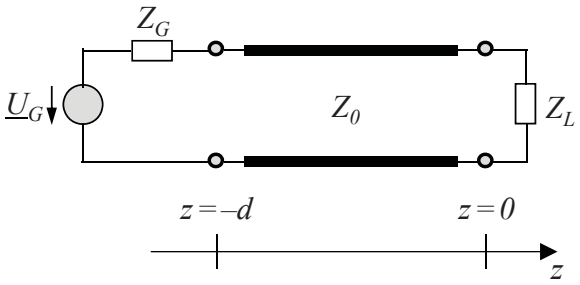
$$\beta = \omega\sqrt{L'C'} \quad , \quad v = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}} \quad , \quad Z_0 = \sqrt{\frac{L'}{C'}}$$

$$\underline{U}(z) = \underline{U}_0^+ \, e^{-j\beta z} + \underline{U}_0^- \, e^{+j\beta z} \quad , \quad \underline{I}(z) = \frac{\underline{U}_0^+}{Z_0} \, e^{-j\beta z} - \frac{\underline{U}_0^-}{Z_0} \, e^{+j\beta z}$$

… this is identical to the propagation of a plane wave !

Reflection coefficient and impedance transformation:

- transmission line with Z_0 „begins“ at $z=-d$. At $z=0$, it is loaded with Z_L .



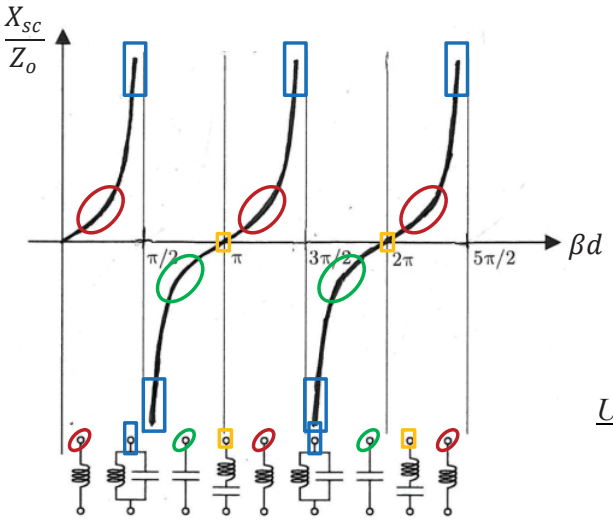
$$\Gamma(z=0)=\Gamma_L=\frac{Z_L-Z_0}{Z_L+Z_0} \quad , \quad \Gamma(z=-d)=\Gamma_L\, e^{-2j\beta d}$$

$$Z(z=-d)=Z_0\,\frac{e^{j\beta d}+\Gamma_L\, e^{-j\beta d}}{e^{j\beta d}-\Gamma_L\, e^{-j\beta d}}=Z_0\,\frac{Z_L+jZ_0\tan\beta d}{Z_0+jZ_L\tan\beta d}$$

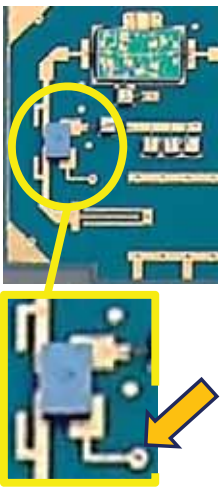
… in analogy to the plane wave (slide 2.18) : special case „λ/4 line“ — d = λ/4 , βd = π/2 , tan βd → ∞ } $Z(z=-d)=\frac{Z_0^2}{Z_L}$

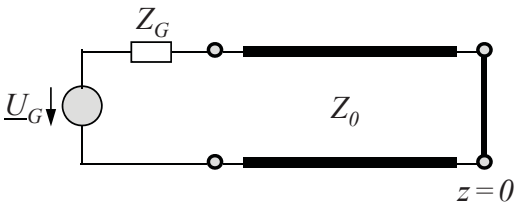
line, short-circuited at its end: $Z_L=0$

- purely imaginary! $Z_{sc}(z=-d)=jZ_0\tan\frac{2\pi d}{\lambda}$



„DC return“:

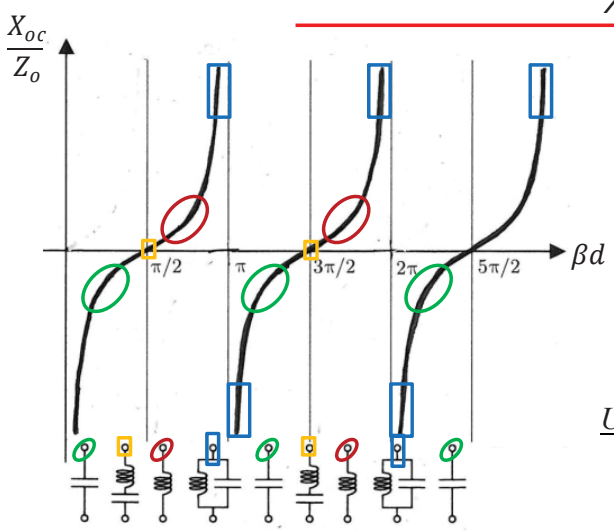


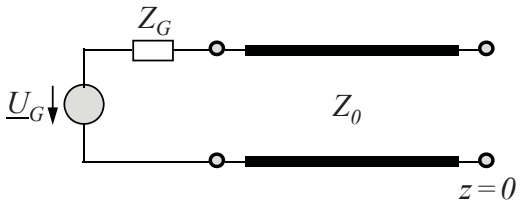


… input reactance of a line, which is short-circuited at its end

line, open-circuited at its end: $Z_L \to \infty$

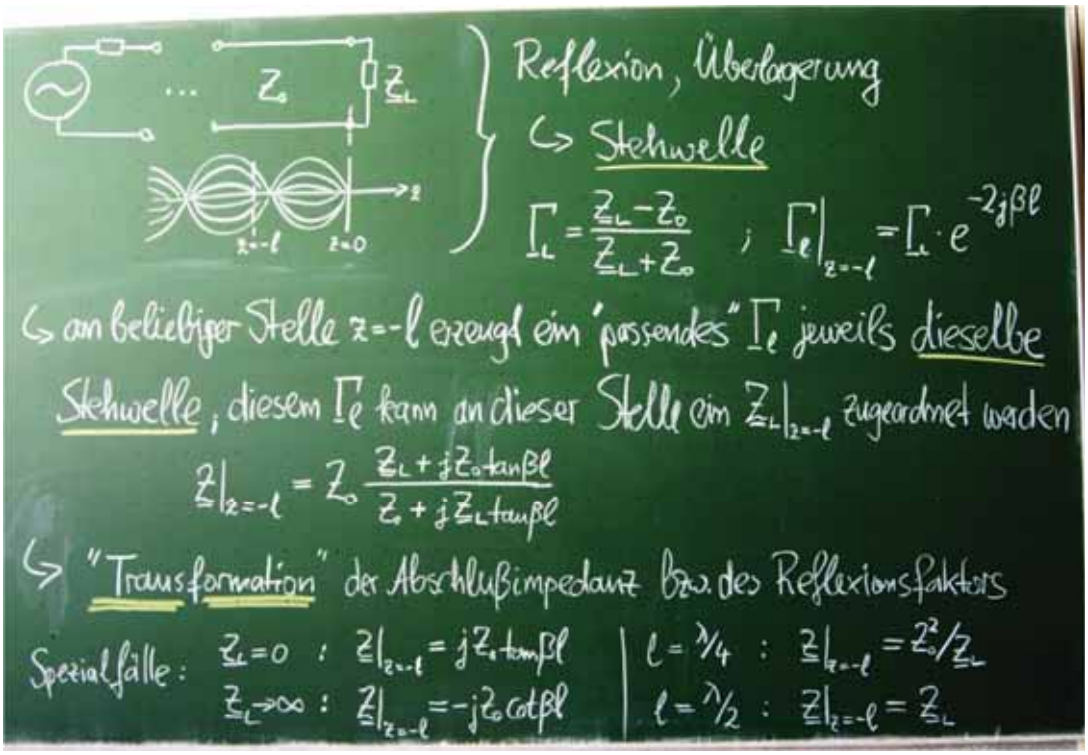
- purely imaginary! $Z_{oc}(z=-d) = \dfrac{-j\,Z_0}{\tan\dfrac{2\pi d}{\lambda}}$





… input reactance of a line, which is open-circuited at its end

… intermediate summary:



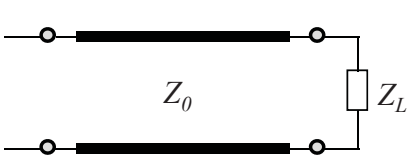
transformation of reflection coeff. and impedance in the Smith chart:

P.H. Smith, Bell Labs 1944

„diagram“

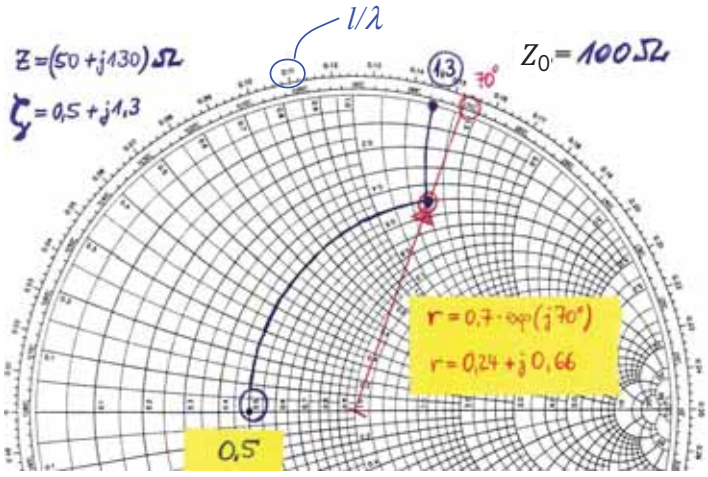
$$(\Gamma=)\ r = \frac{\frac{Z_L}{Z_0} - 1}{\frac{Z_L}{Z_0} + 1} = \frac{\zeta - 1}{\zeta + 1} = 1 - \frac{2}{\zeta + 1}$$

$$\zeta = \frac{Z_L}{Z_0}$$

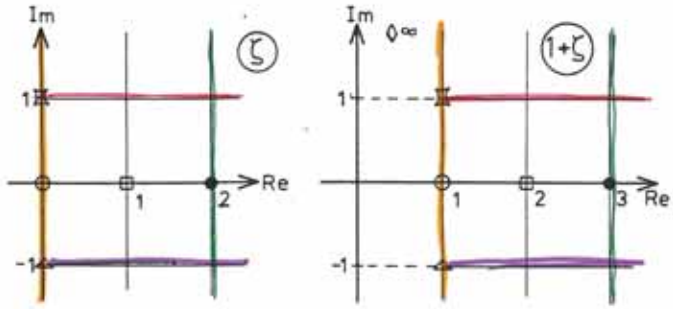


- the Smith chart realises a conformal mapping $\boldsymbol{r \leftrightarrow \zeta}$

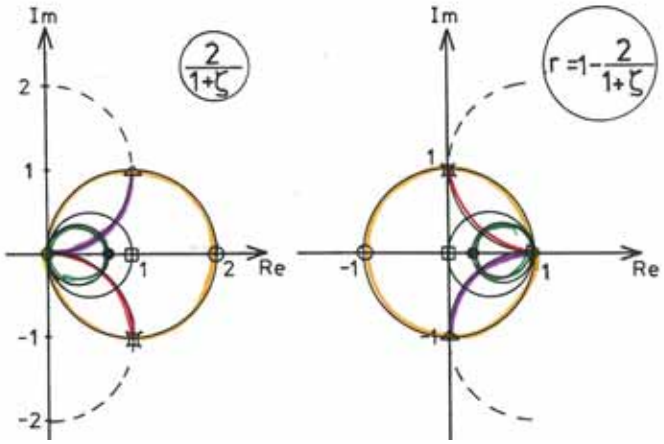
(see next slide)



$Re\{\zeta\} > 0$



... the complex reflection coefficient $\boldsymbol{r}$ in the cartesian coordinate system and the normalized impedance $\zeta = \frac{Z_L}{Z_0}$ located „on the circles“



$|r| < 1$

transformation of reflection coeff. and impedance in the Smith chart:

$$\underline{\Gamma}_L = \frac{\underline{z}-1}{\underline{z}+1}$$

Conformal mapping: Möbius transformation

„invisible“: cartesian coordinate system for $\underline{\Gamma}_L$

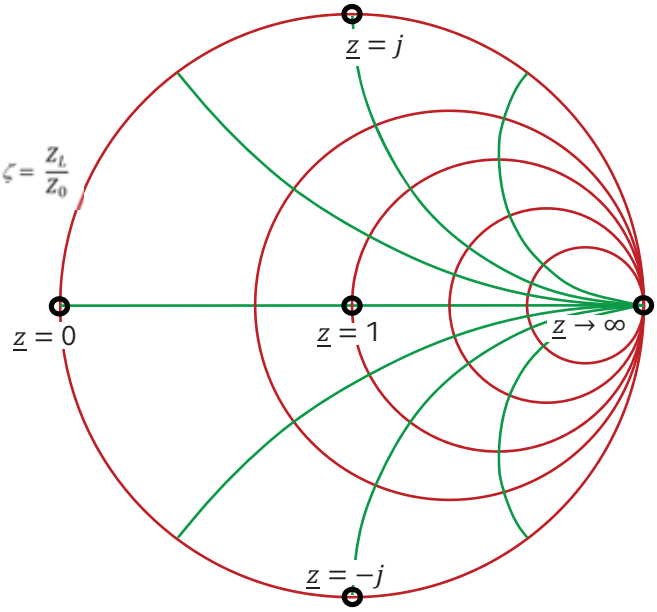
„transformed“ coordinate system for $\underline{z}$:

... $\underline{z}$ named before: $\zeta = \frac{Z_L}{Z_0}$

constant Re($\underline{z}$) : red circles

constant Im($\underline{z}$) : green circles

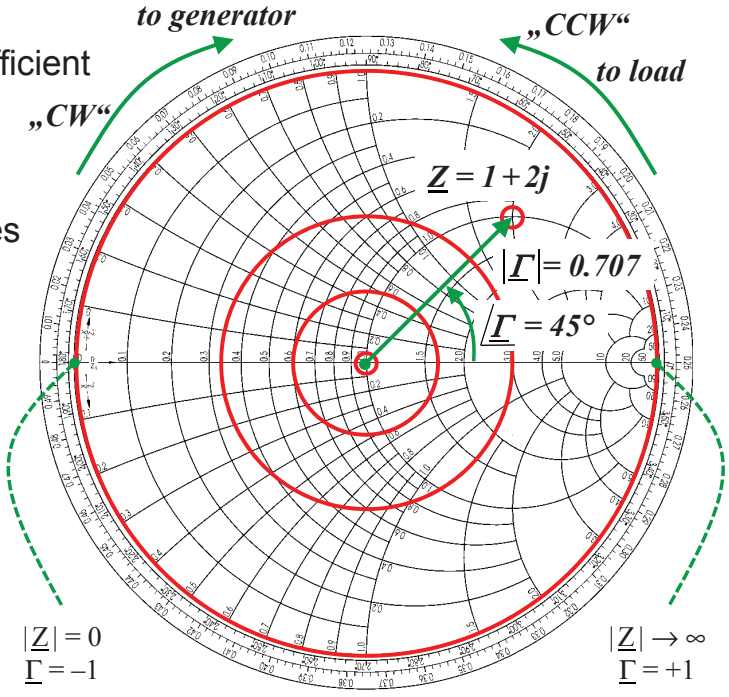
> „overlay“ of the two coordinate systems, and then „removal“ of the cartesian coordinate system for clarity & visibility



transformation of the reflection coefficient in the Smith chart:

- concentric circles: constant magnitude of reflection coefficient
- transformation along the line means: phase of the reflection coefficient changes but its magnitude remains constant
- CW: transformation towards generator
- CCW: transformation towards load

$$\underline{\Gamma}_{z=-l} = \underline{\Gamma}_{z=0}\, e^{-2j\beta_1 l}$$



conversion impedance ↔ admittance in the Smith chart:

… also equals a transformation over 90° line length

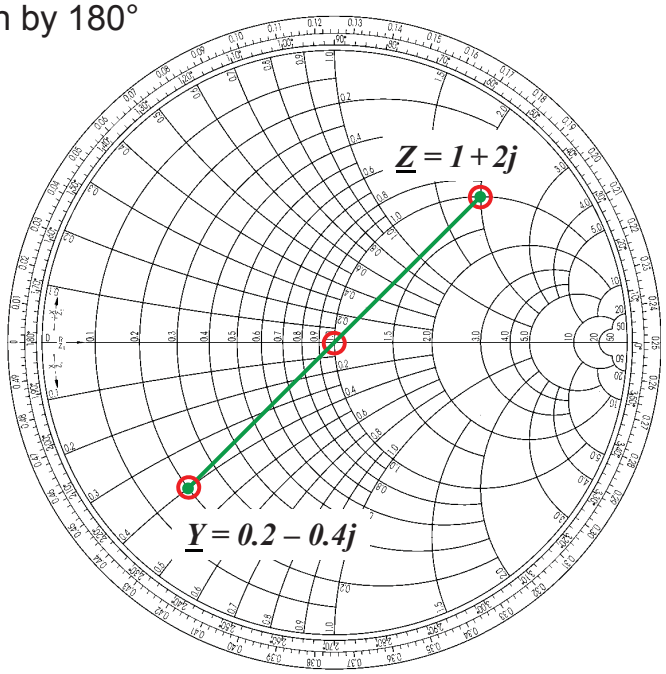
Y ↔ Z corresponds to a rotation by 180°

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\frac{Z_L}{Z_0} = \frac{1+\Gamma_L}{1-\Gamma_L} \qquad \frac{Y_L}{Y_0} = \frac{1-\Gamma_L}{1+\Gamma_L}$$

$$e^{j\pi} = -1$$

normalization: from Z_0 to $1/Y_0$!



impedance transformation in Smith chart:

… $Z(z=-d) = Z_0 \dfrac{Z_L + jZ_0 \tan\beta d}{Z_0 + jZ_L \tan\beta d}$

(i) Z(z=0) → Γ(z=0)

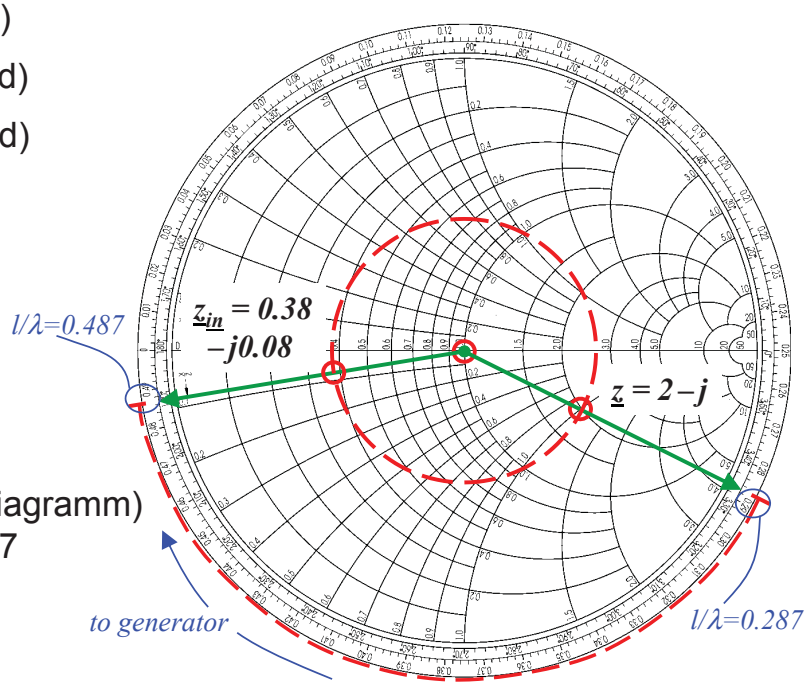
(ii) Γ(z=0) → Γ(z=–d)

(iii) Γ(z=–d) → Z(z=–d)

example:

$Z_0 = 75\,\Omega$
$Z_L = (150 - j75)\,\Omega$
$d/\lambda = 0.2$

$z = 2 - j$
$d'/\lambda = 0.287$ (aus Diagramm)
… CW, +0.2 → 0.487
$z_{in} = 0.38 - j0.08$
$Z_{in} = (28.5 - j6)\,\Omega$



series connection of a line and a λ/4 transformer:

eg: matching $Z_L = (25+j75)\,\Omega$ to 50 Ω

(i) use a line to „rotate“ to a purely-real impedance;

(ii) use λ/4 transformer to 50 Ω

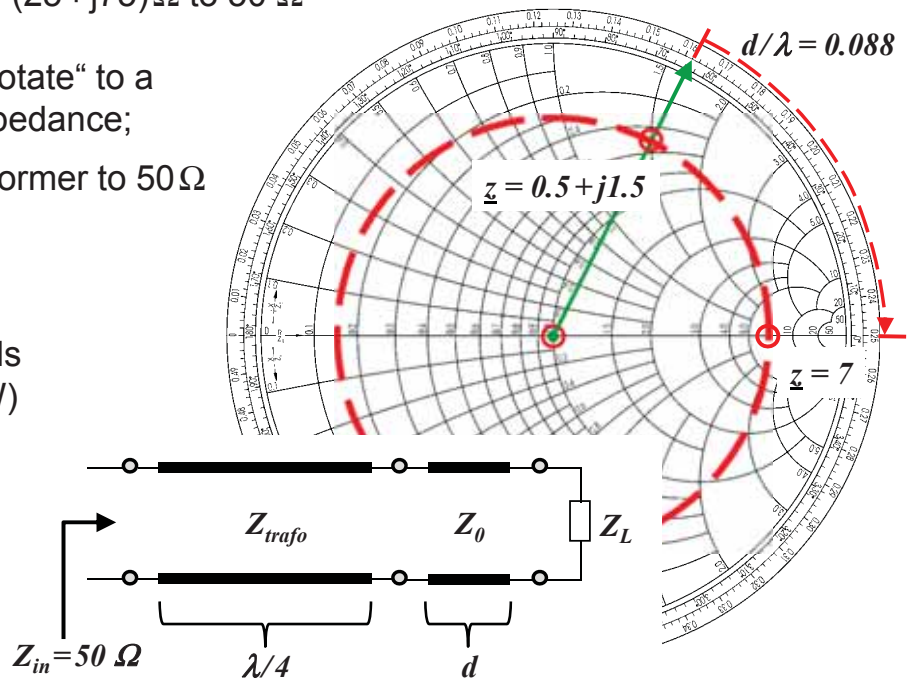
$Z_0 = 50\,\Omega$

$\underline{z} = 0.5 + j1.5$

„rotate“ towards generator (CW)

$z(-d) = 7$
$Z(-d) = 350\ \Omega$

$Z_{trafo} = 132\ \Omega$



matching by series – C :

eg: matching $Z_L = (30-j50)\,\Omega$ to 50 Ω

(i) line of length *d* transforms the real part of impedance to one;

(ii) *C* compensates then the imaginary part

$Z_0 = 50\,\Omega$

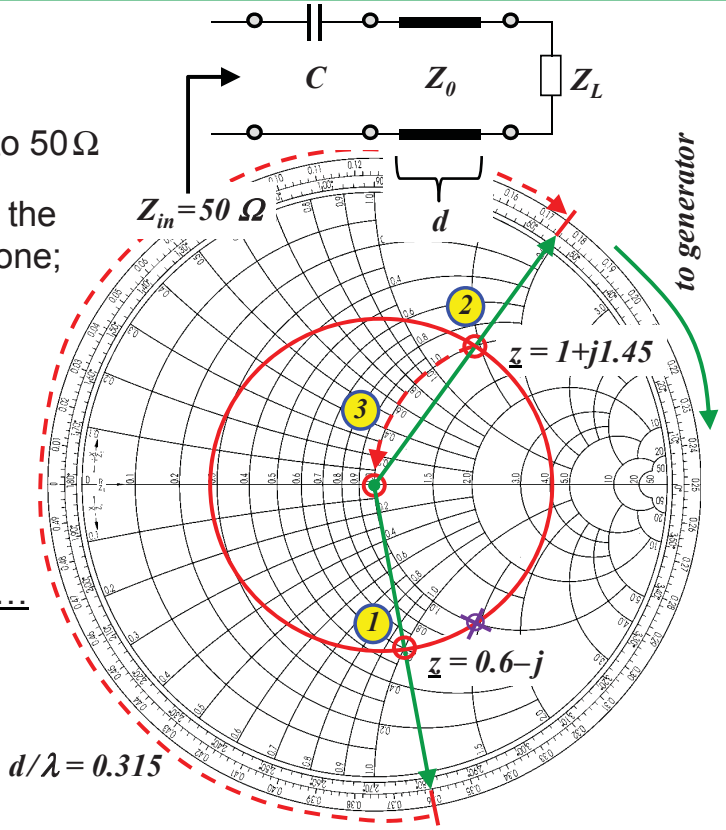
(1) $\underline{z} = 0.6 - j$

rotate towards generator (CW) by $d/\lambda = 0.315 \rightarrow \underline{d = \ldots}$

(2) $X_d/Z_0 = j1.45$

(3) $-j1.45 = \dfrac{1}{j\omega C Z_0}$

$\rightarrow \underline{C = \ldots}$



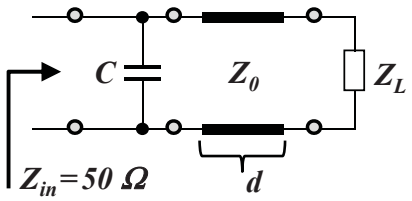
matching by shunt-C in the Smith chart:

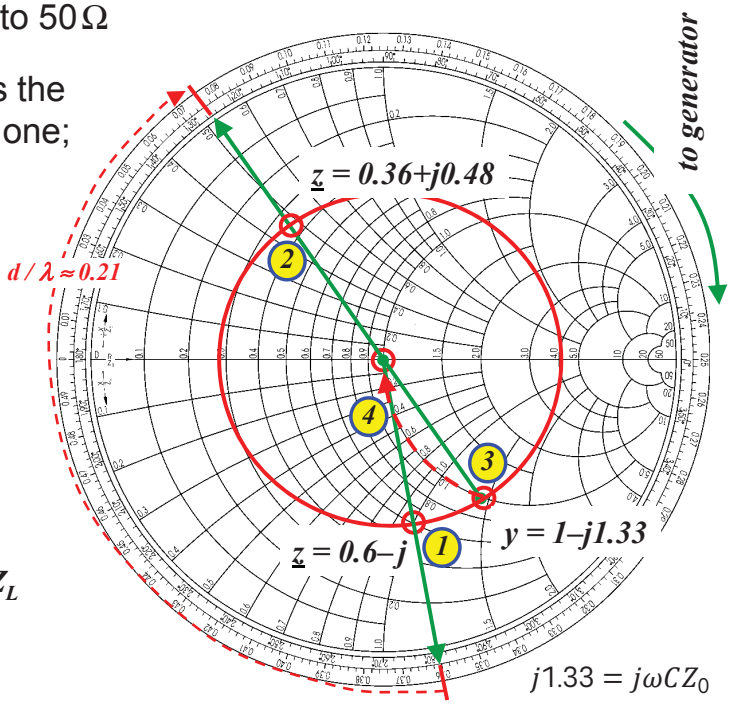
eg: matching $Z_L = (30 - j50)\,\Omega$ to $50\,\Omega$

(i) line of length *d* transforms the real part of admittance to one;

(ii) *C* compensates then the imaginary part

1. $Z_0 = 50\,\Omega$; $\underline{z} = 0.6 - j$
2. rotation (CW)
3. inversion $\underline{Z} \rightarrow \underline{Y}$
4. addition $jb_C = j\omega C Z_0$





matching by shunt-L in the Smith chart:

eg: matching $Z_L = (30 - j50)\,\Omega$ to $50\,\Omega$

(i) line of length *d* transforms the real part of admittance to one;

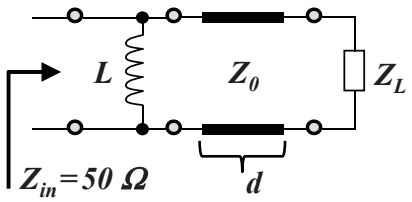
(ii) *L* compensates then the imaginary part

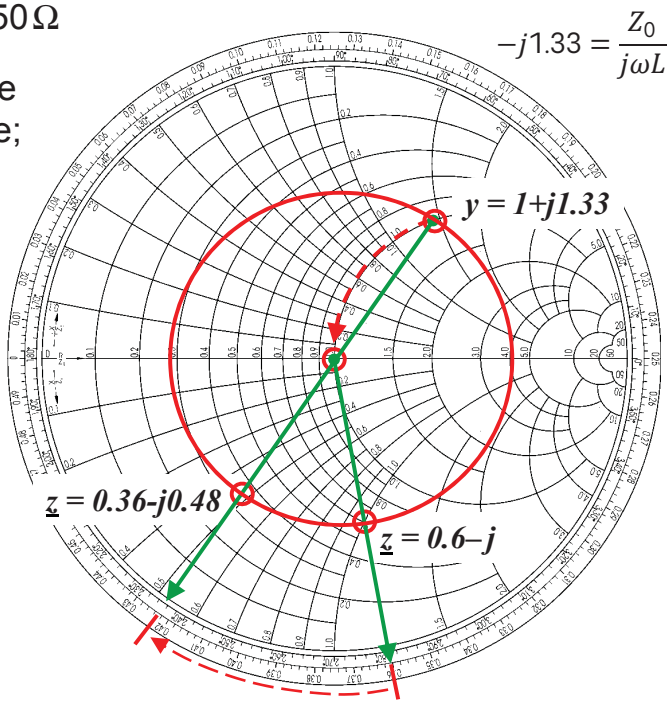
$Z_0 = 50\,\Omega$; $\underline{z} = 0.6 - j$

rotation (CW)

inversion $\underline{Z} \rightarrow \underline{Y}$

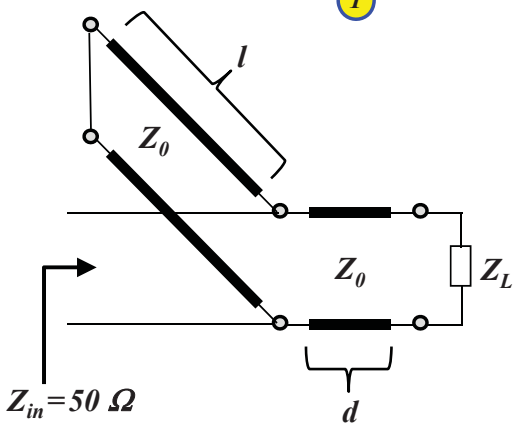
addition $B_L = Z_0 / j\omega L$

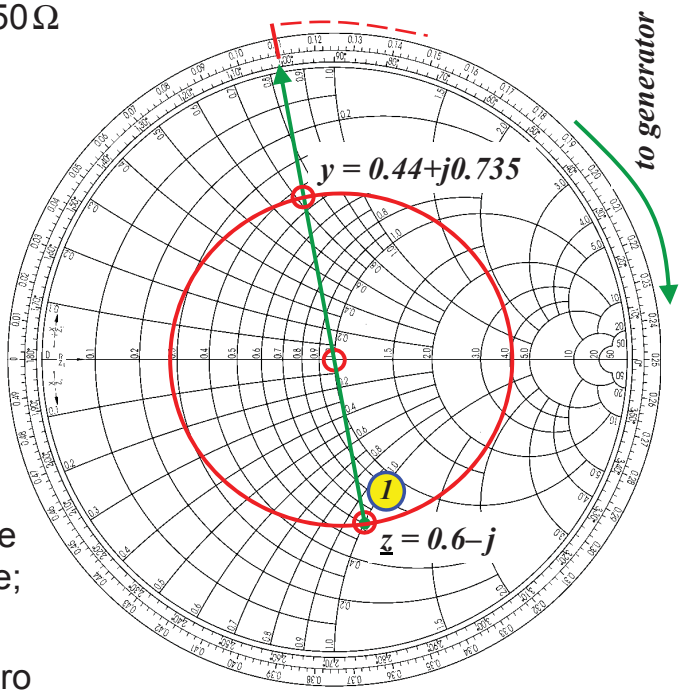




matching by a parallel shorted stub-line in the Smith chart (1/2):

eg: matching $Z_L = (30 - j50)\,\Omega$ to $50\,\Omega$

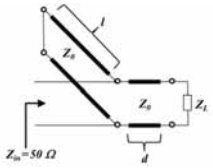




(i) line of length *d* transforms the real part of admittance to one;

(ii) line of length *l* compensates then the imaginary part to zero

matching by a parallel shorted stub-line in the Smith chart (2/2):



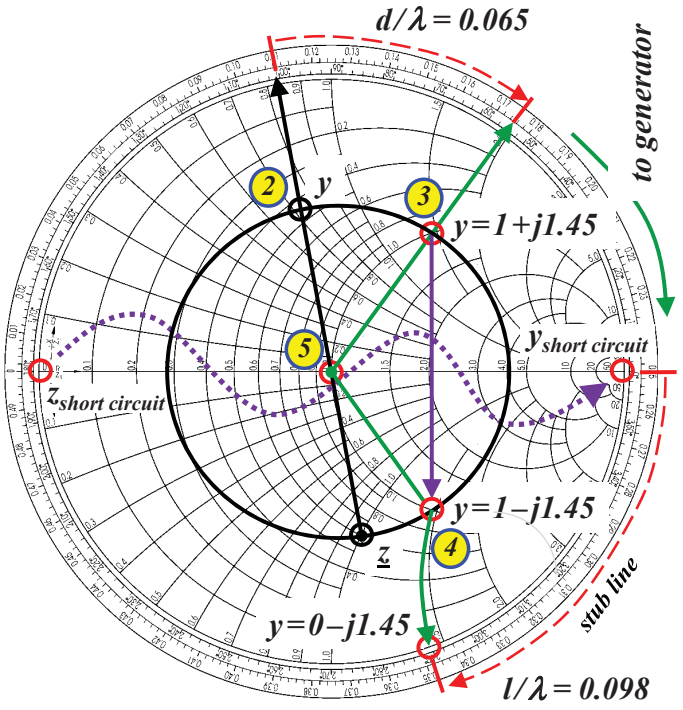
$Z_0 = 50\,\Omega$; z = 0.6– j

2 inversion $Z \rightarrow Y$

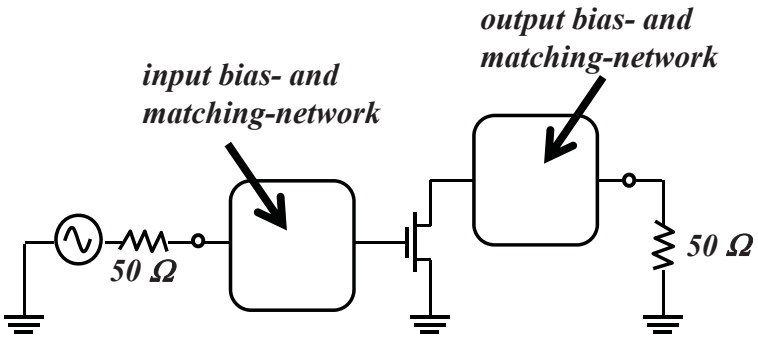
3 rotation (CW) until real part (normalized) becomes one

4 the admittance at the stub node will become real, when the input admittance of the shorted stub line compensates the

5 imaginary part



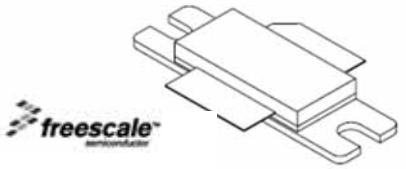
matching in the Smith chart — example (1/2):



realization of the input matching-network:

(i) the transistor presents impedance Z^*_{source}

(ii) matching with 50Ω line, followed by a stub line



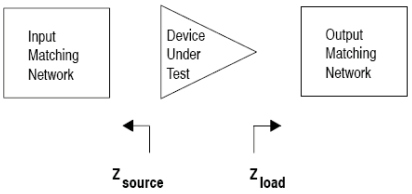
MRF6V12250HR3

V_{DD} = 50 Vdc, I_{DQ} = 100 mA, P_{out} = 275 W Peak

f MHz	Z_{source} Ω	Z_{load} Ω
1030	2.30 – j3.51	4.0 – j2.14

Z_{source} = Test circuit impedance as measured from gate to ground.

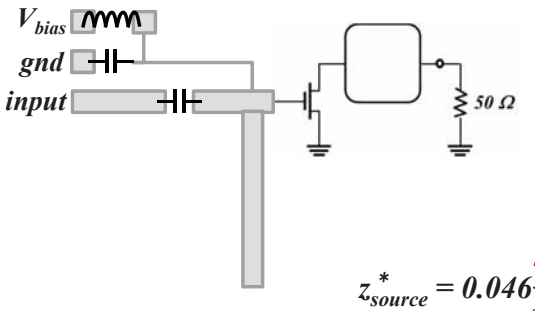
Z_{load} = Test circuit impedance as measured from drain to ground.

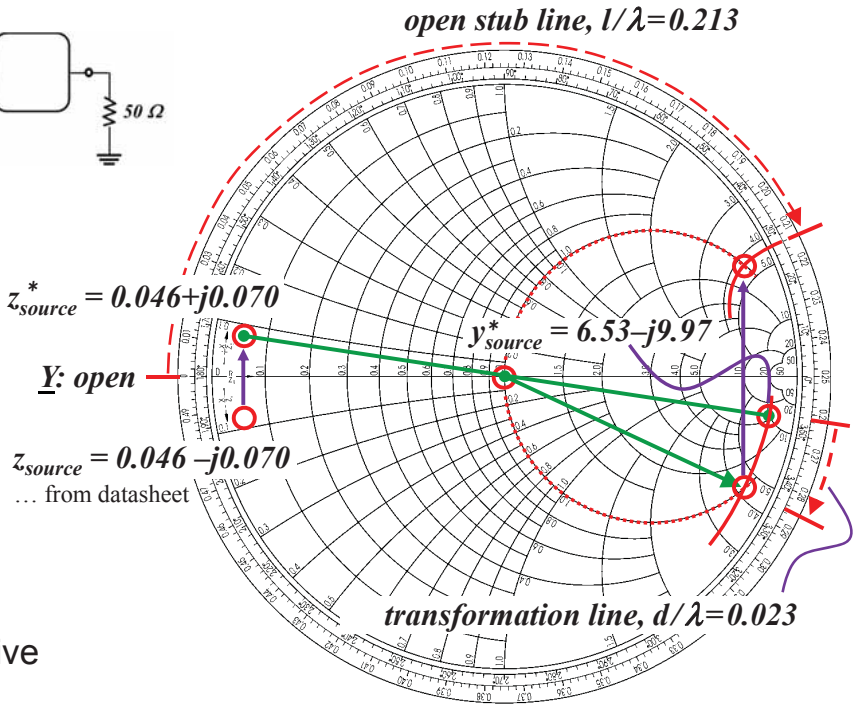


$Z_{source} = (2.30 - j3.51)\Omega$

$Z^*_{source} = (2.30 + j3.51)\Omega$

matching in the Smith chart — example (2/2):



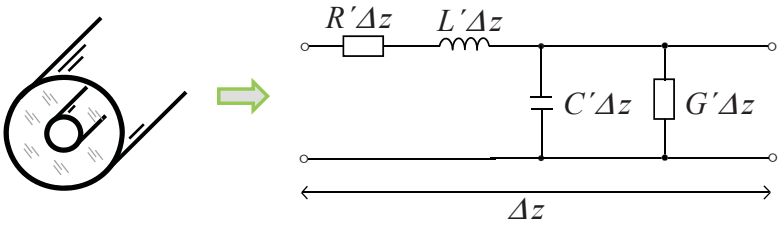


practical drawbacks:

(i) long lines:
- narrow band
- lossy

(ii) flat crossings:
- tolerance sensitive
- narrow band

transmission line with loss:



$$\frac{\partial^2 \underline{U}(z)}{\partial z^2} = \gamma^2 \, \underline{U}(z) \qquad , \qquad \frac{\partial^2 \underline{I}(z)}{\partial z^2} = \gamma^2 \, \underline{I}(z)$$

$$\gamma^2 = (R' + j\omega L')(G' + j\omega C') = (\alpha + j\beta)^2$$

$$\underline{Z}_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

$$\underline{U}(z) = \underline{U}_0^+ \, e^{-\alpha z} \, e^{-j\beta z} + \underline{U}_0^- \, e^{+\alpha z} \, e^{+j\beta z}$$

$$\underline{I}(z) = \frac{\underline{U}_0^+}{\underline{Z}_0} \, e^{-\alpha z} \, e^{-j\beta z} - \frac{\underline{U}_0^-}{\underline{Z}_0} \, e^{+\alpha z} \, e^{+j\beta z}$$

- voltage and current are attenuated,
- voltage and current are not in phase.

power flow in direction +z :

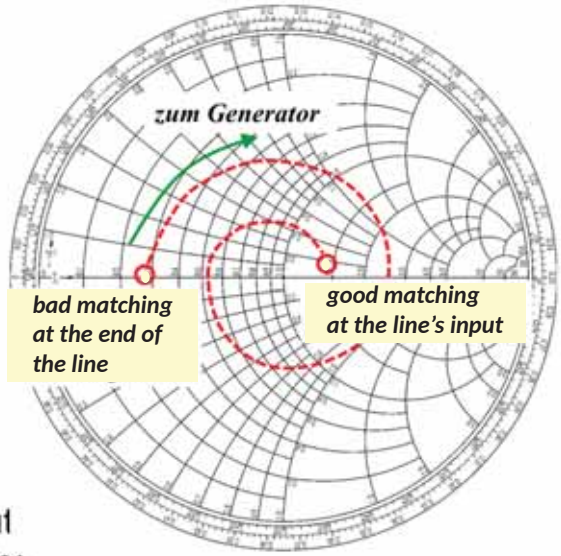
$$P_{+z}(z) = \frac{1}{2} \operatorname{Re}\{\underline{U}(z)\underline{I}^*(z)\} = \frac{|\underline{U}_0^+|^2}{2\,|\underline{Z}_0|} \, e^{-2\alpha z} \cos \phi_{Z_0}$$

$$U(z) \sim e^{-\alpha z} \quad \rightarrow \quad P(z) \sim [U(z)]^2 \sim e^{-2\alpha z}$$

transformation of the reflection coefficient (towards generator):

$$\Gamma(z=-d) = \Gamma_{L,z=0}\, e^{-2\alpha d}\, e^{-2j\beta d}$$

$$|\Gamma(z=-d)| = \left|\Gamma_{L,z=0}\right| e^{-2\alpha d}$$

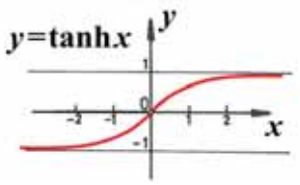


- the reflection coefficient at the input of a lossy line is (considering magnitude) smaller (mostly meaning: „better“) than at the end of that line

transformation of the impedance (towards generator):

$$Z(z=-d) = Z_0\,\frac{e^{\gamma d} + \Gamma_L\, e^{-\gamma d}}{e^{\gamma d} - \Gamma_L\, e^{-\gamma d}} = Z_0\,\frac{Z_L + Z_0 \tanh \gamma d}{Z_0 + Z_L \tanh \gamma d}$$

$$\tanh(a+jb) = \frac{\tanh a + j \tan b}{1 + j \tanh a \tan b}$$



- <u>open-ended line</u> ($Y_L = 0$):

$$Z_{\text{in,open}}(z=-d) = \frac{\underline{Z}_0}{\tanh \gamma d} = \underline{Z}_0\,\frac{1 + j \tanh \alpha d \tan \beta d}{\tanh \alpha d + j \tan \beta d}$$

- <u>short-ended line</u> ($Z_L = 0$):

$$Z_{\text{in,sc}}(z=-d) = \underline{Z}_0\,\frac{\tanh \alpha d + j \tan \beta d}{1 + j \tanh \alpha d \tan \beta d}$$

$$\text{d}=\lambda/4: \quad Z_{\text{in,sc}}\left(z=-\lambda/4\right)=\frac{\underline{Z}_0}{\tanh \alpha d}\approx\frac{\underline{Z}_0}{\alpha d}$$

$$\text{d}=\lambda/2: \quad Z_{\text{in,sc}}\left(z=-\lambda/2\right)=\underline{Z}_0 \tanh \alpha d\approx \underline{Z}_0\, \alpha d$$

- <u>example:</u> a line, short-cicuited at its end, operating at 10 GHz shall compensate a reactance of –j300Ω (at Z_0 =50Ω, α = 0.08 dB/cm). Calculate line length, equivalent circuit, Q-factor.

An estimate of αd : $\alpha=0.08\,\text{dB/cm}=0.92\,\text{Np/m}$; $d_{max}=\lambda/4=7.5\,\text{mm}$
$\alpha d|_{max}=0.007$ … this is small! E.g. ($\tanh x\approx x$):

$$Z_{\text{in,sc}}=\underline{Z}_0\,\frac{\tanh \alpha d+j\tan \beta d}{1+j\tanh \alpha d\tan \beta d}\approx\ \underline{Z}_0\,(\alpha d+j\tan \beta d)$$

tan(βd)=6, βd=80°, d=6.7 mm, Z_{in}=(0.345+j300)Ω …
series connection of R and L, Q=ωL/R=300/0.345=870

Content:

Repeating Maxwell‘s equations

Plane waves

Waveguides

Lumped components

Waves on transmission lines

Wave matrices

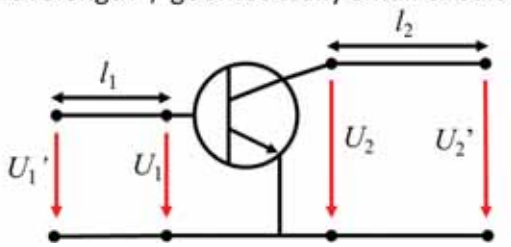
Transmission line circuits

Impedance transformers and filters

Wave matrices

Description of circuits using ***I***, ***U***, ***Z***, ***Y*** parameter is mostly suitable for (geometrically) „small“ circuits.

↳ low frequency / large wavelength / geometrically small circuit

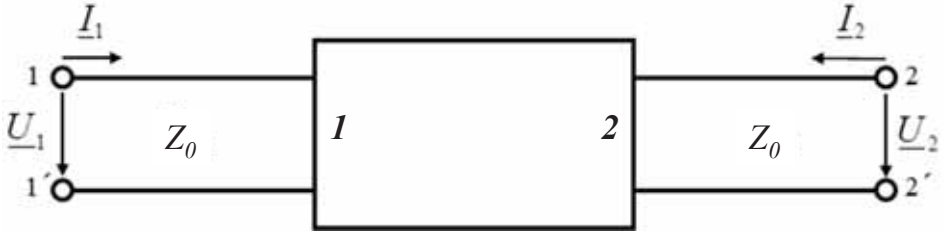


Several problems at high frequency:

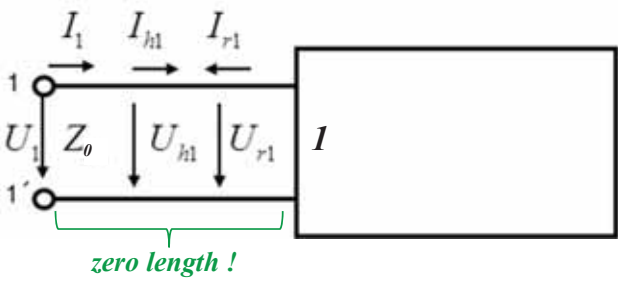
- probes and cables contribute significant inductance
- (mis-) matching and standing waves in the circuit get affected by the probe
- impedance transformation (phase shift & standing waves) along connection lines
- states „open“ and „shorted“ may not be allowed (for active circuits)
- impedance not (uniquely) defined (e.g., waveguide, optical fiber)

Description of circuits using wave matrices assumes to connect the ports of the circuit with defined, suitable load impedances:

- consider a two-port (= four-pole) network, where each port is connected to a line with impedance Z_0
- on the attached lines, current- and voltage-waves travel in both directions, going into the two-port and originate out of the two-port



- input / port 1: on the line at port 1 there is wave traveling towards the two-port device, with U_{h1} and I_{h1} , and another wave traveling away from the two-port device, with U_{r1} und I_{r1}. The superposition of these two waves results in the „total“ current and voltage at port 1.

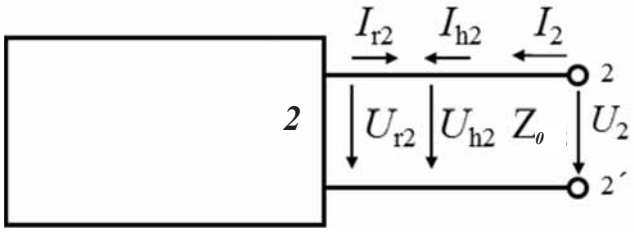


$$\frac{U_{h1}}{I_{h1}} = Z_0 \qquad \frac{U_{r1}}{I_{r1}} = Z_0$$

$$U_1 = U_{h1} + U_{r1} \qquad U_{h1} = \frac{1}{2}(U_1 + I_1 \, Z_0)$$

$$I_1 = I_{h1} - I_{r1} = \frac{1}{Z_0}(U_{h1} - U_{r1}) \qquad U_{r1} = \frac{1}{2}(U_1 - I_1 \, Z_0)$$

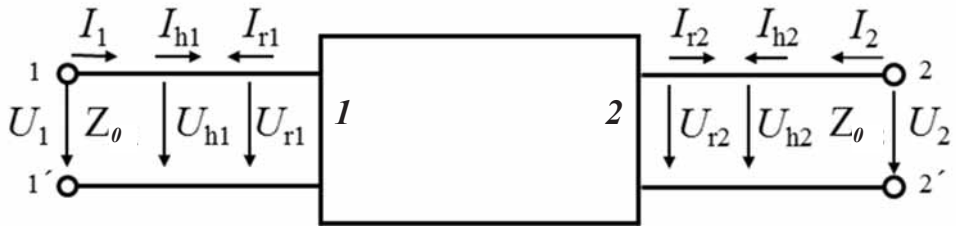
- output / port 2: … similar …



$$U_2 = U_{h2} + U_{r2} \qquad U_{h2} = \frac{1}{2}(U_2 + I_2 \, Z_0)$$

$$I_2 = I_{h2} - I_{r2} = \frac{1}{Z_0}(U_{h2} - U_{r2}) \qquad U_{r2} = \frac{1}{2}(U_2 - I_2 \, Z_0)$$

- description of the (linear) two-port device by the interaction of the complex voltage magnitudes $\underline{U}_{h1}$, $\underline{U}_{r1}$, $\underline{U}_{h2}$, $\underline{U}_{r2}$ at the ports:



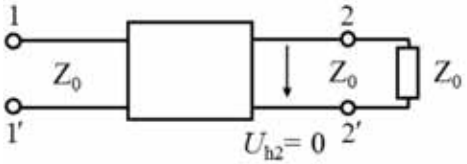
$$U_{r1} = S_{11} U_{h1} + S_{12} U_{h2}$$
$$U_{r2} = S_{21} U_{h1} + S_{22} U_{h2}$$

- scattering matrix [S]:

$$\begin{pmatrix} \underline{U}_{r1} \\ \underline{U}_{r2} \end{pmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{pmatrix} \underline{U}_{h1} \\ \underline{U}_{h2} \end{pmatrix} = [\mathrm{S}] \begin{pmatrix} \underline{U}_{h1} \\ \underline{U}_{h2} \end{pmatrix}$$

- S_{11} … input reflection coefficient at port 1 (while matched load at port 2):

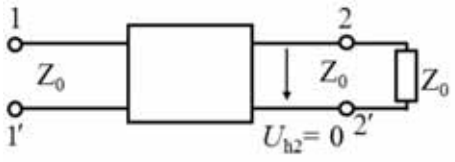
$$S_{11} = \left. \frac{\underline{U}_{r1}}{\underline{U}_{h1}} \right|_{\underline{U}_{h2}=0}$$



matched load at port 2

- S_{21} … transmission coefficient from port 1 to port 2:

$$S_{21} = \left. \frac{\underline{U}_{r2}}{\underline{U}_{h1}} \right|_{\underline{U}_{h2}=0}$$

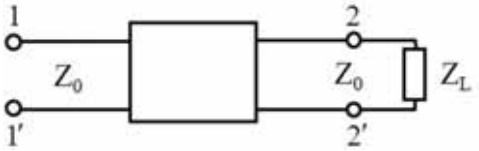


matched load at port 2

- S_{22} … input reflection coefficient at port 2 (while matched load at port 1)
- S_{12} … transmission coefficient from port 2 to port 1

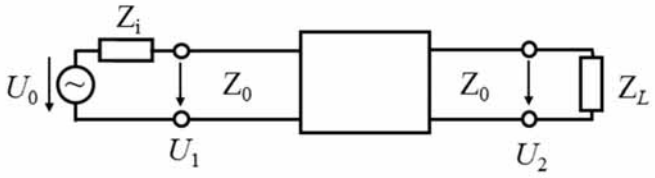
matched load at port 1

- transformation of the reflection coefficient „trough a two-port“:



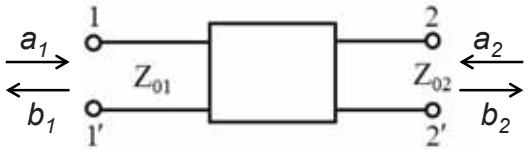
$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \qquad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

- voltage transduction:



$$\frac{U_2}{U_1} = \frac{S_{21}(1 + \Gamma_L)}{(1 - S_{22}\Gamma_L)(1 + \Gamma_{in})}$$

- so far, same reference impedance at all ports — generalization:



$$a_i = \frac{U_{hi}}{\sqrt{Z_{0i}}} = I_{hi}\sqrt{Z_{0i}} \qquad b_i = \frac{U_{ri}}{\sqrt{Z_{0i}}} = I_{ri}\sqrt{Z_{0i}} \qquad [\sqrt{\text{W}}]$$

$$u_{i,normiert} = \frac{U_{hi} + U_{ri}}{\sqrt{Z_{0i}}} = a_i + b_i \qquad [\sqrt{\text{W}}]$$

$$i_{i,normiert} = (I_{hi} - I_{ri})\sqrt{Z_{0i}} = a_i - b_i \qquad [\sqrt{\text{W}}]$$

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = [\text{S}] \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

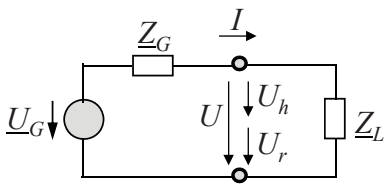
- generalized reflection coefficient: theory:

In two scenarios, considering the reflection coefficient of a plane wave at a boundary (chapter 2), and considering the reflection coefficient on transmission line with the Smith chart description – in both cases the reference impedance (of medium 1, line 1) was a real number.

In the case of connecting an arbitrary (complex) load impedance to an arbitrary (complex) source impedance, one can define the **„generalized reflection coefficient",** Γ_p .

The special case, $\Gamma_p = 0$, is well-known as conjugate-complex match.

For the general case of mismatch, the description using the „generalized reflection coefficient" is useful.



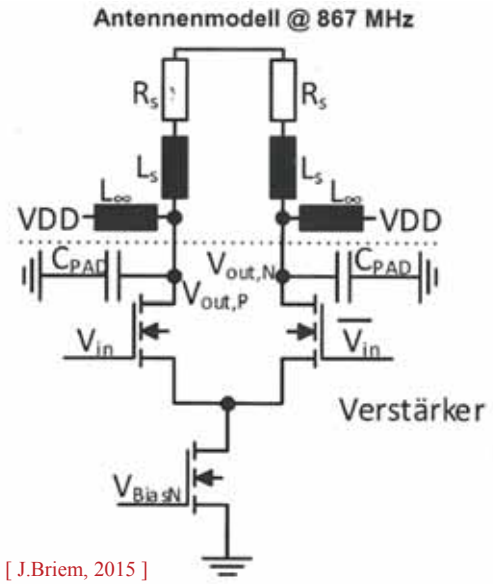
definition of *power waves* a_p, b_p : (see slides 6.3, 6.8)

$$a_p = \frac{U_h}{\sqrt{Re(\underline{Z}_G)}} = \frac{1}{2}\frac{(U + I\,\underline{Z}_G)}{\sqrt{Re(\underline{Z}_G)}} \quad , \quad b_p = \frac{U_r}{\sqrt{Re(\underline{Z}_G)}} = \frac{1}{2}\frac{(U - I\,\underline{Z}_G^*)}{\sqrt{Re(\underline{Z}_G)}}$$

$$\Gamma_p = \frac{b_p}{a_p} = \frac{U - I\,\underline{Z}_G^*}{U + I\,\underline{Z}_G} = \frac{U/I - \underline{Z}_G^*}{U/I + \underline{Z}_G} = \frac{\underline{Z}_L - \underline{Z}_G^*}{\underline{Z}_L + \underline{Z}_G}$$

example:

conjugate-complex match between amplifier and connected antenna



[J.Briem, 2015]

reflection coeffcient measured into one of the two balanced amplifier outputs with Zref = 50 Ω @ 867 MHz:

$$\Gamma_{amp\,,single_ended\,,50Ohm} = 0.72 - j0.57$$

... accordingly, the generator source impedance of one amplifier output @ 867 MHz:

$$\underline{Z}_{G,single_ended} = 50\,\Omega\,\frac{1+\Gamma_{amp\,,single_ended\,,50Ohm}}{1-\Gamma_{amp\,,single_ended\,,50Ohm}}$$

... generator source impedance of both amplifier outputs in balanced mode @ 867 MHz:

$$\underline{Z}_{G,balanced} = 2\,\underline{Z}_{G,single_ended} = (38{,}9 - j\,282{,}7)\,\Omega$$

→

$$\underline{Z}_{G,balanced} = 2\,\underline{Z}_{G,single_ended} = (38{,}9 - j\,282{,}7)\,\Omega$$

... required antenna input impedance (balanced) at conjugate-complex match @ 867 MHz:

$$\underline{Z}_L = \underline{Z}_G^* = (38{,}9 + j\,282{,}7)\,\Omega$$

... represented as series of L and R :

$$L = 2L_s = 51{,}9\,\text{nH} \qquad R = 2R_s = 38{,}9\,\Omega$$

→ input reflection coefficient (balanced) of the antenna at reference 50 Ω @ 867 MHz:

$$\Gamma_{ant\,,50Ohm} = \frac{\underline{Z}_L - 50\Omega}{\underline{Z}_L + 50\Omega} =$$

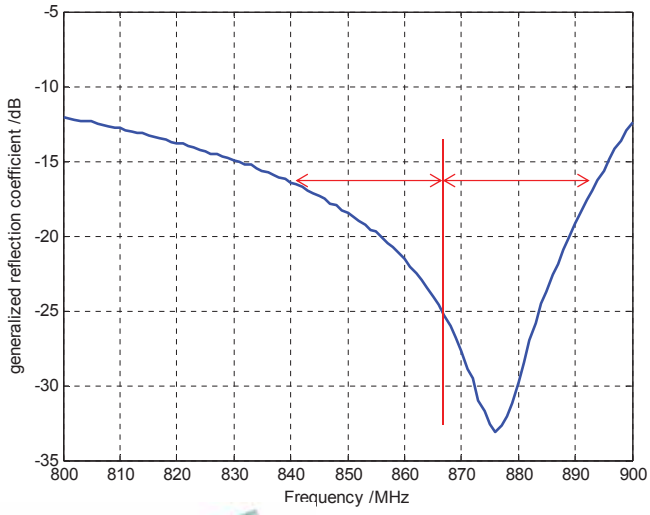
$$= 0.90 + j0.32 = 0.955\exp(j\,19.71^\circ)$$

→ so, match @ 867 MHz, but how much mismatch at other frequencies?

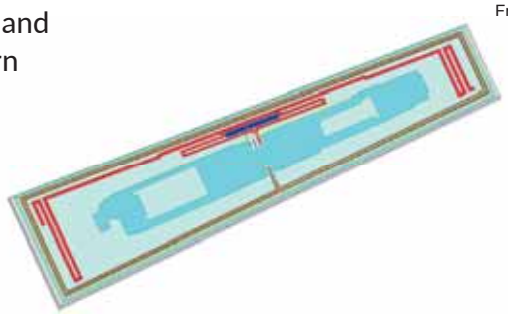
Antennenmodell @ 867 MHz
[J.Briem, 2015]

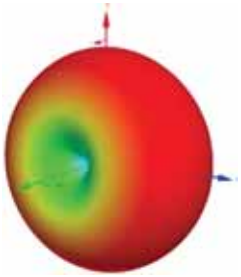
magnitude (in dB) of the generalized reflection coefficient Γ_p :

(for optimium bandwidth, the „perfect match“ at 867 MHz has been slightly de-tuned)



antenna layout and radiation pattern (dipole-like):





[project KoSiF, 2015]

- scattering matrix [S] :

  the main diagonal elements S_{ii} are input reflection coefficients (at port i)

  off-diagonal elements S_{ij} are transmission coefficients (from port j to port i)

  advantages: 1) coefficients easier to measure at high frequency (match at other ports easier to realize than short or open, power easier to measure than current or voltage)

  2) „clear“ in representing power-based wave quantities

  3) can be defined for non-TEM waveguides (rectangular waveguide, dielectric waveguide) and canbe measured based on (even scalar) power measurements

- scattering matrix – property „reciprocity“:

  passive multi-ports (few exceptions, such as ferrites and plasmas) are reciprocal: $S_{ij} = S_{ji}$

- scattering matrix – property of the loss-less, reciprocal two-port:

$$|S_{21}|^2 = 1 - |S_{11}|^2$$

- scattering matrix – unitarity of loss-less multi-ports:

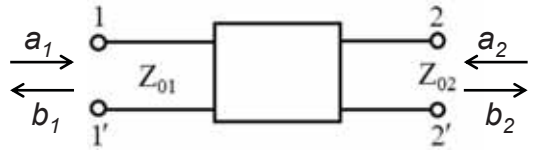
$$[S^*]^T [S] = [E]$$

[derivation, e.g., in: Zinke/Brunswig, 4.11.5, p. 181]

conjugate-transposed (=transjugated) scattering matrix

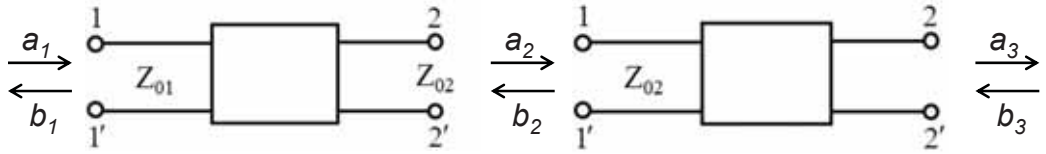
wave chain matrix == <u>transmission matrix</u> [T] :

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = [S] \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$



$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = [T] \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

- [T]-matrices of series-connected two-ports are simply multiplied:



$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = [T_1] \begin{pmatrix} b_2 \\ a_2 \end{pmatrix} \qquad [T] = [T_1][T_2] \qquad \begin{pmatrix} b_2 \\ a_2 \end{pmatrix} = [T_2] \begin{pmatrix} b_3 \\ a_3 \end{pmatrix}$$

- relation [S] — [T] :

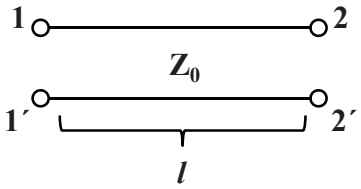
$$[T] = \begin{bmatrix} [S]_{12} - [S]_{11}[S]_{21}^{-1}[S]_{22} & [S]_{11}[S]_{21}^{-1} \\ -[S]_{21}^{-1}[S]_{22} & [S]_{21}^{-1} \end{bmatrix}$$

$$[S] = \begin{bmatrix} [T]_{12}[T]_{22}^{-1} & [T]_{11} - [T]_{12}[T]_{22}^{-1}[T]_{21} \\ [T]_{22}^{-1} & -[T]_{22}^{-1}[T]_{21} \end{bmatrix}$$

Examples of scattering matrices (1/3):

- matched, loss-less line

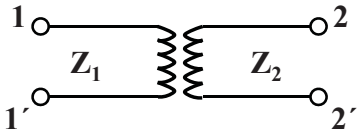
$$[S] = \begin{bmatrix} 0 & \exp(-j\beta l) \\ \exp(-j\beta l) & 0 \end{bmatrix}$$



- step-of-impedance / ideal transformer

$$\rho = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

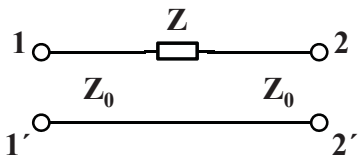
$$[S] = \begin{bmatrix} \rho & \sqrt{1-\rho^2} \\ \sqrt{1-\rho^2} & -\rho \end{bmatrix}$$



Examples of scattering matrices (2/3):

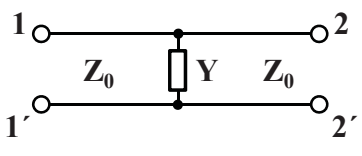
- series impedance

$$[S] = \begin{bmatrix} \frac{Z}{2Z_0 + Z} & \frac{2Z_0}{2Z_0 + Z} \\ \frac{2Z_0}{2Z_0 + Z} & \frac{Z}{2Z_0 + Z} \end{bmatrix}$$



- shunt admittance

$$[S] = \begin{bmatrix} -\frac{YZ_0}{2 + YZ_0} & \frac{2}{2 + YZ_0} \\ \frac{2}{2 + YZ_0} & -\frac{YZ_0}{2 + YZ_0} \end{bmatrix}$$



Examples of scattering matrices (3/3):

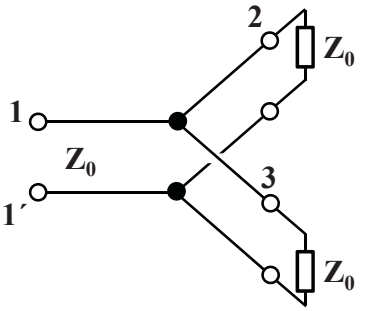
- loss-less divider junction

$$S_{11} = S_{22} = S_{33} = -\frac{1}{3}$$

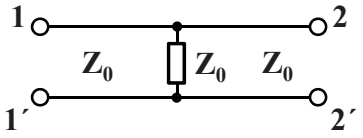
$$S_{11} = \frac{Z_0 || Z_0 - Z_0}{Z_0 || Z_0 + Z_0} = \frac{-\frac{1}{2}Z_0}{\frac{3}{2}Z_0} = -\frac{1}{3}$$

$$S_{ij} = \frac{2}{3}$$

$$[S] = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$$



$$S_{21} = \frac{2}{3}$$:

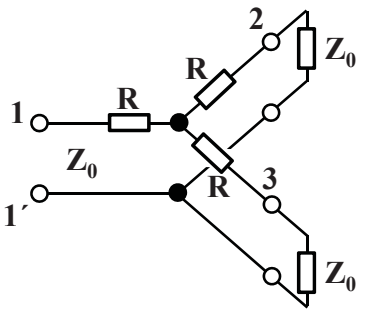


$$1 = |S_{11}|^2 + |S_{21}|^2 + |S_{31}|^2$$

- the „loss-less symmetric three-port“ is not matched
- the „lossy matched symmetric three-port“ is possible
- the „loss-less matched non-symmetric 3-port“ is possible (→ circulator)

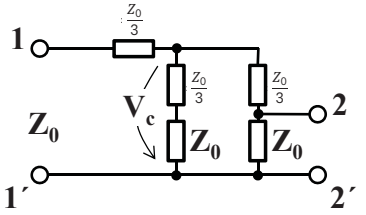
lossy symmetric divider junction:

- three equal (because of symmetry) resistors included:



- match is achieved if:

$$Z_0 = R + \frac{1}{2}(R + Z_0) \quad \rightarrow \quad R = \frac{Z_0}{3}$$



- S-matrix:

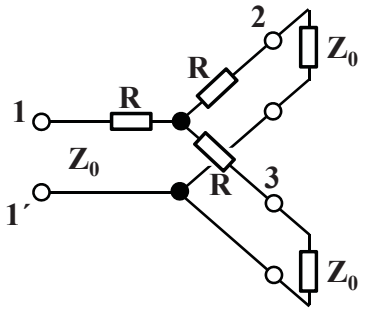
$$[S] = \frac{1}{2}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \leftarrow \quad V_{22'} = \frac{1}{2}V_{11'}$$

$$V_{22'} = V_c \frac{Z_0}{\frac{4}{3}Z_0} = V_c \frac{3}{4}$$

$$V_c = V_{11'} \frac{\frac{2}{3}Z_0}{Z_0} = V_{11'} \frac{2}{3}$$

lossy symmetric divider junction:





The Keysight 11636C provides a symmetrical 6 dB power division.

The Keysight 11636C power divider uses three 16 2/3 Ω resistors

Insertion loss	6.5 dB, DC to 18 GHz
	7 dB, 18 GHz to 26.5 GHz
	8 dB, 26.5 GHz to 40 GHz
	8.5 dB, 40 GHz to 50 GHz

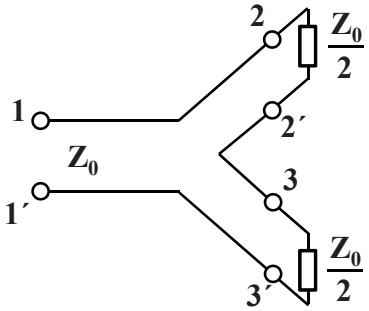
$$[S] = \frac{1}{2}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

loss-less divider junction w. different port reference impedances (1/7)

determine S_{11} :

feed at port 1 :

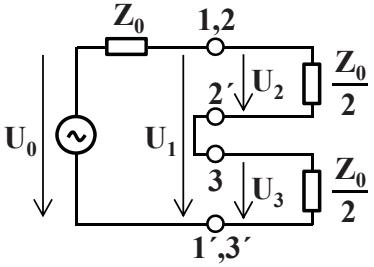
… match : $S_{11} = 0$



loss-less divider junction w. different port reference impedances (2/7)

determine S_{21} :

all quantities referred to U_0 :



$$S_{21} = \left.\frac{b_2}{a_1}\right|_{a_2=a_3=0}$$

$$a_i = \frac{U_{hi}}{\sqrt{Z_i}} \quad : \quad a_1 = \frac{U_{h1}}{\sqrt{Z_0}} = \frac{\frac{U_0}{2}}{\sqrt{Z_0}}$$

$$b_i = \frac{U_{ri}}{\sqrt{Z_i}} \quad : \quad b_2 = \frac{U_{r2}}{\sqrt{\frac{Z_0}{2}}} = \frac{\frac{U_0}{4}}{\sqrt{\frac{Z_0}{2}}}$$

$$S_{21} = \frac{1}{\sqrt{2}}$$

$U_1 = \frac{U_0}{2}$, $U_{r1} = 0$, $U_1 = U_{h1} + U_{r1}$ → $S_{11} = 0$, $a_2 = 0$, $a_3 = 0$

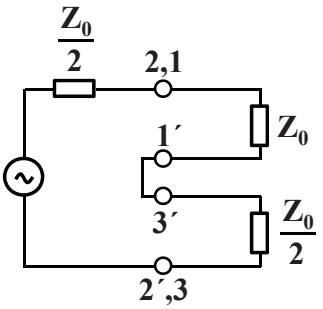
$U_2 = \frac{U_0}{4}$, $U_2 = U_{h2} + U_{r2}$, $U_{h2} = 0$ → $a_2 = 0$

for symmetry reasons: $S_{31} = S_{21}$

loss-less divider junction w. different port reference impedances (3/7)

determine S_{22} :

feed at port 2 :



$$S_{22} = \frac{\frac{3}{2}Z_0 - \frac{Z_0}{2}}{\frac{3}{2}Z_0 + \frac{Z_0}{2}} \rightarrow S_{22} = \frac{1}{2}$$

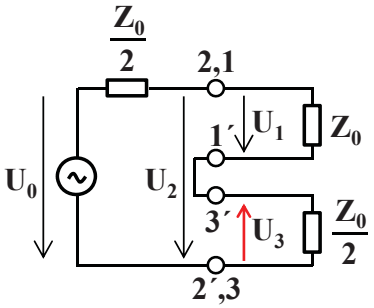
for symmetry reasons: $S_{22} = S_{33}$

loss-less divider junction w. different port reference impedances (4/7)

determine S_{12} :

all quantities referred to U_0 :

feed at port 2 :



$$S_{12} = \left.\frac{b_1}{a_2}\right|_{a_1=a_3=0}$$

$$b_1 = \frac{U_{r1}}{\sqrt{Z_0}} \left\{ \begin{array}{l} a_1 = 0 \\ U_{h1} = 0 \\ U_{r1} = U_1 \\ U_1 = \frac{U_0}{2} \end{array} \right\} \rightarrow b_1 = \frac{\frac{U_0}{2}}{\sqrt{Z_0}}$$

… b1 is the at port 1 (matched!) outgoing wave

$$a_2 = \frac{U_{h2}}{\sqrt{\frac{Z_0}{2}}} = ?$$

… a2 is the at port 2 (not matched!) incident wave

$U_2 = U_{h2} + U_{r2}$ **&** $U_2 = \frac{3U_0}{4}$ … neither U_{h2} nor U_{r2} are zero!

loss-less divider junction w. different port reference impedances (5/7)

$$a_2 = \frac{U_{h2}}{\sqrt{\frac{Z_0}{2}}} = ?$$

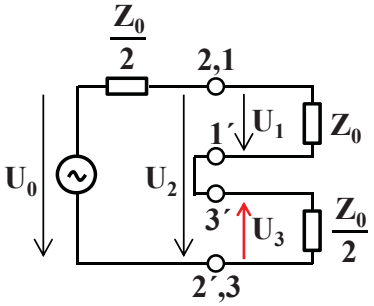
$U_2 = U_{h2} + U_{r2}$

$U_{h2} = U_2 - U_{r2}$

$$S_{22} = \left.\frac{b_2}{a_2}\right|_{a_1=a_3=0} = \left.\frac{U_{r2}}{U_{h2}}\right|_{a_1=a_3=0}$$

$U_{h2} = U_2 - S_{22}\, U_{h2}$

$$U_{h2} = \frac{U_2}{1 + S_{22}} = \frac{2U_2}{3} \quad \& \quad U_2 = \frac{3U_0}{4}$$



$$a_2 = \frac{U_{h2}}{\sqrt{\frac{Z_0}{2}}} = \cdots = \frac{\frac{2}{3}U_2}{\sqrt{\frac{Z_0}{2}}} = \frac{\frac{2}{3}\frac{3}{4}U_0}{\sqrt{\frac{Z_0}{2}}} = \frac{U_0}{\sqrt{2}\sqrt{Z_0}}$$

$$S_{12} = \frac{1}{\sqrt{2}}$$

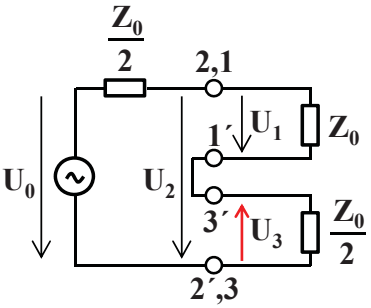
for symmetry reasons: $S_{12} = S_{13}$

loss-less divider junction w. different port reference impedances (6/7)

determine S_{32} :

all quantities referred to U_0 :

feed at port 2 :



$$S_{32} = \frac{b_3}{a_2}\Big|_{a_1=a_3=0}$$

$$a_2 = \frac{U_0}{\sqrt{2}\sqrt{Z_0}}$$

$$b_3 = \frac{U_{r3}}{\sqrt{\frac{Z_0}{2}}}$$

$$U_{r3} = U_3$$

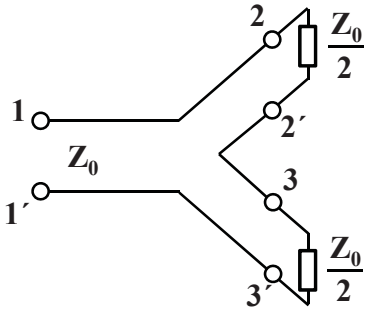
$$U_3 = -\frac{U_0}{4}$$

$$S_{32} = -\frac{1}{2}$$

for symmetry reasons: $S_{32} = S_{23}$

loss-less divider junction w. different port reference impedances (7/7)

result :



$$[S] = \frac{1}{2}\begin{bmatrix} 0 & \sqrt{2} & \sqrt{2} \\ \sqrt{2} & 1 & -1 \\ \sqrt{2} & -1 & 1 \end{bmatrix}$$

… loss-less!

Content:

Repeating Maxwell‘s equations

Plane waves

Waveguides

Lumped components

Waves on transmission lines

Wave matrices

Transmission line circuits

Impedance transformers and filters

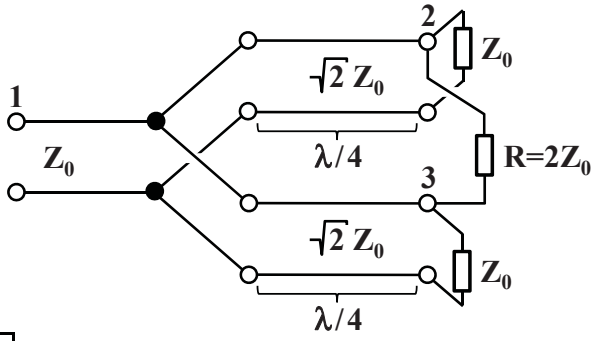
Transmission line circuits — overview:

- Wilkinson-divider — 3-port
- concept of even-odd-analysis
- 90°-hybrid — 4-port
- applications of hybrids and example of a planar circuit
- 180°-hybrid
- balun
- balanced circuits
- transmission line coupler
- measurement of S-parameters using directional couplers

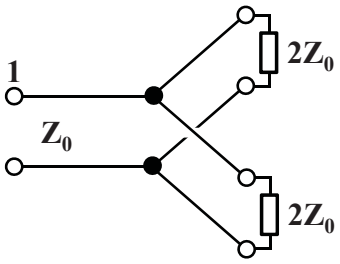
Transmission line circuits

Wilkinson power divider

- a 3-port which is matched on all ports,
- symmetric, reciprocal, and lossy
- determine the S-parameters:



$S_{11} = 0$

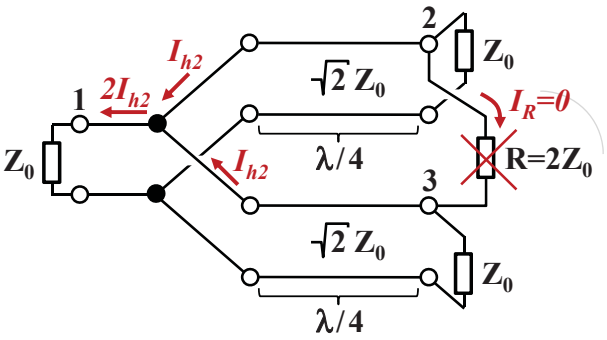


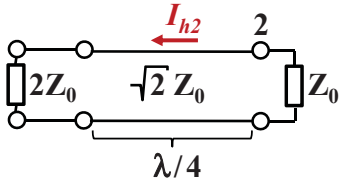
… for symmetry reasons: $S_{21} = S_{31} = \frac{1}{\sqrt{2}} \exp\left(-j\beta\frac{\lambda}{4}\right) = \frac{-j}{\sqrt{2}}$

… determine $S_{22} = S_{33}$ and $S_{23} = S_{32}$:

… $a_1 = 0$: $b_2 = S_{22}a_2 + S_{23}a_3$ and $b_3 = S_{23}a_2 + S_{22}a_3$

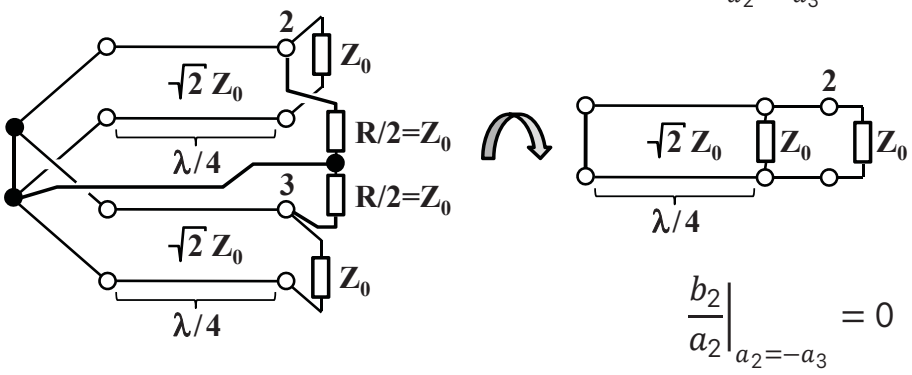
… „even“ excitation: : $a_2 = a_3$ $\left.\frac{b_2}{a_2}\right|_{a_2=a_3} = S_{22} + S_{23}$





$\left.\frac{b_2}{a_2}\right|_{a_2=a_3} = 0$

… „odd“ excitation: $a_2 = -a_3$ $\left.\frac{b_2}{a_2}\right|_{a_2=-a_3} = S_{22} - S_{23}$



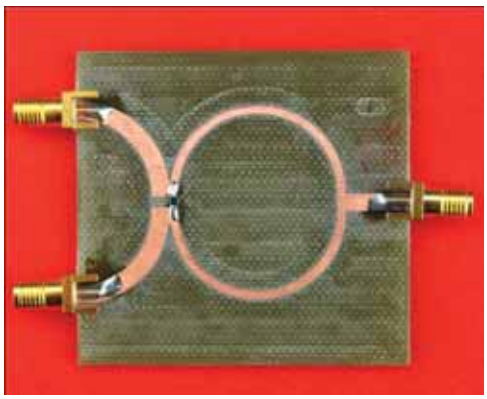
$\left.\frac{b_2}{a_2}\right|_{a_2=-a_3} = 0$

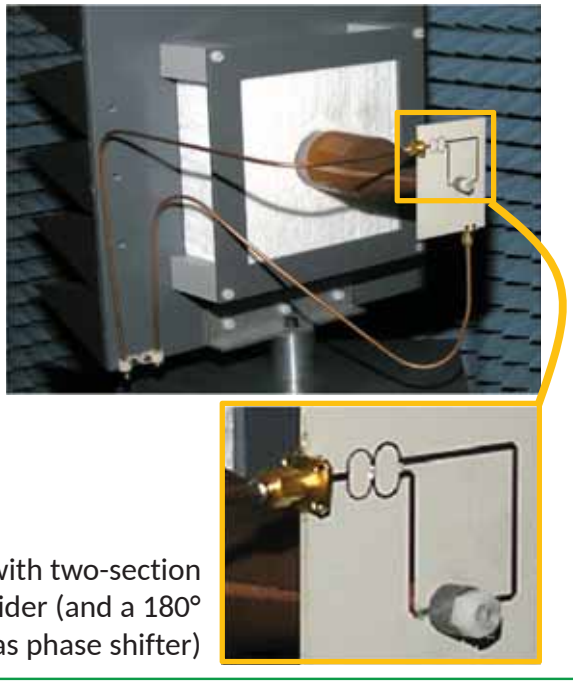
… system of equations: $S_{22} + S_{23} = 0$, $S_{22} - S_{23} = 0$ $\Rightarrow S_{22} = S_{23} = 0$

$$[S] = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

… without the 100-Ohms-resistor, the ports 2 and 3 would not be matched and not be decoupled/isolated!

Wilkinson-divider, examples:



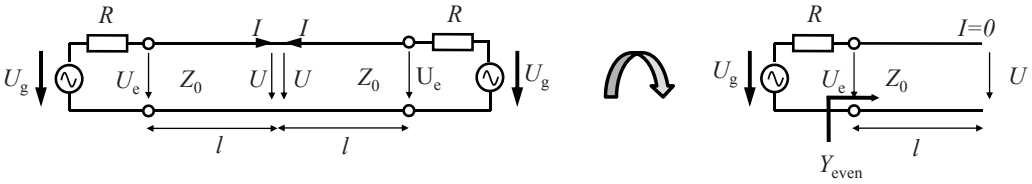


→ symmetric antenna feed with two-section (broadband) Wilkinson divider (and a 180° line as phase shifter)

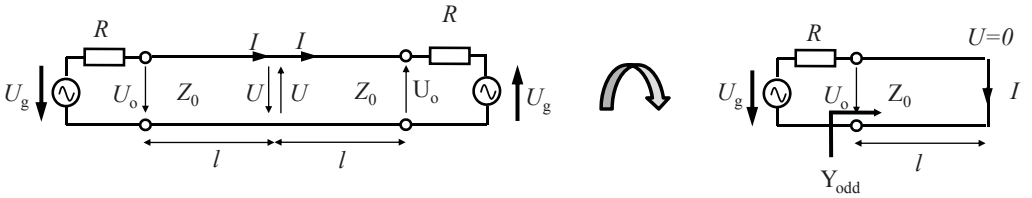
Even-mode and odd-mode excitation:

- symmetric and anti-symmetric excitation can split a symmetric circuit in two simler circuit parts, each of them can possibly be analyzed in a much simpler way (two times an anylysis of a half-sized circuit is cheaper than one analysis of the full-size circuit).

- by even-mode excitation at two ports of a symmetric two-port, the currents cancel in the symmetry plane – then, it is enough to consider the line crossing the symmetry plane as „open“ and consider only half of the line:



- by odd-mode excitation at two ports of a symmetric two-port, the voltage nulls in the symmetry plane – then, it is enough to consider the line crossing the symmetry plane as „shorted“ and consider only half of the line:



Transmission-line coupler:

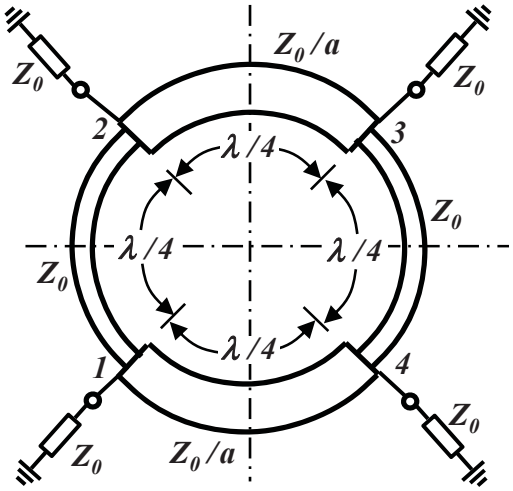
basic structure:
two lines are coupled by their interleaving fields.

special structure:
two lines are coupled by crossing line section(s).
→ named: hybrid couplers or hybrides

90° hybrid:

also : 90° hybrid coupler,
90° ring hybrid,
branchline coupler

two-fold symmetry – can be characterized by two-fold application of even-odd-mode approach (first, ports 1+2, then, ports 3+4)



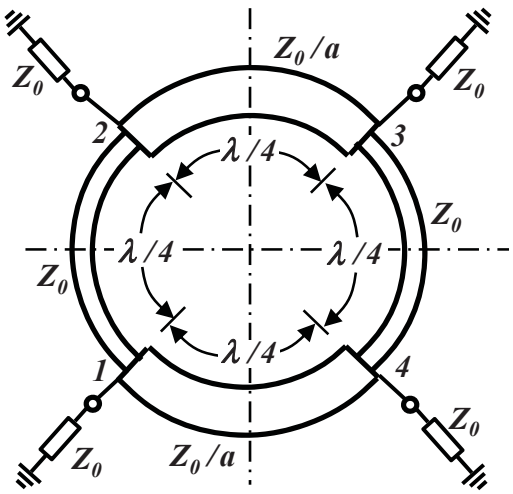
note: phase rotation over λ/4-line: βL=90° :

$$e^{-j\beta L} = e^{-j\,90^\circ} = \cos(-90^\circ) + j\sin(-90^\circ) = -j$$

90° hybrid – characteristics:

- by feeding port 1: the power splits equally between ports 3 and 4;
- the output signal at port 3 is 90 degrees later than the output at port 4;
- port 2 is decoupled/isolated (zero signal);
- all ports are matched to Z_0 ;
- loss-less structure;
- response for feeding the other ports is similar.

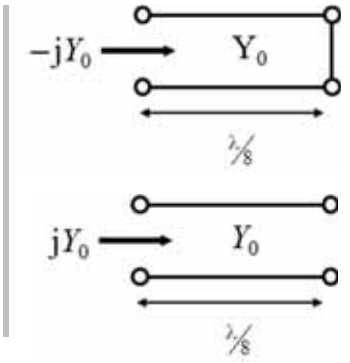
$$a = \sqrt{2}$$



$$[S] = -\frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 0 & 1 & j \\ 0 & 0 & j & 1 \\ 1 & j & 0 & 0 \\ j & 1 & 0 & 0 \end{bmatrix}$$

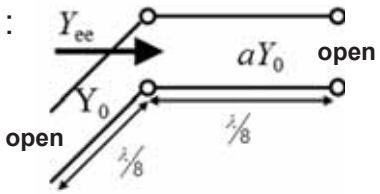
90° hybrid – even-odd-analysis (1/4):

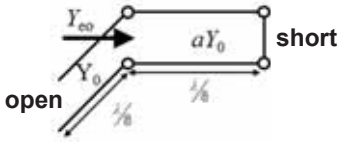
analysis of port impedances for the four different symmetries:



„i“ = even / „j“ = even :

$$Y_{ee} = j\,Y_0\,(1+a)$$

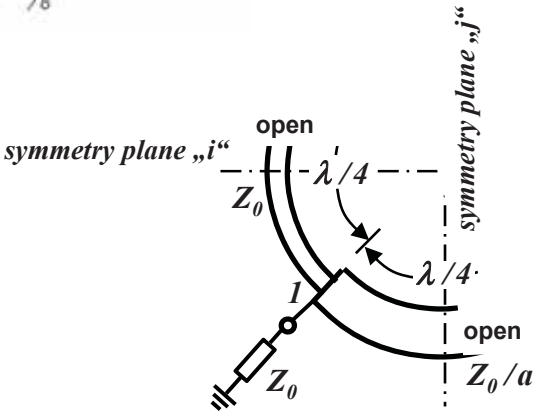




$$Y_{eo} = j\,Y_0\,(1-a)$$

$$Y_{oe} = j\,Y_0\,(-1+a)$$

$$Y_{oo} = j\,Y_0\,(-1-a)$$



90° hybrid – even-odd-analysis (2/4):

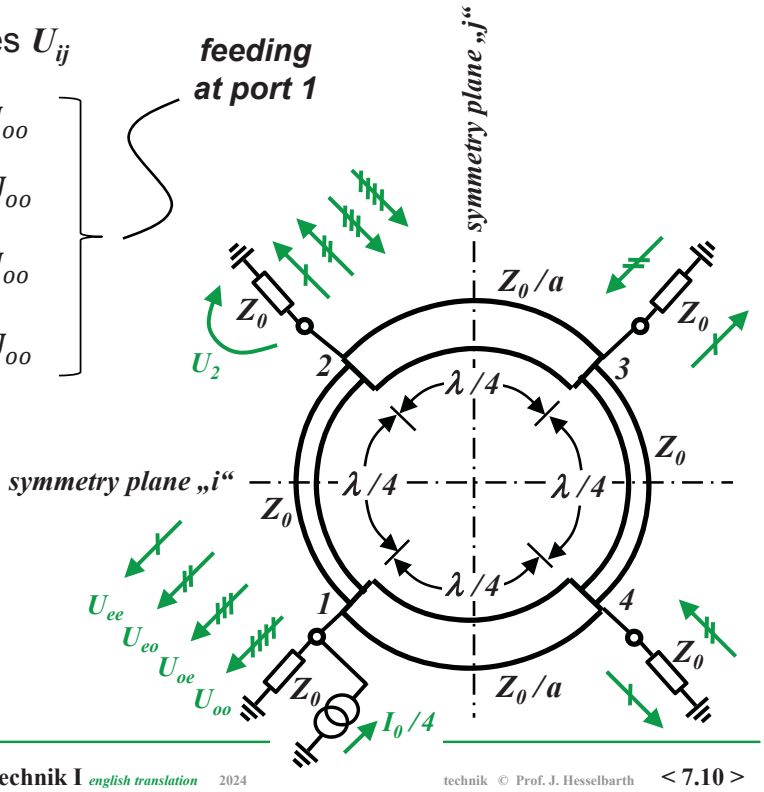
- analysis of partial voltages U_{ij}

$$U_1 = U_{ee} + U_{eo} + U_{oe} + U_{oo}$$
$$U_2 = U_{ee} + U_{eo} - U_{oe} - U_{oo}$$
$$U_3 = U_{ee} - U_{eo} - U_{oe} + U_{oo}$$
$$U_4 = U_{ee} - U_{eo} + U_{oe} - U_{oo}$$

feeding at port 1

$$U_{ij} = \frac{I_0}{4}\left(Z_0 \| Z_{ij}\right)$$

$$= \frac{I_0}{4}\,\frac{1}{Y_0 + Y_{ij}}$$

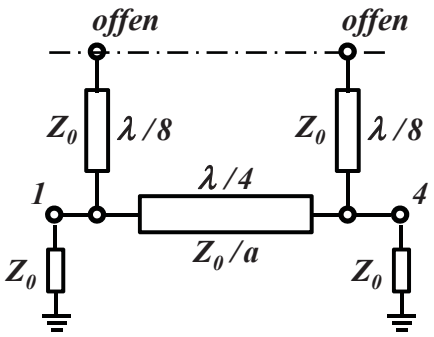


90° hybrid – even-odd-analysis (3/4):

- match at port 1 :

if port 1 is matched under condition „even“ and also under condition „odd“, then is $S_{11}=0$ … the „even“-excitation case is as follows:

„even“ :



input admittance of a $\lambda/8$ -line which is open at the end: $Y_{open}=+jY_0$

normalized admittance at node „4“: $y_4=1+j$

at the input of the $\lambda/4$-trafo: a^2/y_4

at port 1 : $y_{in}=j+a^2/y_4$

→ $y_{in}=1$ (match) for: $a=\sqrt{2}$

„odd“ : …

- because of symmetry: $S_{11}=S_{22}=S_{33}=S_{44}=0$

90° hybrid – even-odd-analysis (4/4):

- isolation: $S_{21}=0$, d.h., $U_2=0$ (if $S_{22}=0$) :

$$U_2=\frac{I_0}{4Y_0}\left(\frac{1}{1+j+ja}+\frac{1}{1+j-ja}-\frac{1}{1-j+ja}-\frac{1}{1-j-ja}\right)=0$$

$$a=\sqrt{2}$$

- transmission & reflection :

$S_{11}=0$, $S_{33}=0$, $S_{44}=0$

$$U_1=\frac{I_0}{2Y_0}$$

$$U_3=-\frac{I_0}{2\sqrt{2}\,Y_0}$$

$$U_4=-j\frac{I_0}{2\sqrt{2}\,Y_0}$$

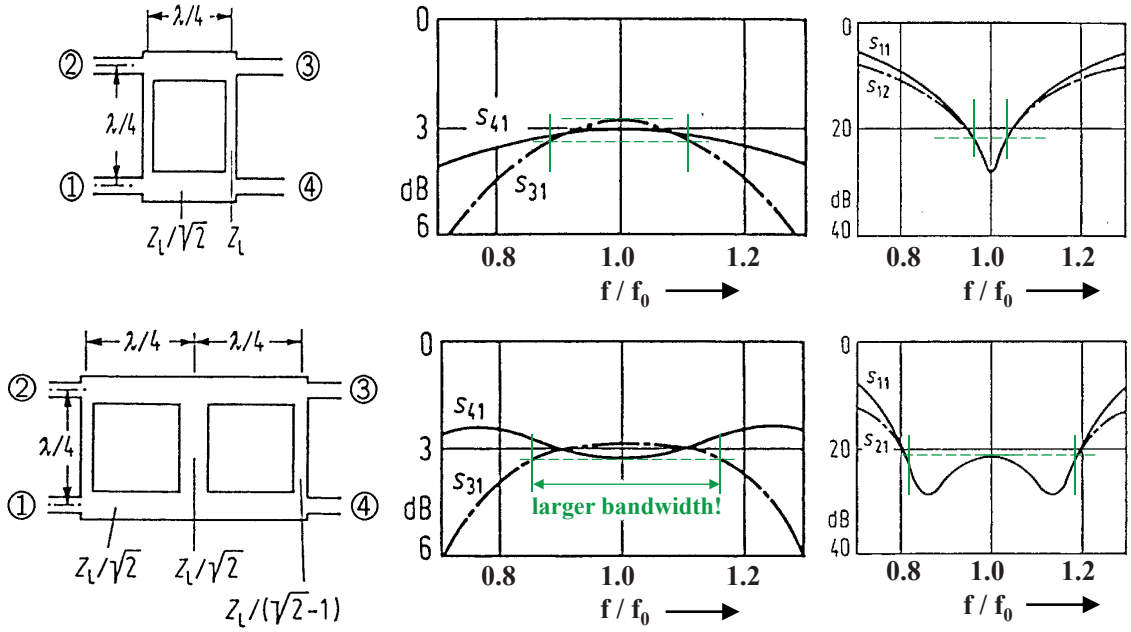
$$\frac{U_3}{U_1}=-\frac{1}{\sqrt{2}}=S_{31}$$

$$\frac{U_4}{U_1}=-\frac{j}{\sqrt{2}}=S_{41}$$

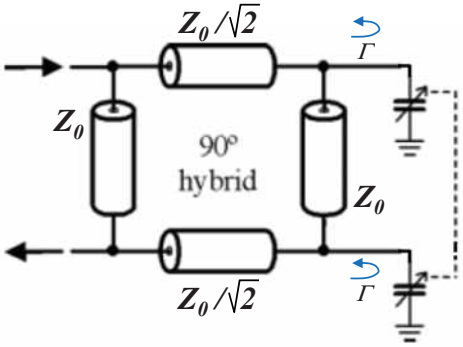
Match at all ports. Input power is split equally to two ouput ports (3dB coupler). Output phases differ by 90 degrees. One port is isolated.

90° hybrid – bandwidth:

- characteristics based on λ/4-trafo, that is, only at a single frequency!



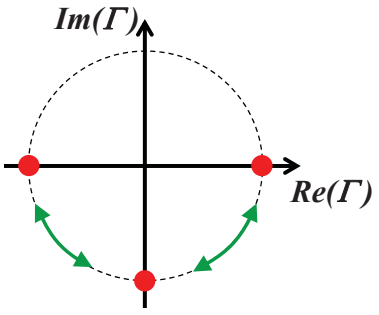
90° hybrid – application as phase modulator:



$C = 0 \quad \rightarrow \quad \Gamma = 1$

$C = \dfrac{1}{\omega Z_0} \quad \rightarrow \quad \Gamma = -j$

$C \rightarrow \infty \quad \rightarrow \quad \Gamma = -1$

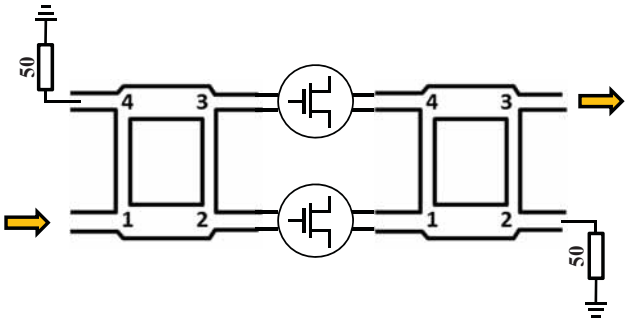


$\angle S_{21} = \angle \Gamma + \varphi_0$

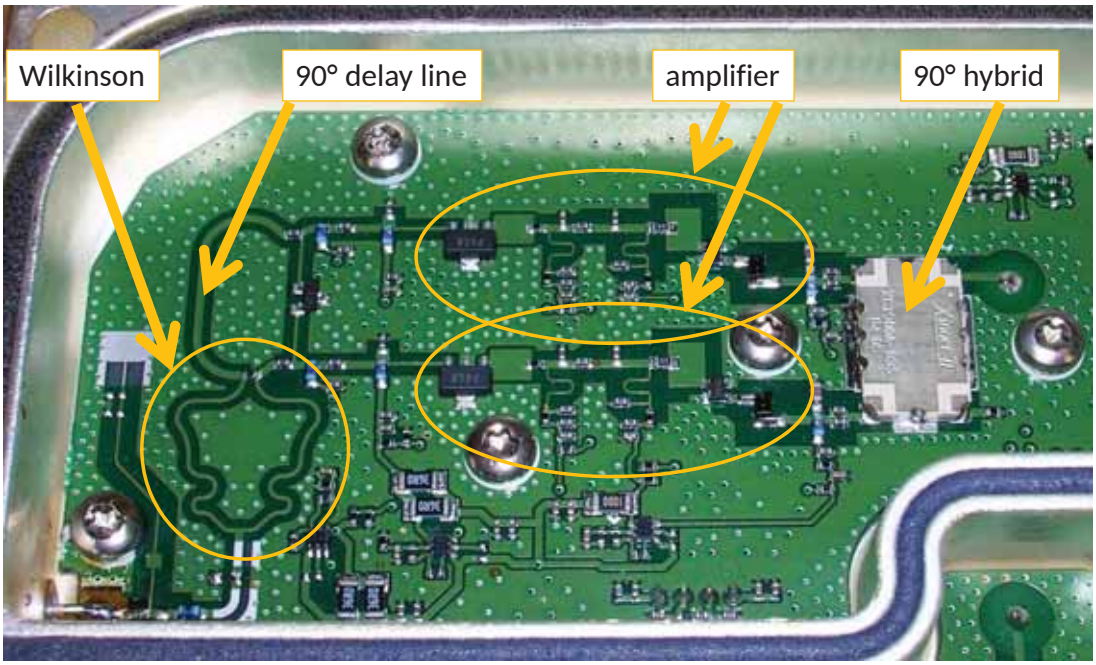
e.g., BPSK

90° hybrid – application for LNA (low noise amplifier):

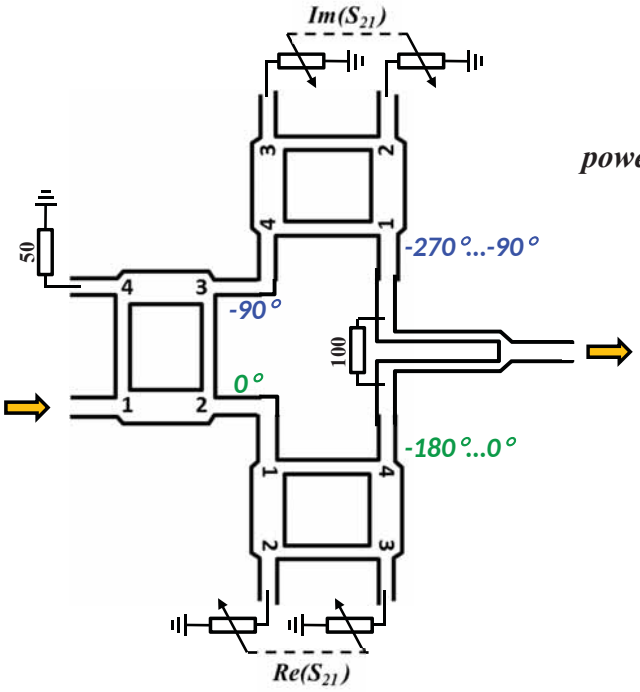
- problem: minimum *noise figure* often does not happen at LNA input match



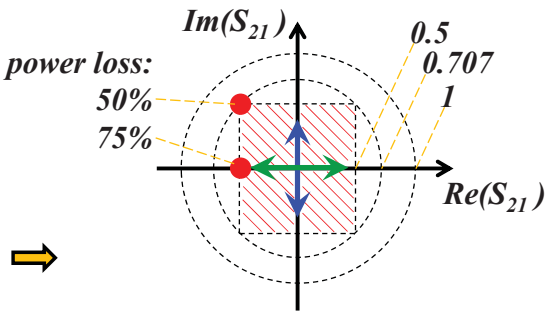
90° hybrid – LNA (low noise amplifier), example:



90° hybrid – application as vector modulator:

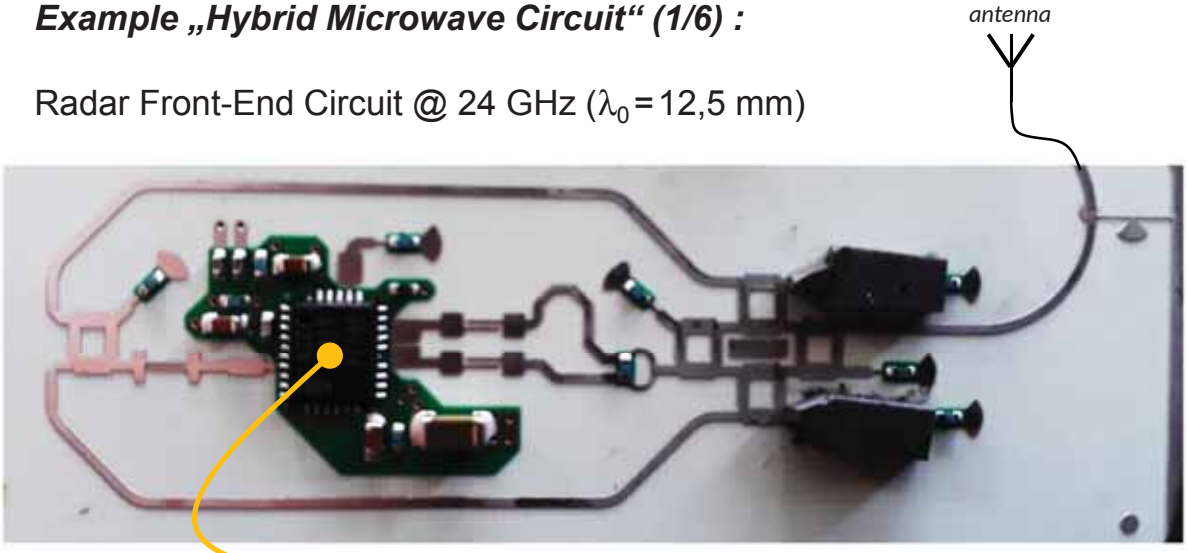


control range:

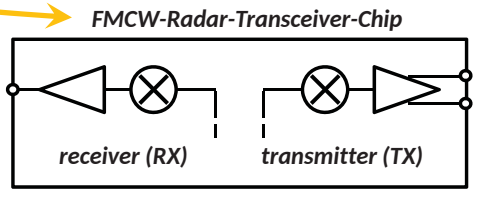


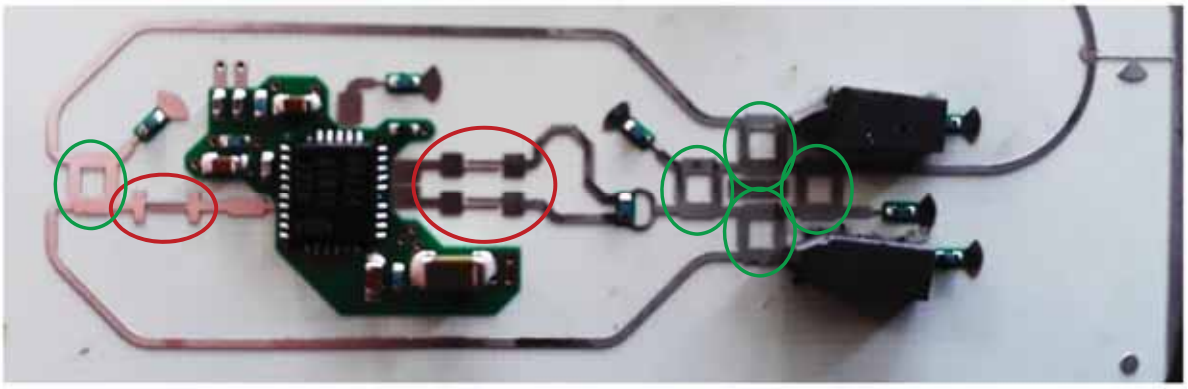
Example „Hybrid Microwave Circuit“ (1/6) :

Radar Front-End Circuit @ 24 GHz (λ_0 = 12,5 mm)



Decoupling/isolation of the transmit signal from the receive input is most critical!



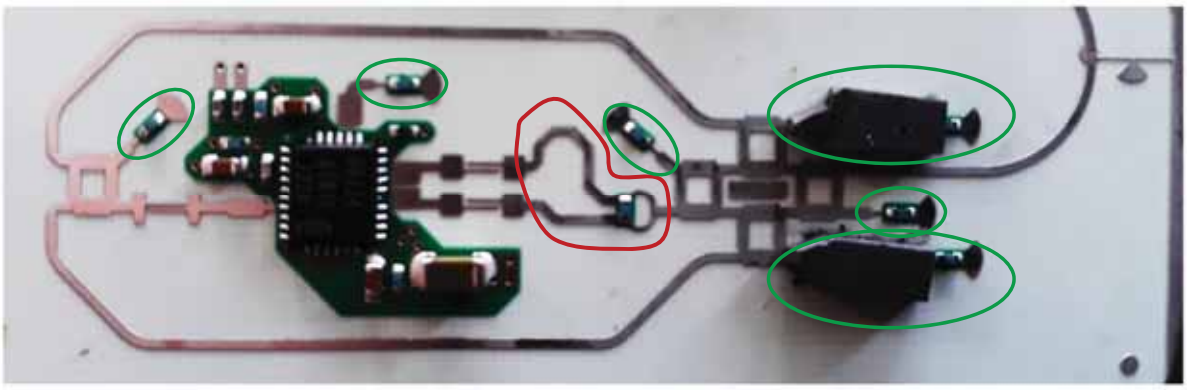


- grey / white circuit board material: typical for RF-grade board dielectrics
- green circuit board surface: typical for solder-stop cover in SMT areas

Series capacitance for de-coupling of DC and chip;

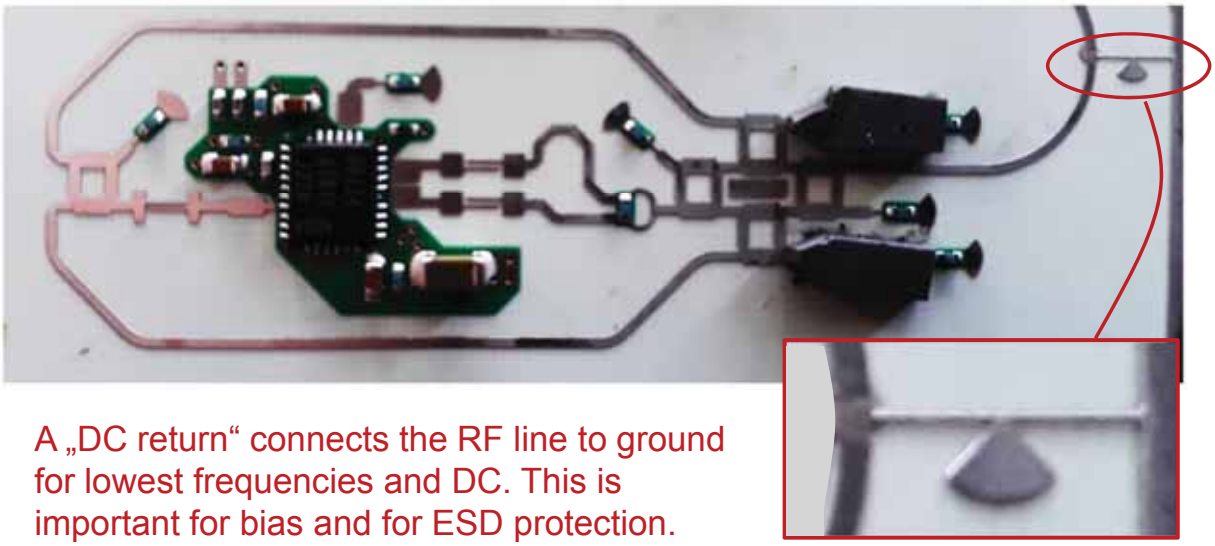
The RF-series inductance of the long, narrow lines will get compensated by capacitances to ground at both ends.

90° - 3dB - hybrid.



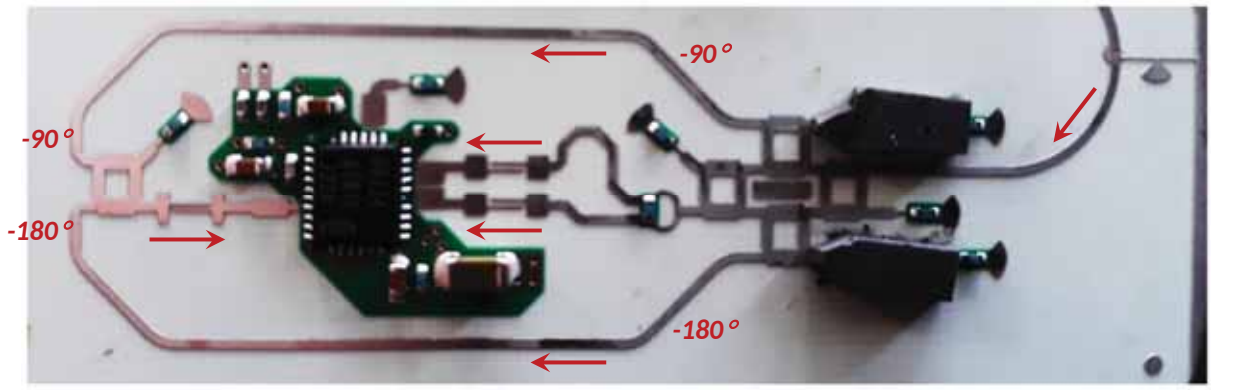
The TX outputs (in odd mode, i.e., 180° phase difference) pass through the $\lambda_g/2$ delay line to become in-phase, then they are added by a 3-port Wilkinson combiner.

The (unwanted) reactance of the 50-Ohm load will get compensated by a series inductance (narrow line section). Instead of a metal via-hole to ground (which would show an inductance), a radial stub is used to provide RF-ground. A piece of absorbing foam/rubber improves the match and/or increases the maximally permitted power dissipation.



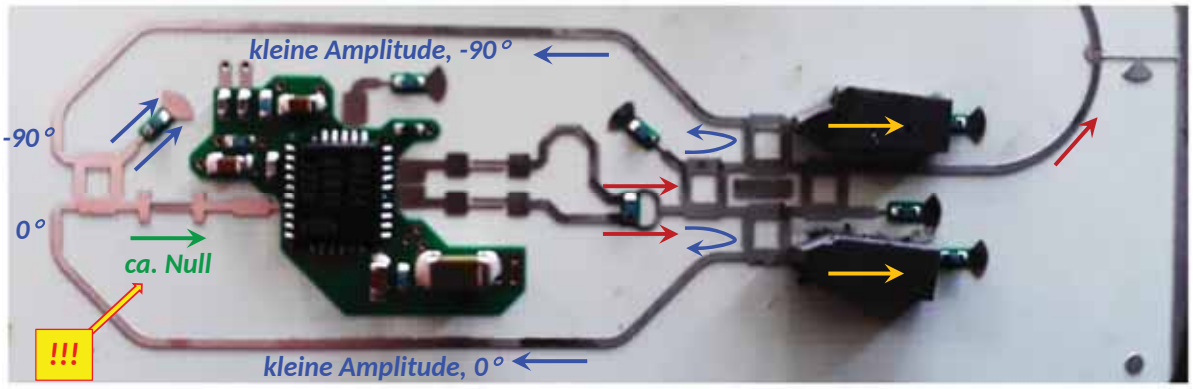
A „DC return“ connects the RF line to ground for lowest frequencies and DC. This is important for bias and for ESD protection.

- A line of minimum width (maximum impedance) causes minimum disturbance at the junction with the RF line (line width at technology limit).
- If the length of the narrow line – from junction to radial stub – is $\lambda_g/4$, then the ground at the radial stub transforms into „open“ at the junction and there is no RF-influence of this structure at all (though narrowband).



Path of a receive signal:

- Half-power of the receive signal travels to the transmit outputs. This is sorry for halving the receive power, but the transmit amplifiers/outputs are not affected by these little signals.
- the other half-power of the receive signal (-3dB) is added in-phase at the receive-input of the chip.

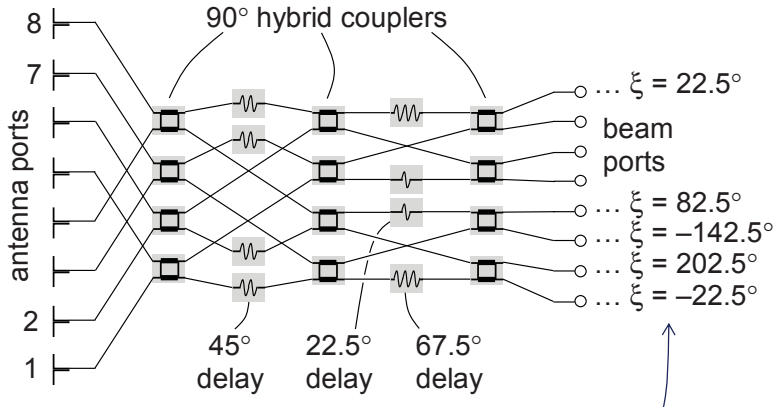


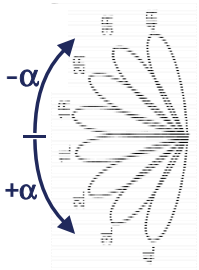
Path of the transmit signal / TX-to-RX-isolation:

- Half-power of the transmit signal travels towards the antenna output, and quarter-power each goes to the two „large“ absorbers (yellow arrow).
- The isolation of the 90°-hybrids is not perfect, therefore there are small signals traveling to the left (blue arrows).
- The 90°-hybrid at the receive input will add the two „small“ signals into the 50-ohms resistor because of their phase relationship. The receive input of the chip then gets only a residual/error signal of ***very small*** magnitude.

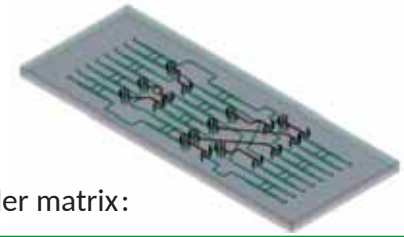
90° Hybrid – application: Butler matrix:

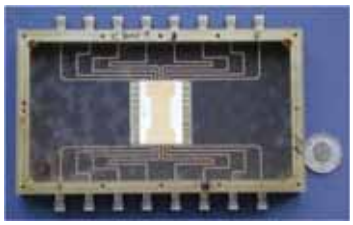
- for feeding of (linear, equidistant) antenna array, while forming (one or several) directive beams





beam-steering angle α is proportional to phase-increment angle ξ



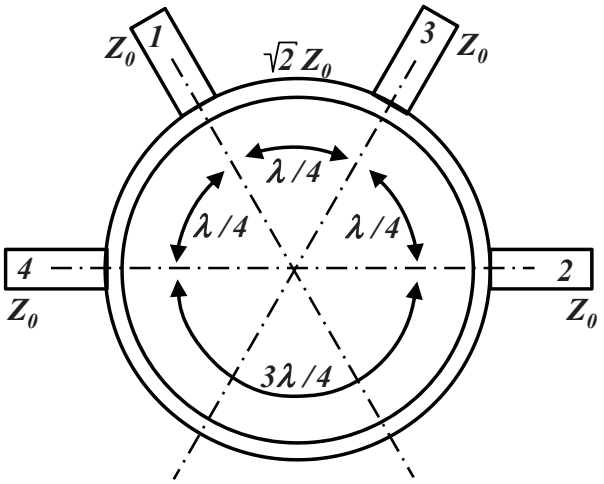


60 GHz 4-layer LTCC Butler matrix:

180° hybrid – „Ratrace“ :

- feeding port 1:
  - . equal power split between port 3 and port 4,
  - . signals at ports 3,4 are in-phase,
  - . port 2 is decoupled/isolated.
- feeding port 2:
  - . equal power split between port 3 and port 4,
  - . signals at ports 3,4 are out-of-phase (180 deg phase difference),
  - . port 1 is decoupled/isolated.

$$[S] = -\frac{j}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$



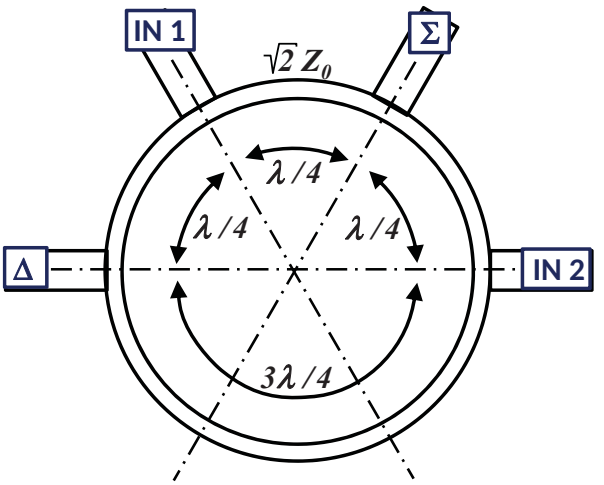
180° hybrid – „Ratrace“ :

- when feeding two signals at the same time at ports 1 and 2, respectively:
  - . no coupling between 1 and 2
  - . the sum of both signals at port 3,
  - . the difference of the two signals at port 4.
- all ports are matched to Z_0;
- loss-less circuit;

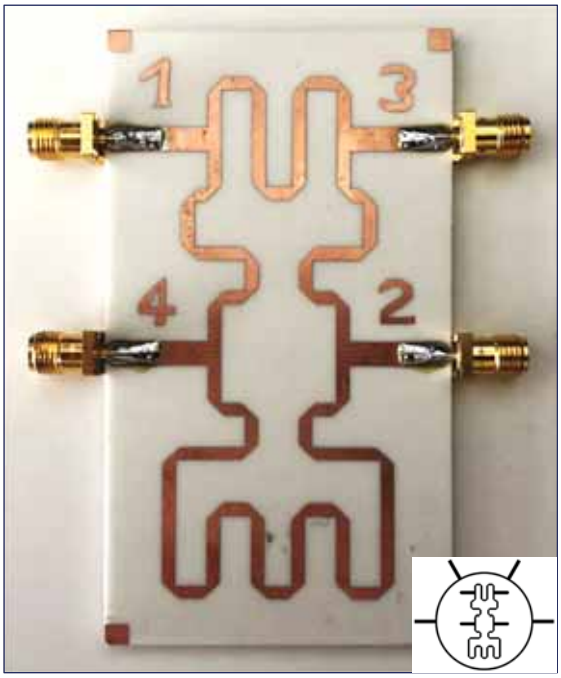
analysis:

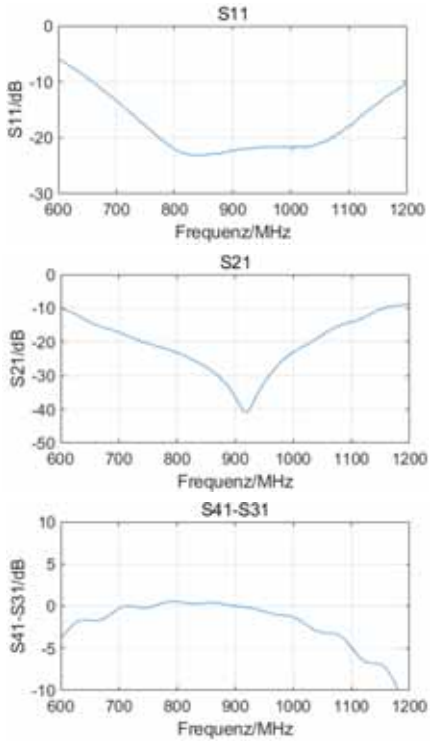
- even/odd with one symmetry …

$$[S] = -\frac{j}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$



180° hybrid – „Ratrace“ , „folded“ realization (saving area):

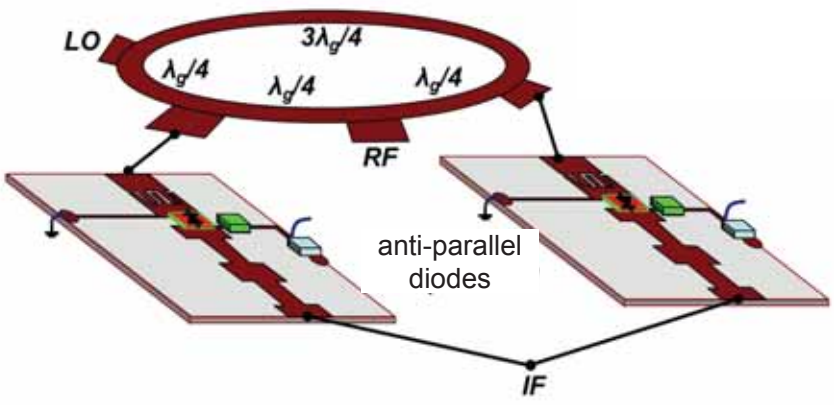




[Yizhang Li 2019]

180° hybrid – „rat race“ , application example 1 :

- **Balanced mixer**
  - with 180 deg hybrid („rat race“)



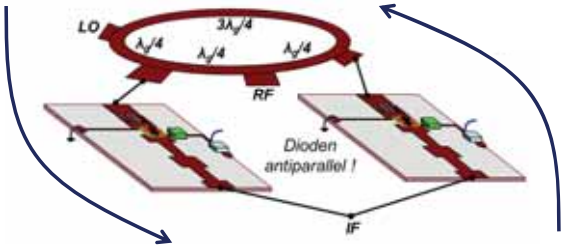
  - advantages: LO-to-RF-isolation, suppression of LO AM noise, suppression of some harmonics/mixing products

180° hybrid – „rat race“ , example „balanced mixer“

down-conversion:

The left diode „sees“ the sum, RF+LO, while the right diode „sees“ the difference, RF-LO. The LO signal is large and is switching the diodes. Since the diodes are anti-parallel, the IF signals from the respective diodes are in-phase and can be added at the mixer output.

Filter: The IF signal does not reach the hybrid because of series-C (high-pass).



Filter: LO signal and RF signal (both having similar, very high frequency) do not reach the IF ports because of the *stepped-impedance* low-pass filters.

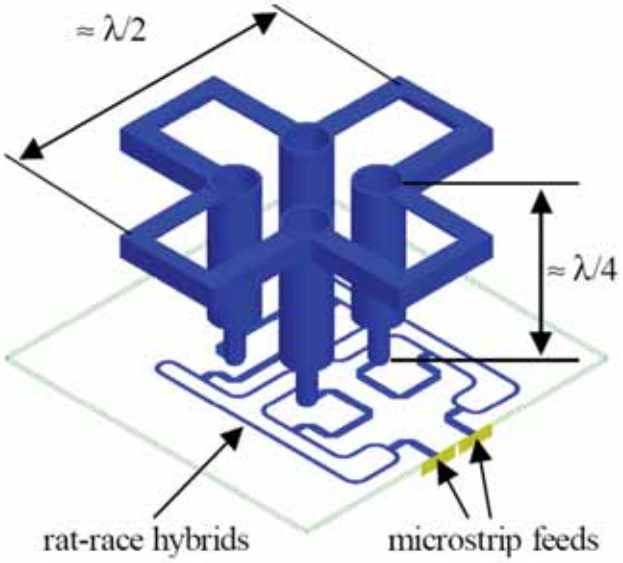
up-conversion:

Each diode „sees“ the IF as in-phase. The diodes will be switched in-phase by the large LO signal (because of the 180° phase difference of the LO plus the diodes being anti-parallel). Therefore, the RF signals coming out of the two diodes are in-phase and combine/add at the RF port of the hybrid.

180° hybrid – „rat race“ , application example 2 :

- feeding „balanced“ circuits (→ balun), e.g., dipole antenna :





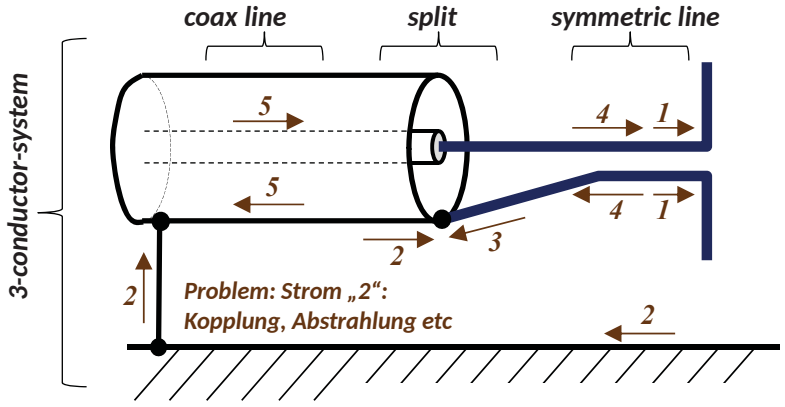
Examples of planar transmission-line circuits

(from „mini-project“ of IHF-lab course „Microwave Components“):

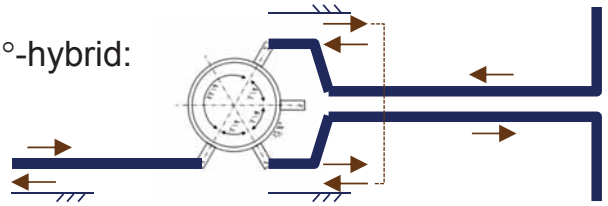


180° hybrid as Balun :

- transition from non-symmetric line to symmetric line:

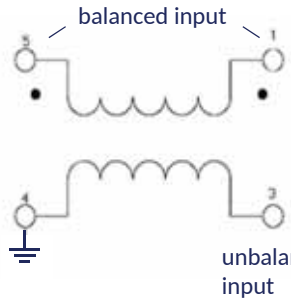


- solution: using 180°-hybrid:



Practical implementations of a Balun

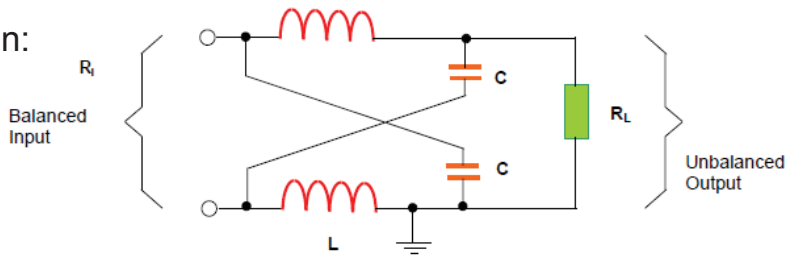
- wire-wound transformator:
  very wide bandwidth
  impedance transformation easily included
  rather low upper freq limit (very few GHz)



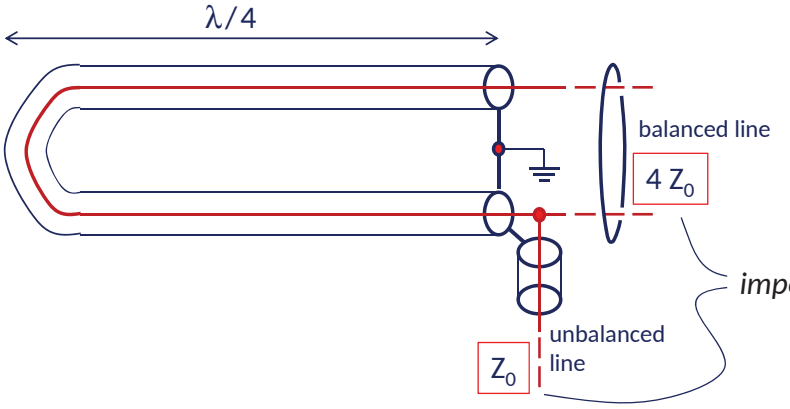

[Macom]

Practical implementations of a Balun

- lattice balun / LC-balun:



- λ/2 – line:



LC-high-pass creates
+90° phase shift

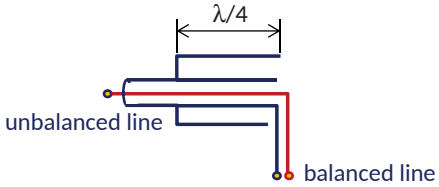
LC-low-pass creates
-90° phase shift

Practical implementations of a Balun

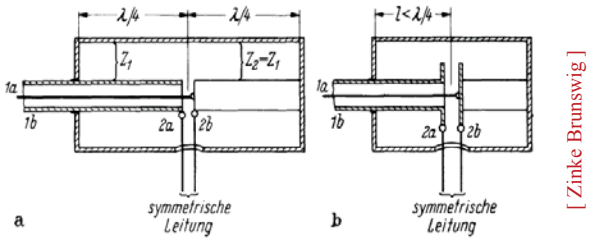
- „choke balun“

  any wave between ground and the outer conductor of a non-symmetric line will be suppressed by the high impedance of a „choke“ (that is, by the open-end of a λ/4-line)

„Bazooka“ or „sleeve“ Balun:
non-symmetric, therefore narrow-band



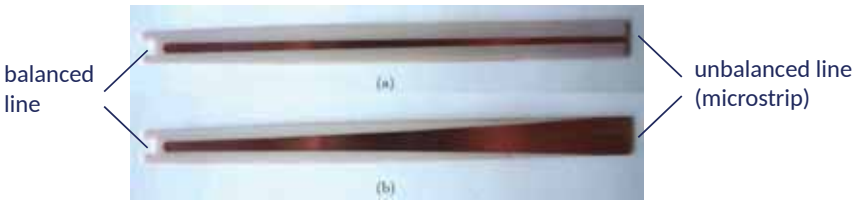
„colinear“ or
symmetric Balun:
broader bandwidth

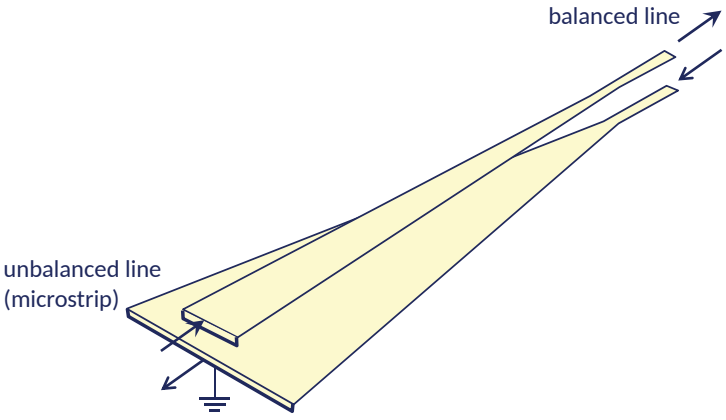


[Zinke Brunswig]

Practical implementations of a Balun

- tapered line:





tapered feedline for
Vivaldi antenna:



Balanced circuits

- three-conductor system: two signal traces plus ground; symmetric
- odd mode: currents on the two signal traces have 180° phase difference; zero current on ground
- even mode: currents on the two signal traces equal and in-phase; double-magnitude current on ground in the opposite direction

… in general, ***any*** current distribution can be represented as a linear combination of even mode and odd mode!

- if only (almost only) the odd mode is used:
  - large immunity to distortions/noise on ground (as it can be caused by current peaks from power supply or clocked digital circuitry)
  - large immunity to EMI (*electromagnetic interference*), i.e., to inductive coupling into wire/current loops (this is because the two signal traces are close to each other, forming small-area loops only, and are sometimes even twisted, and ground is „far away“)

Balanced circuits

- many advantages regarding circuit-performance. But:

  complicated measurements („reasonable“ connectors, cables, and calibration standards do not exist)

- two-port & 2 modes per port → 4-port S-matrix

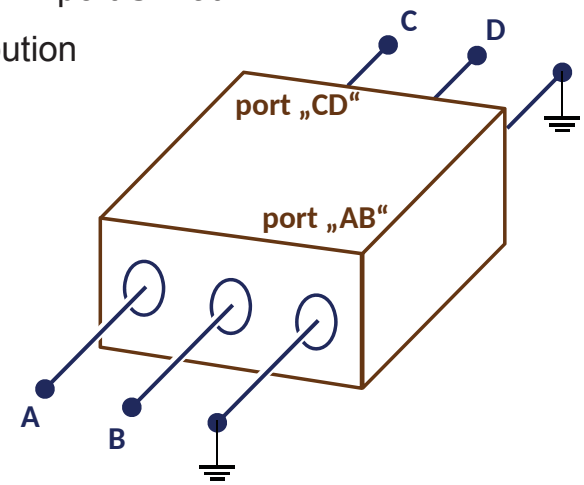
  the generalized current distibution can always be described as linear combination of

  differential mode

  (= odd mode) und

  common mode

  (= even mode)



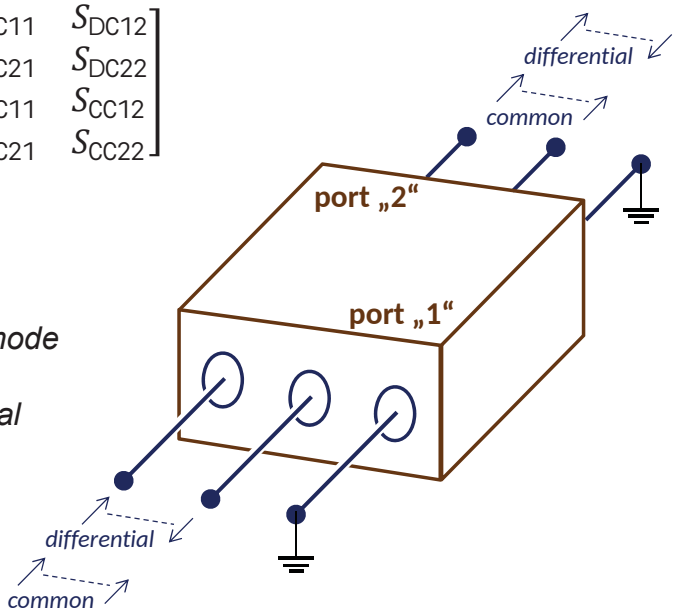
Balanced circuits

- „*mixed-mode*“ S-Matrix:

$$[S_{\text{balanced}}] = \begin{bmatrix} S_{DD11} & S_{DD12} & S_{DC11} & S_{DC12} \\ S_{DD21} & S_{DD22} & S_{DC21} & S_{DC22} \\ S_{CD11} & S_{CD12} & S_{CC11} & S_{CC12} \\ S_{CD21} & S_{CD22} & S_{CC21} & S_{CC22} \end{bmatrix}$$

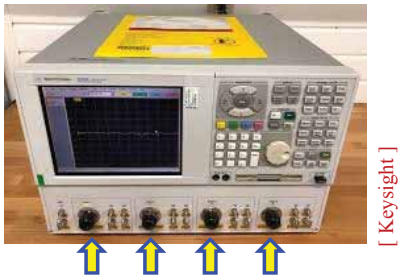
… this column: excitation with the *differential mode* at port 1:

- reflection into the *differential mode* at port 1 (S_{DD11})
- transmission into the *differential mode* at port 2 (S_{DD21})
- reflection into port 1 into the *common mode* (S_{CD11})
- transmission into the *common mode* at port 2 (S_{CD21})



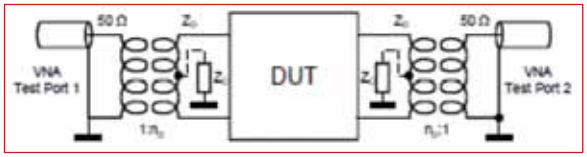
Balanced circuits

- measurement of the „*mixed-mode*“ 2-port S-matrix can be accomplished with the (very expensive!) 4-port „*single-ended*“ network analyzer:


[Keysight]


[Keysight]

- a simpler measurement uses Baluns – but this measurement is „incomplete“ *) even when the Baluns are „known“ a priori and can be de-embedded mathematically



[Anritsu]

*) multiple measurements using different (known!) baluns can result in the complete „mixed-mode“ S-matrix

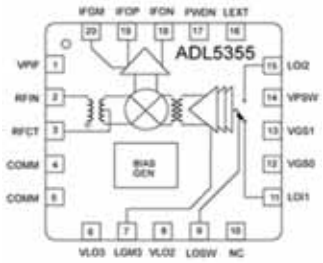
Balanced circuits – examples*:*

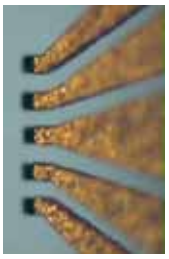
- twisted two-wire line (*twisted pair*):



[Dätwyler]

- active integrated circuit:







[FormFactor]

- dipole antenna (see also: Baluns)

- crosstalk in *multi-conductor* lines (*NEXT, FEXT*)

Transmission-line coupler — single-hole coupler :

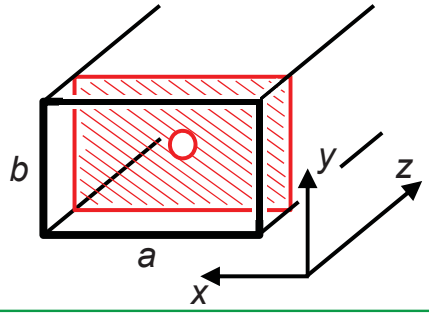
- electric and magnetic field couple through a small hole in a conducting plate:



- e.g.: small hole in a metallic wall accross a rectangular waveguide cross-section:

  . coupling by Hx only

  . corresponds to a shunt inductance



- small hole in the center of the top/bottom wall of rectangular waveguide:

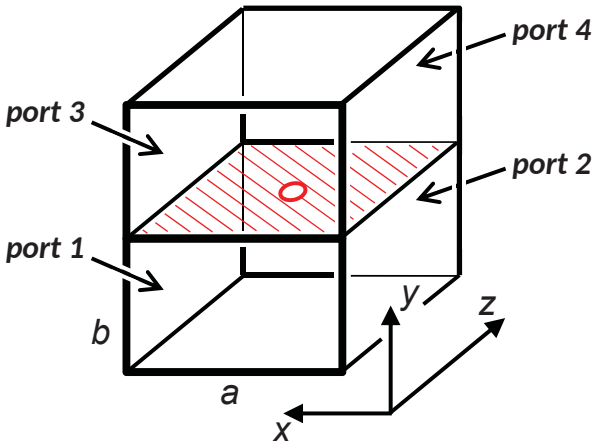
  . couples two waveguide by E_y and H_x

  . for small hole is: $S_{11} \approx 0$ und $S_{21} \approx 1$

  . for thin wall (hole radius r):

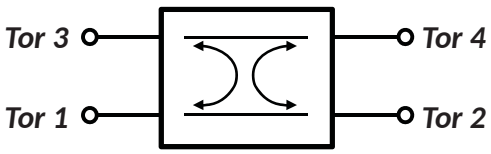
$$S_{31} = -j\frac{2\omega r^3}{v_{ph} 3ab}\left(2 + \frac{1}{1-\left(\frac{\lambda}{2a}\right)^2}\right)$$

$$S_{41} = j\frac{2\omega r^3}{v_{ph} 3ab}\left(2 - \frac{1}{1-\left(\frac{\lambda}{2a}\right)^2}\right)$$



  . thus for $f = \frac{c}{a\sqrt{2}} = \sqrt{2} \cdot f_{cutoff}$ follows $S_{41} = 0$ → directional coupler

- directional coupler (backward coupler) – definitions:



$$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} & (S_{14}) \\ S_{12} & 0 & (S_{14}) & S_{13} \\ S_{13} & (S_{14}) & 0 & S_{12} \\ (S_{14}) & S_{13} & S_{12} & 0 \end{bmatrix}$$

  . ideally $S_{14} = 0$

  . transmission coefficient S_{12}

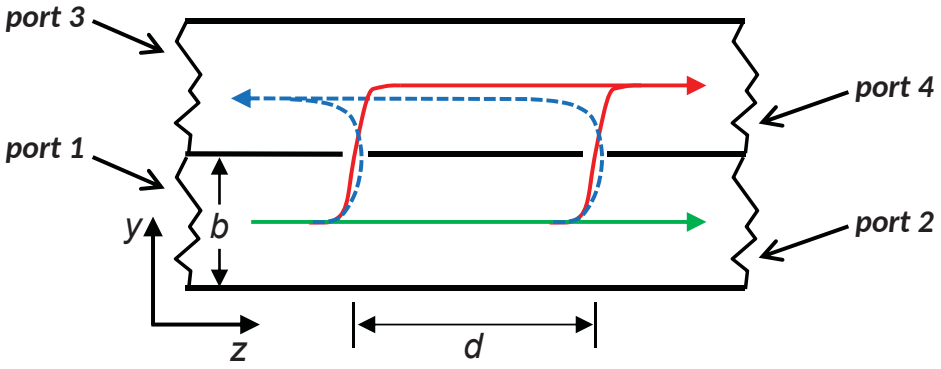
  . coupling coefficient S_{13}

  . directivity $A_R = \frac{S_{13}}{S_{14}}$, $A_R|_{dB} = 20 \log \frac{S_{13}}{S_{14}}$

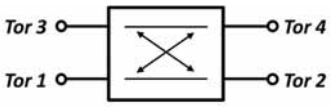
Transmission-line coupler — two-hole coupler :

- disadvantages of the single-hole coupler:
  narrow bandwidth and small coupling coefficients only

- two-hole coupler:



. for $d = \frac{\lambda_g}{4}$ follows (for small coupling): $S_{31} = 0$

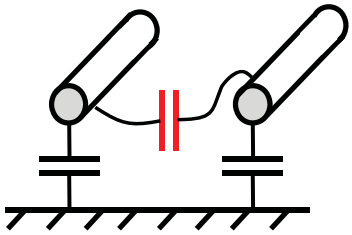


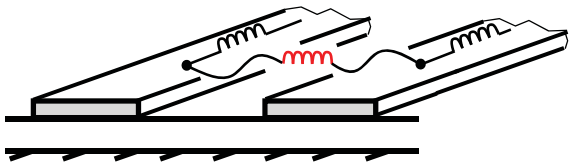
Transmission-line coupler using coupled lines, variant 1 :

- the following analysis concept applies to couplers
  with „distributed coupling“ between two transmission lines

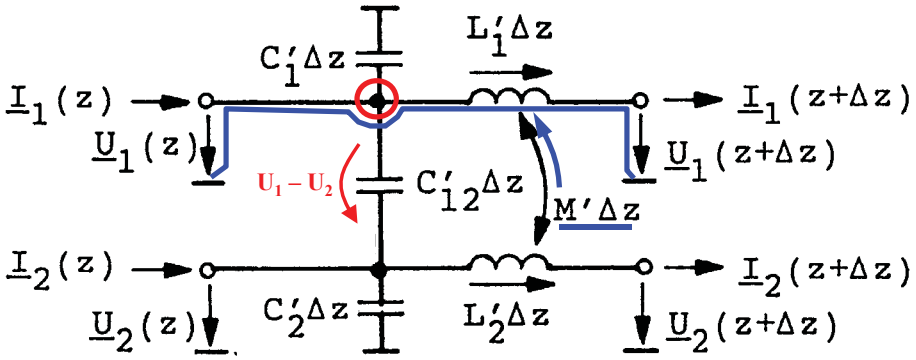
- also important for the analysis of unwanted coupling:
  . field of EMI / EMC,
  . cross-talk along long, weakly-coupled lines
    - NEXT (*near-end cross-talk*)
    - FEXT (*far-end cross-talk*)

- modelled by coupling capacitances and coupling inductances:





- assuming a short piece of coupled transmission lines with constant (along length) cross-section:



node: $I_1(z) = j\omega C_1' \Delta z U_1 + j\omega C_{12}' \Delta z (U_1 - U_2) + I_1(z + \Delta z)$

loop: $U_1(z) = U_1(z + \Delta z) + j\omega L_1' \Delta z I_1(z + \Delta z) + j\omega M' \Delta z I_2(z + \Delta z)$

weak coupling: $I_{1,2}(z + \Delta z) \approx I_{1,2}(z)$

- short piece of coupled transmission lines :

$$\frac{I_1(z + \Delta z) - I_1(z)}{\Delta z} = -j\omega(C_1 + C_{12})U_1 + j\omega C_{12} U_2$$

... follows from Kirchhoff's junction rule, see above

$$\frac{dI_1}{dz} = -j\omega(C_1' + C_{12}')U_1 + j\omega C_{12}' U_2$$

$$\frac{dI_2}{dz} = -j\omega(C_2' + C_{12}')U_2 + j\omega C_{12}' U_1$$

$$\frac{dU_1}{dz} = -j\omega L_1' I_1 - j\omega M' I_2$$

$$\frac{dU_2}{dz} = -j\omega L_2' I_2 - j\omega M' I_1$$

… these are coupled differential equations; solution/simplification using technique of even/odd modes:

$$U_e = U_1 + U_2$$

$$U_o = U_1 - U_2$$

$$I_e = I_1 + I_2$$

$$I_o = I_1 - I_2$$

… simplification: identical lines

$$\frac{dI_e}{dz} = -j\omega C' U_e \qquad \frac{dI_o}{dz} = -j\omega (C' + 2C'_{12}) U_o$$

$$\frac{dU_e}{dz} = -j\omega (L' + M') I_e \qquad \frac{dU_o}{dz} = -j\omega (L' - M') I_o$$

$$Z_{\text{even}} = \sqrt{\frac{L' + M'}{C'}} \qquad Z_{\text{odd}} = \sqrt{\frac{L' - M'}{C' + 2C'_{12}}}$$

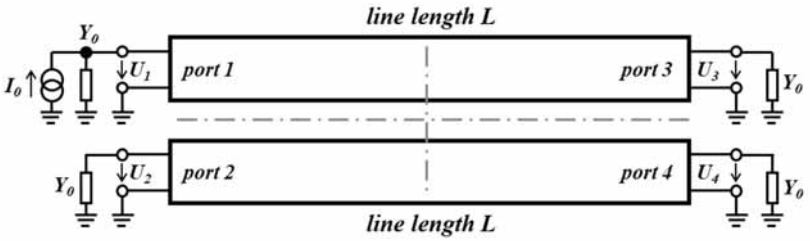
$$v_e = \frac{1}{\sqrt{(L' + M')C'}} \qquad v_o = \frac{1}{\sqrt{(L' - M')(C' + 2C'_{12})}}$$

- when $v_e = v_o$: $\dfrac{Z_{\text{even}} - Z_{\text{odd}}}{Z_{\text{even}} + Z_{\text{odd}}} = \dfrac{M'}{L'} = \dfrac{C'_{12}}{C' + C'_{12}} \overset{!}{=} k$ … coupling factor

$$Z_0 = \sqrt{\frac{L'}{C'}} \quad ; \quad Z_{\text{even}} = Z_0 \sqrt{\frac{1+k}{1-k}} \quad ; \quad Z_{\text{odd}} = Z_0 \sqrt{\frac{1-k}{1+k}} \quad ; \quad Z_0 = \sqrt{Z_{\text{even}} Z_{\text{odd}}}$$

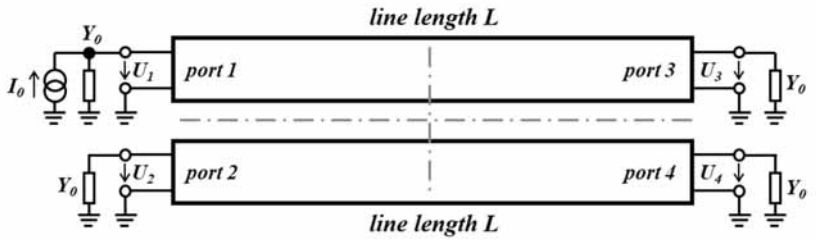
The following, detailed analysis aims at the determination of the 4-port S-matrix of the ***coupled line directional coupler***.

For simplification, a structure is assumed which has two symmetry planes (dash-dotted lines in the figure).



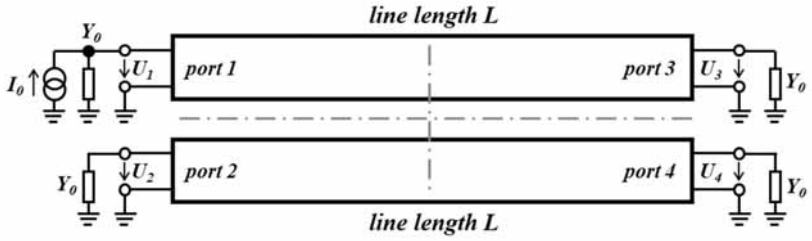
Such analysis could be done by connecting a source to port 1, and connecting matched loads (no sources) to all other ports. Then, …

… Then one will calculate the signals „traveling out“ of ports 1,2,3,4. This allows to calculate one row of the S-matrix. The remaining parameters are found from symmetry considerations.



However, this procedure is cumbersome. It is cheaper to analyse, four times, one quarter of the structure each. Because of the two symmetries of the structure, this is a valid approach, and the „quarter“ turns out to be quite simple to analyse: This is called a **even-mode-odd-mode-analysis**.

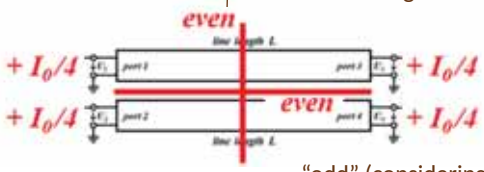
In the **even-mode-odd-mode-analysis**, 4 sources are connected to the 4 ports at the same time. Magnitiudes (and source impedmances) of the 4 sources are equal, and the respective phases of the 4 sources are chose such that the symmetry planes are realized either as „short“ (zero voltage or E-field at symmetry plane) or as „open“ (zero current or max E-field a the symmetry plane). Since there are two symmetry planes with two potential states, there are 4 resulting scenarios of „circuit with sources“.



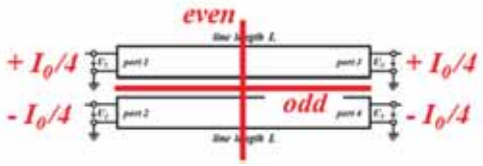
The addition of the respective sources at the 4 ports of the 4 circuits (shown below) results in the earlier scenario with a single source at port 1 and no sources – they add/subtract to zero – at the other 3 ports. This is indicated below by the respective signs („*“ and „-“) of the source currents at the ports.

scenario „ee“:

"even" (considering voltages) equals "open circuit", i.e. max. voltage and zero current

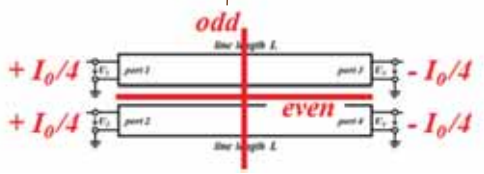


scenario „oe“:

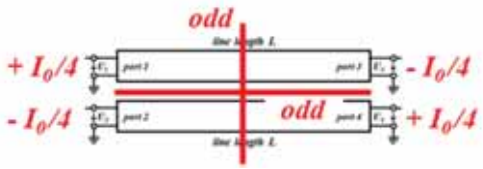


scenario „eo“:

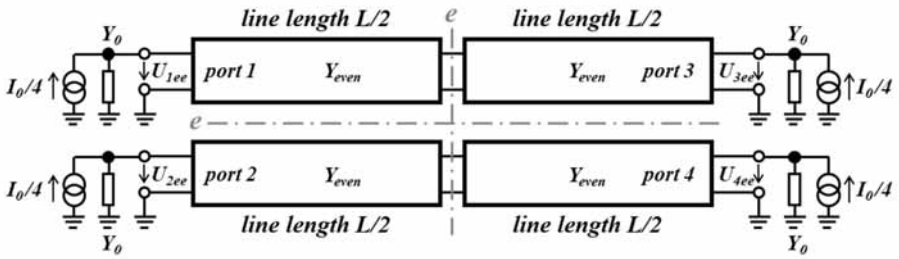
"odd" (considering voltages) equals "short circuit", i.e. max. current and zero voltage



scenario „oo“:

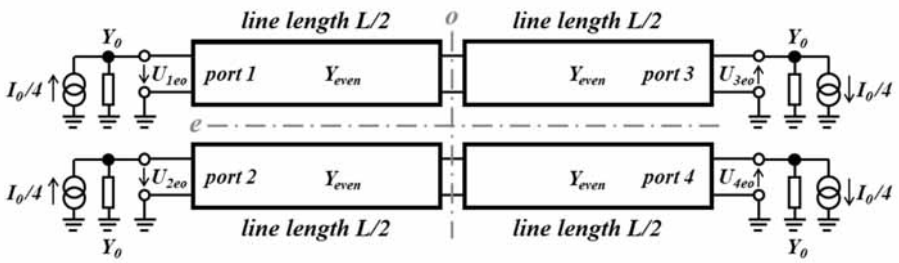


scenario „*even-even*“



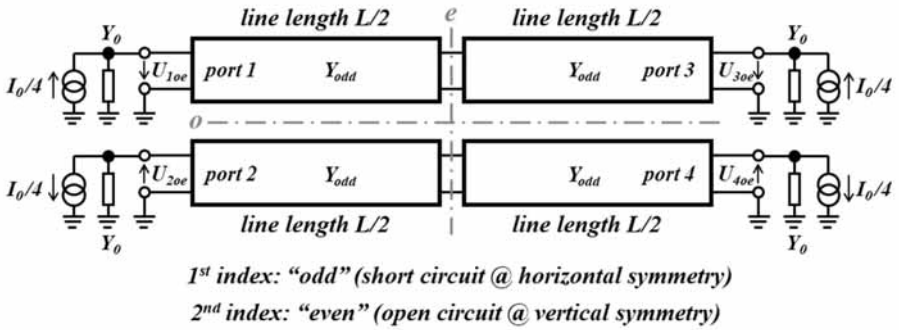
1st index: "even" (open circuit @ horizontal symmetry)
2nd index: "even" (open circuit @ vertical symmetry)

scenario „*even-odd*“



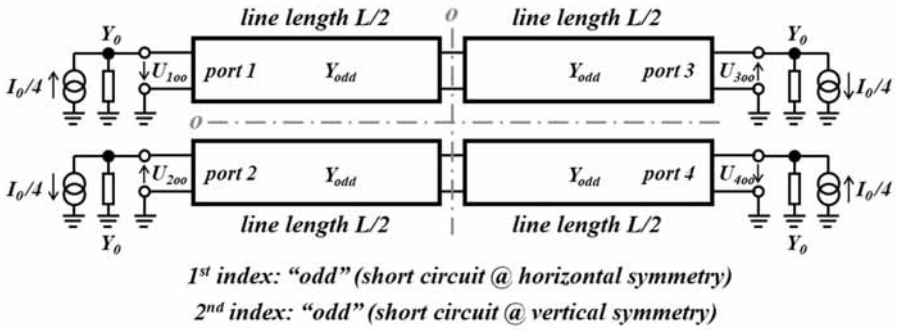
1st index: "even" (open circuit @ horizontal symmetry)
2nd index: "odd" (short circuit @ vertical symmetry)

scenario „*odd-even*“



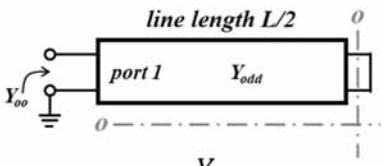
1st index: “odd” (short circuit @ horizontal symmetry)
2nd index: “even” (open circuit @ vertical symmetry)

scenario „*odd-odd*“



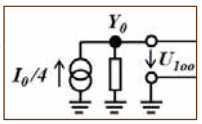
1st index: “odd” (short circuit @ horizontal symmetry)
2nd index: “odd” (short circuit @ vertical symmetry)

The excitation at all 4 ports at the same time causes specific currents and voltages in the symmetry planes, such that a specific „quarter“ of the circuit can be analyzed easily. For the example case of „scenario „oo““ it is found:



$$Y_{oo} = \frac{Y_{odd}}{t}$$

$$U_{i,oo} = \frac{I_0}{4(Y_0 + Y_{oo})} \quad , \quad i = 1,2,3,4$$



… here, quantity ***t*** describes the impedance transformation from the line end (at the symmetry plane) to the input of the line (at the port):

Example: input admittance Y_{in} of a line of characteristic admittance Y_{line} of length ***d*** which is **short-circuited** at its end:

$$Y_{in} = -jY_{line}\cot(\beta d)$$

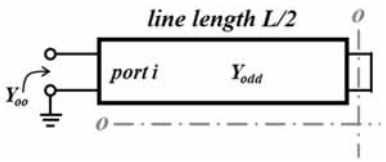
… Substitution: $d = \frac{L}{2}$ & $t = j\tan(\beta d)$ → $Y_{in} = \frac{Y_{line}}{t}$

… input admittance Y_{in} of a line of characteristic admittance Y_{line} of length ***d*** which is **open-circuited** at its end:

$$Y_{in} = jY_{line}\tan(\beta d)$$

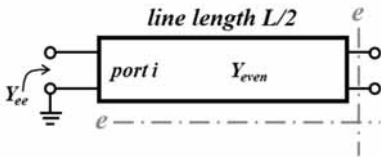
… substitution: $d = \frac{L}{2}$ & $t = j\tan(\beta d)$ → $Y_{in} = Y_{line}t$

For all 4 scenarios for the 4 cases of symmetry plane-combinations results:

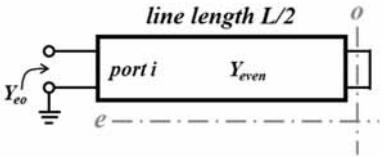


→ $Y_{oo} = \frac{Y_{odd}}{t}$ → $U_{i,oo} = \frac{I_0}{4(Y_0 + Y_{oo})}$, $i = 1,2,3,4$

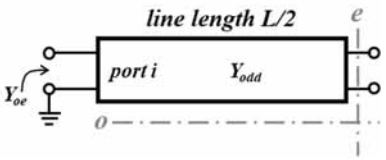
…



→ $Y_{ee} = Y_{even}t$ → $U_{i,ee} = \frac{I_0}{4(Y_0 + Y_{ee})}$, $i = 1,2,3,4$



→ $Y_{eo} = \frac{Y_{even}}{t}$ → $U_{i,eo} = \frac{I_0}{4(Y_0 + Y_{eo})}$, $i = 1,2,3,4$



→ $Y_{oe} = Y_{odd}t$ → $U_{i,oe} = \frac{I_0}{4(Y_0 + Y_{oe})}$, $i = 1,2,3,4$

As shown above for coupled lines, even- and odd-coupled impedances (and admittances) were defined and a parameter ***k*** was introduced (***k*** will turn out being the coupling coefficient):

$$Y_{even} = Y_0\sqrt{\frac{1-k}{1+k}}$$

$$Y_{odd} = Y_0\sqrt{\frac{1+k}{1-k}}$$

For simplification, a quantity ***a*** is defined as:

$$a = \sqrt{\frac{1-k}{1+k}} \quad \rightarrow \quad Y_{even} = aY_0 \qquad Y_{odd} = \frac{Y_0}{a}$$

… then all 16 parameters at the ports are defined as:

$$U_{i,ee} = \frac{I_0}{4Y_0}\frac{1}{1+at} \quad , \quad U_{i,eo} = \frac{I_0}{4Y_0}\frac{1}{1+\frac{a}{t}} \quad , \quad U_{i,oe} = \frac{I_0}{4Y_0}\frac{1}{1+\frac{t}{a}} \quad , \quad U_{i,oo} = \frac{I_0}{4Y_0}\frac{1}{1+\frac{1}{at}}$$

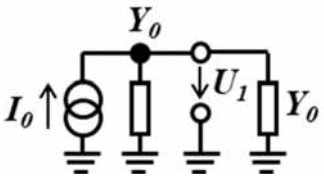
$$i = 1,2,3,4$$

At each port, the total voltage is found by adding the respective four voltages even/odd-scenarios (with the correct respective signs!). That is, superposition of “ee”, “eo”, “oe”, “oo”:

port 1:

$$U_1 = U_{1,ee} + U_{1,eo} + U_{1,oe} + U_{1,oo} = \frac{I_0}{4Y_0}\left(\frac{1}{1+at} + \frac{1}{1+\frac{a}{t}} + \frac{1}{1+\frac{t}{a}} + \frac{1}{1+\frac{1}{at}}\right) = \frac{I_0}{2Y_0}$$

… the expression $U_1 = \frac{I_0}{2Y_0}$ means, that the source (source admittance Y_0) at port 1, while „looking into“ port 1, „sees“ an admittance Y_0 : Therefore, port 1 is matched!



port 4:

$$U_4 = U_{4,ee} - U_{4,eo} - U_{4,oe} + U_{4,oo} = \frac{I_0}{4Y_0}\left(\frac{1}{1+at} - \frac{1}{1+\frac{a}{t}} - \frac{1}{1+\frac{t}{a}} + \frac{1}{1+\frac{1}{at}}\right) = 0$$

… the expression $U_4 = 0$ means, that, with source at port 1, no signal "leaves" port 4 (also no signal enters port 4, because port 4 is matched and without source). Therefore, port 4 is isolated (source at port 1)!

port 2:

$$U_2 = U_{2,ee} + U_{2,eo} - U_{2,oe} - U_{2,oo} = \frac{I_0}{4Y_0}\left(\frac{1}{1+at} + \frac{1}{1+\frac{a}{t}} - \frac{1}{1+\frac{t}{a}} - \frac{1}{1+\frac{1}{at}}\right)$$

… this simplifies in a special case: $L = \frac{\lambda}{4} \rightarrow d = \frac{\lambda}{8} \rightarrow t = j\tan(\beta d) = j \rightarrow U_2 = \cdots = k\frac{I_0}{2Y_0}$

Since: $S_{21} = \frac{U_2}{U_1}$ this is because there is no outgoing wave at port 1 (port 1 matched, see above) and no incoming wave at port 2 (matched, no source) (otherwise, in general, total voltage is the sum of incoming plus outgoing).

Thus: $S_{21} = k$

port 3:

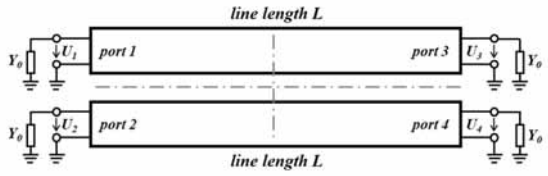
$$U_3 = U_{3,ee} - U_{3,eo} + U_{3,oe} - U_{3,oo} = \frac{I_0}{4Y_0}\left(\frac{1}{1+at} - \frac{1}{1+\frac{a}{t}} + \frac{1}{1+\frac{t}{a}} - \frac{1}{1+\frac{1}{at}}\right) = \cdots$$

… this simplifies in a special case: $L = \frac{\lambda}{4} \rightarrow U_3 = \cdots = -j\sqrt{1-k^2}\frac{I_0}{2Y_0}$

Thus (reasoning similar to port 2): $S_{31} = \frac{U_3}{U_1} \rightarrow S_{31} = -j\sqrt{1-k^2}$

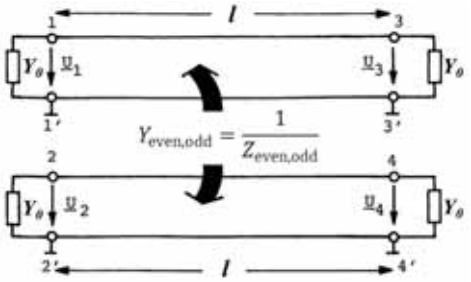
The 4-port S-matrix for $L = \frac{\lambda}{4}$ is (applying symmetries and reciprocity):

$$[S] = \begin{bmatrix} 0 & k & -j\sqrt{1-k^2} & 0 \\ k & 0 & 0 & -j\sqrt{1-k^2} \\ -j\sqrt{1-k^2} & 0 & 0 & k \\ 0 & -j\sqrt{1-k^2} & k & 0 \end{bmatrix}$$



… feeding the coupler at port 1 causes a signal split between ports 2 and 3 according to the coupling coefficient ***k*** . The output signals at ports 2 and 3 show a phase difference of 90°.

$$[S] = \begin{bmatrix} 0 & k & -j\sqrt{1-k^2} & 0 \\ k & 0 & 0 & -j\sqrt{1-k^2} \\ -j\sqrt{1-k^2} & 0 & 0 & k \\ 0 & -j\sqrt{1-k^2} & k & 0 \end{bmatrix}$$



However the above results are for quarter-wave length of the coupling section. The general case of length ***l*** reads:

$$[S] = \begin{bmatrix} 0 & S_{21} & S_{31} & 0 \\ S_{21} & 0 & 0 & S_{31} \\ S_{31} & 0 & 0 & S_{21} \\ 0 & S_{31} & S_{21} & 0 \end{bmatrix}$$

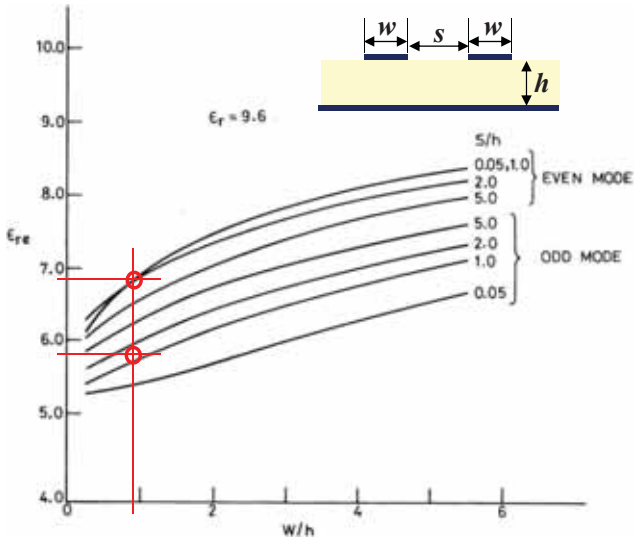
$$S_{21} = \frac{j k \sin(\beta l)}{\sqrt{1-k^2}\cos(\beta l) + j\sin(\beta l)}$$

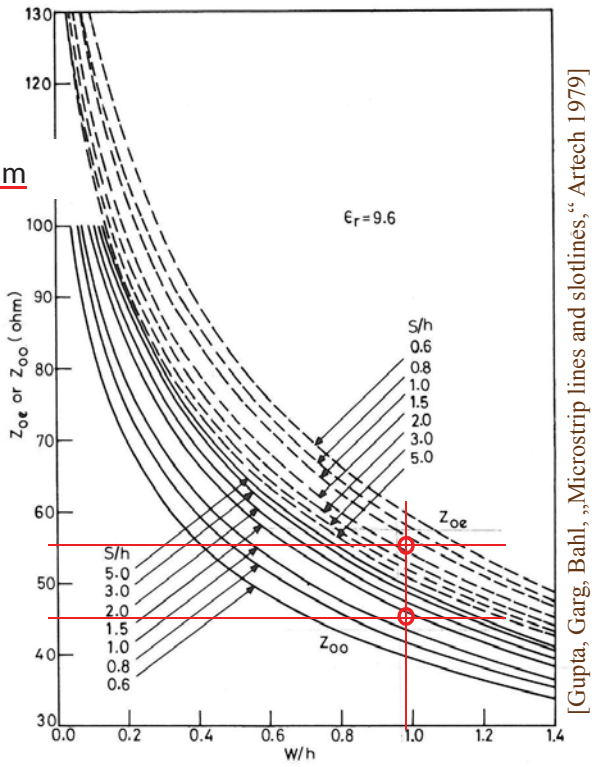
$$S_{31} = \frac{\sqrt{1-k^2}}{\sqrt{1-k^2}\cos(\beta l) + j\sin(\beta l)}$$

- Characteristics of transmission-line couplers:
  - . one-hole coupler, TEM line coupler → backward coupler
  - . two-hole coupler → forward coupler
  - . 90° phase difference between through port and coupled port
  - . frequency dependency
  - . directivity (*dt.*: Richtfaktor) as a measure of the coupling vs isolation
- Design process of TEM transmission-line couplers:
  - . selection of suitable coupled-line geometry (type, x-section)
  - . calculation of x-section geometry, parameters Z_{even}, Z_{odd}
- Problems:
  - . required coupling cannot be realized (for strong coupling, e.g., 3dB)
  - . “real” non-TEM transmission-line (e.g., microstrip): limited directivity
  - . required bandwidth too large

- example: microstrip coupler:

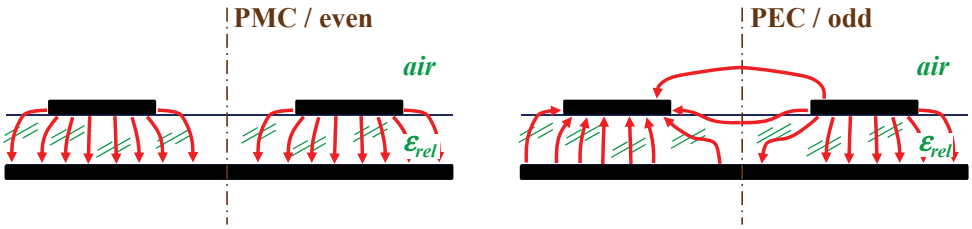
20dB-coupler for 2.5 GHz, Z_0=50Ω, → k=0.1,
Z_{oe}=55.3Ω, Z_{oo}=45.2Ω, s/h=1.3, w/h=0.98,
→ h=0.762mm, s=0.99mm, w=0.74mm,
$\varepsilon_{eff,even}$=6.8, $\varepsilon_{eff,odd}$=5.8 → length=11.5...12.5 mm





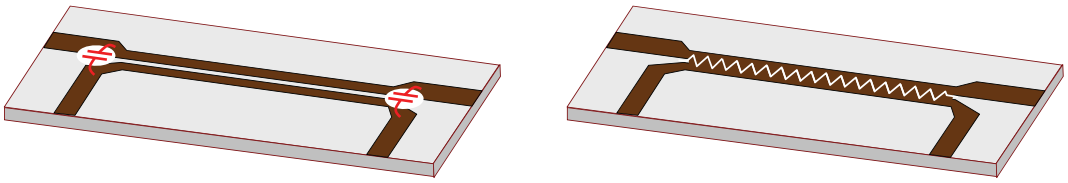
[Gupta, Garg, Bahl, „Microstrip lines and slotlines," Artech 1979]

- problem: different phase velocity of v_e and v_o limits the directivity (resulting in non-perfect isolation of the „isolated" port)



$\varepsilon_{eff,even} > \varepsilon_{eff,odd}$ ⇨ $v_{even} < v_{odd}$

- solution:



increased delay for the odd-mode only

increase of line length for the odd-mode only

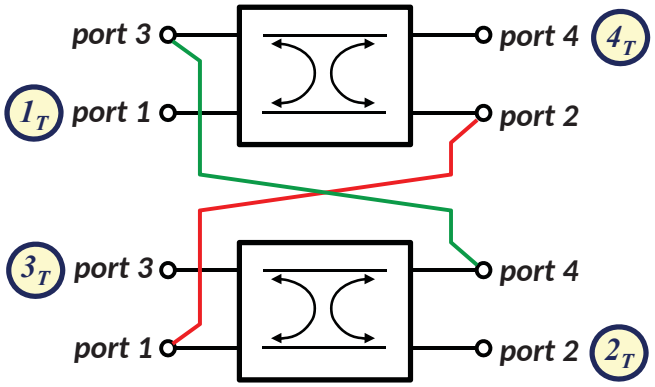
- problem: strong coupling practically impossible

  . would require too small distance between coupled lines

- solution:

  . tandem coupler

$$[S] = \begin{bmatrix} 0 & S_{21} & S_{31} & 0 \\ S_{21} & 0 & 0 & S_{31} \\ S_{31} & 0 & 0 & S_{21} \\ 0 & S_{31} & S_{21} & 0 \end{bmatrix}$$



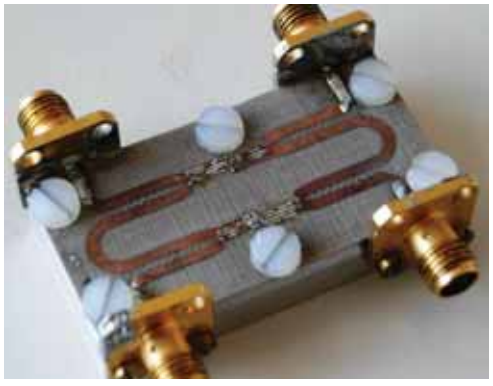
$$S_{21,T} = S_{21}^2 + S_{31}^2$$

$$S_{31,T} = 2\, S_{21}\, S_{31}$$

$$|S_{31,T}| = 2\,|S_{31}|\,\sqrt{1 - |S_{31}|^2}$$

$$k_T = 2\,k\,\sqrt{1 - k^2}$$

. for $k_T = 0.707$ (3 dB coupler) results $k = 0.383$ (8.34 dB)



tandem coupler (3 dB) for 2…8 GHz



Lange coupler (8.34 dB)

- Lange coupler: strong coupling and quite large bandwidth



Transmission-line coupler using coupled lines, variant 2 :

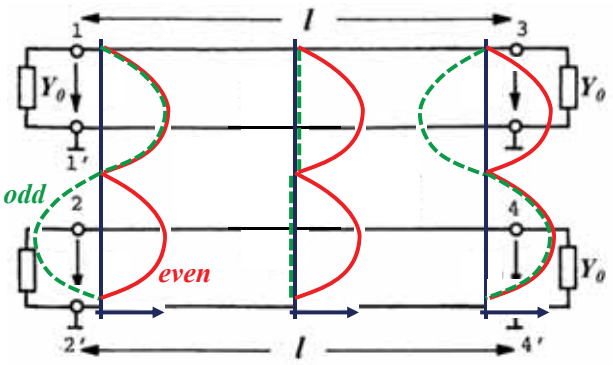
- forward coupler using property of different v_e and v_o
- very strong overall coupling feasible
  . 0 dB coupler
- very long couplers (with locally very weak coupling) possible
  . optical fiber coupler

$v_{\text{even}} \neq v_{\text{odd}} \Rightarrow \beta_{\text{even}} l \neq \beta_{\text{odd}} l$

- 0 dB coupler:

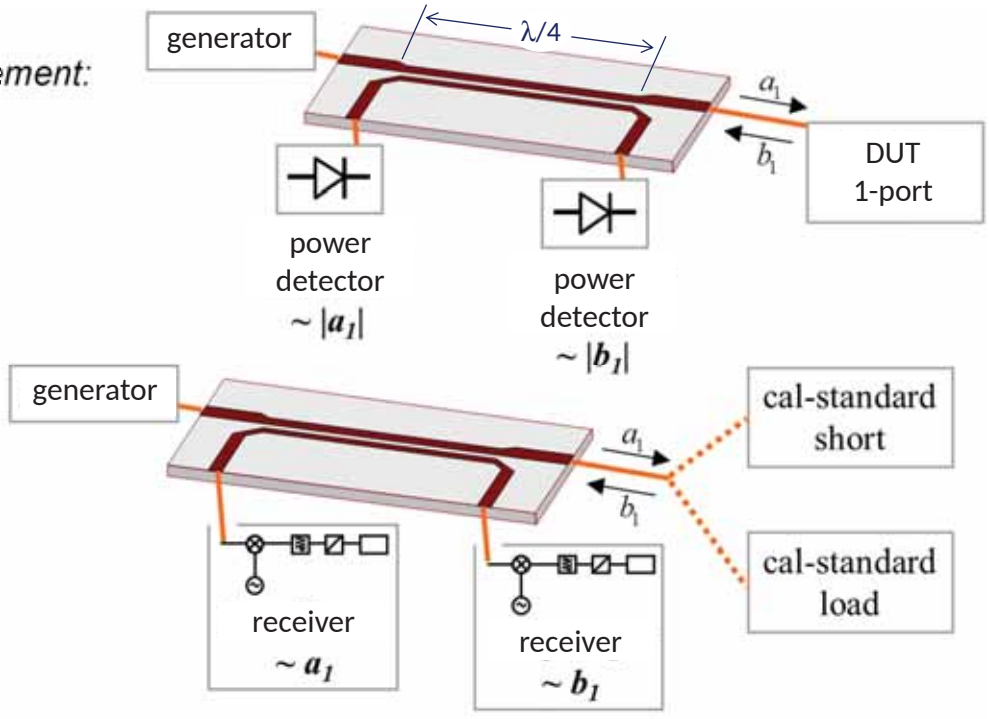
$$\beta_{\text{even}}\, l_{0\text{dB}} - \beta_{\text{odd}}\, l_{0\text{dB}} = \pi$$

$$l_{0\text{dB}} = \frac{\pi}{\beta_{\text{even}} - \beta_{\text{odd}}}$$



Application of directional coupler — S-parameter measurement :

- *1-port measurement:*



. measurement with (power-) detectors gives amplitude information only

. down-conversion using a common reference oscillator signal preserves the (relative) phase information

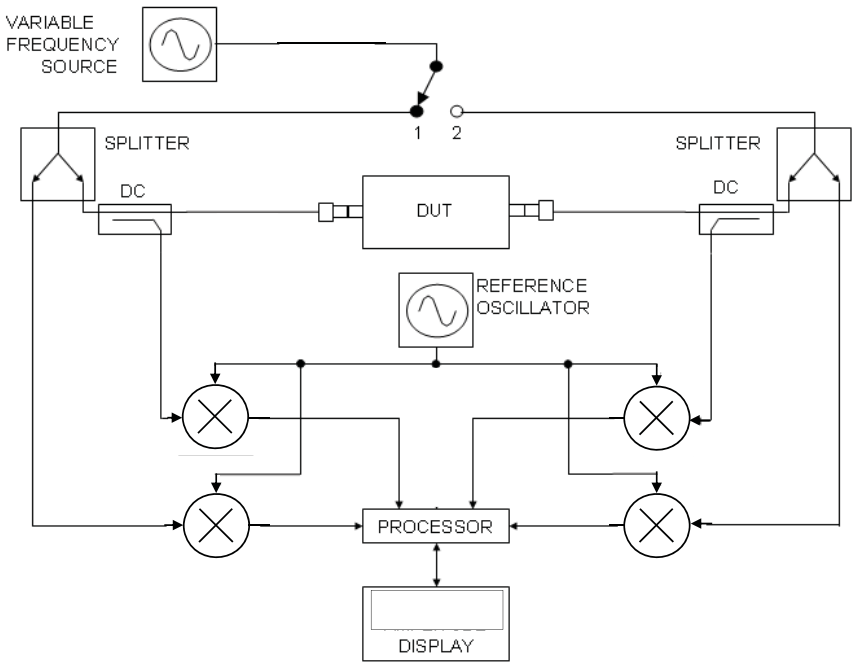
. non-perfect hardware (impedance steps, non-ideal couplers, variations over frequency) can be corrected by additional measurement of „known“ DUT

- short-circuit
- „open“
- 50 Ω load

- „calibration“

. important are linearity and reproducibility

- *2-port measurement* — the „vector network analyzer“:



- *2-port measurement* — the „vector network analyzer“:

1970: HP8542A (0.1 – 12.4 / 18 GHz)



[Hewlett Packard Journal, April 1971]

today: – 110 GHz (+/– 1 THz)



→ **more about RF measurements: „Radio Frequency Technology“ & Lab Course**

Content:

Repeating Maxwell‘s equations

Plane waves

Waveguides

Lumped components

Waves on transmission lines

Wave matrices

Transmission line circuits

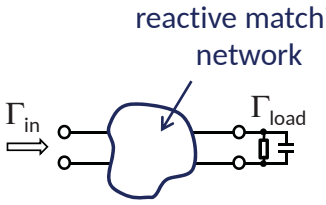
Impedance transformers and filters

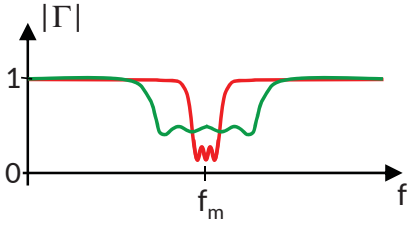
Multi-section impedance transformers and filters

Multi-section impedance transformers:

- the loss-less match of a (reactive) load, and the (loss-less) transformation of line impedances …

… always shows band-pass behaviour:



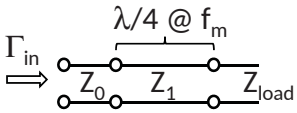


- the bandwidth of a quarter-wave (λ/4) line transformer depends on the „height“ of the impedance step and on the maximum allowed reflection

center frequency: $f_m = \frac{1}{2}(f_1 + f_2)$, bandwidth: $w = \frac{f_2 - f_1}{f_m}$

… bandwidth w = 0.66 is one octave (2:1). w = 1 means 3:1 bandwidth!

- example: λ/4 – transformer:



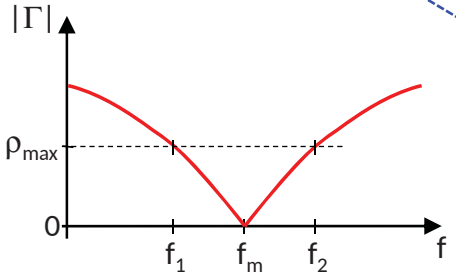
. impedance ratio: $R = \frac{Z_{load}}{Z_0}$

$VSWR = \frac{1+|\Gamma|}{1-|\Gamma|}$ $\quad |\Gamma| = \frac{VSWR-1}{VSWR+1}$

[Matthaei, Young, Jones, McGraw Hill, 1964, S. 267]

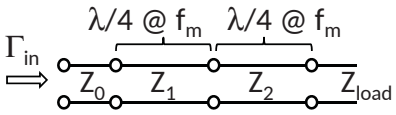
IMPEDANCE RATIO, R	BANDWIDTH, W					
	0.2	0.4	0.6	0.8	1.0	1.2
1.25	1.03	1.07	1.11	1.14	1.17	1.20
1.50	1.06	1.13	1.20	1.27	1.33	1.39
1.75	1.09	1.19	1.30	1.39	1.49	1.57
2.00	1.12	1.24	1.38	1.51	1.64	1.76
2.50	1.16	1.34	1.53	1.73	1.93	2.12
3.00	1.20	1.43	1.68	1.95	2.21	2.47
4.00	1.26	1.58	1.95	2.35	2.76	3.15
5.00	1.32	1.73	2.21	2.74	3.30	3.83
6.00	1.37	1.86	2.45	3.12	3.82	4.50
8.00	1.47	2.11	2.92	3.86	4.86	5.84
10.00	1.55	2.35	3.37	4.58	5.88	7.16
12.50	1.65	2.63	3.92	5.47	7.15	8.81
15.00	1.75	2.90	4.47	6.36	8.41	10.46
17.50	1.84	3.17	5.01	7.25	9.67	12.10
20.00	1.92	3.43	5.54	8.11	10.93	13.74

maximum VSWR



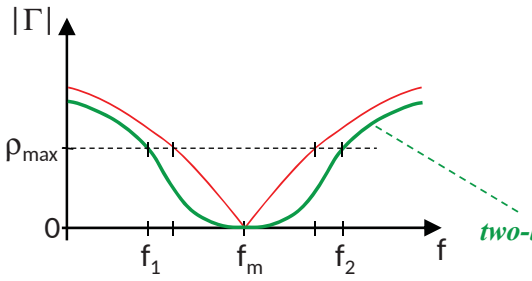
example: $Z_0 = 50\,\Omega$, $Z_{load} = 300\,\Omega$ → R = 6
$Z_1 = (50\,\Omega \times 300\,\Omega)^{1/2} = 122.5\,\Omega$
bandwidth w = 20%
→ $VSWR_{max} = 1.37$ ($\triangleq -16$ dB)

- reduction of the *worst case* reflection (or, bandwidth increase) by insertion of an additional (or, multiple) line section — two-line λ/4 transformer:



- different gradings (Z_0 –) Z_1 – Z_2 (– Z_{load}) are possible:

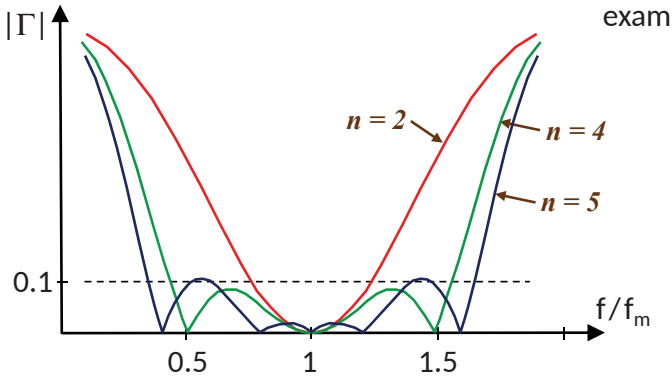
eg, geometric grading: $\ln\frac{Z_{i+1}}{Z_i} = \frac{\ln R}{n+1}$



example: $Z_0 = 50\Omega$, $Z_{load} = 300\Omega$ → R = 6
n = 2
$Z_1 = 50\Omega \times 1.817 = 90.9\Omega$
$Z_2 = 90.9\Omega \times 1.817 = 165.1\Omega$
$Z_{load} = 165.1\Omega \times 1.817 = 300\Omega$

two-line λ/4 trafo, geometric grading (qualitatively)

geometric distribution with n > 2 leads to non-constant *ripple*:



example: $Z_0 = 50\Omega$, $Z_{load} = 200\Omega$ → R = 4
n = 2,4,5

Пафнутий Львович Чебышёв,
1821 – 1894
mathematician

engl.: Chebyshev, Tchebyscheff

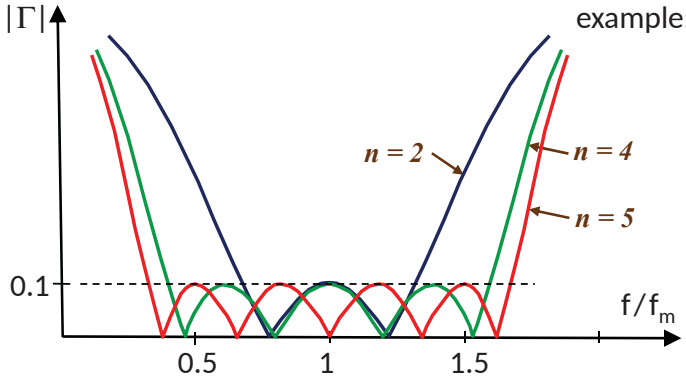
distribution with constant *ripple* → Tschebyscheff distribution:

n = 2:

$$\frac{Z_1}{Z_0} = \sqrt[4]{R}\,\sqrt[4]{\frac{1+|\Gamma|_{f_m}}{1-|\Gamma|_{f_m}}} \quad , \quad \frac{Z_1}{Z_0} = \frac{Z_{load}}{Z_2}$$

example:
$Z_0 = 50\Omega$, $Z_{load} = 300\Omega$ → R = 6
n = 2, $|\Gamma|_{fm} = 0.1$
$Z_1 = 50\Omega \times 1.646 = 82.3\Omega$
$Z_2 = 300\Omega / 1.646 = 182.3\Omega$

Tschebyscheff distribution leads to constant ripple:



- Tschebyscheff – polynoms of first order $T_n(x)$:

. solutions of the T. differential equation: $(1-x^2)y'' - xy' + n^2 y = 0$

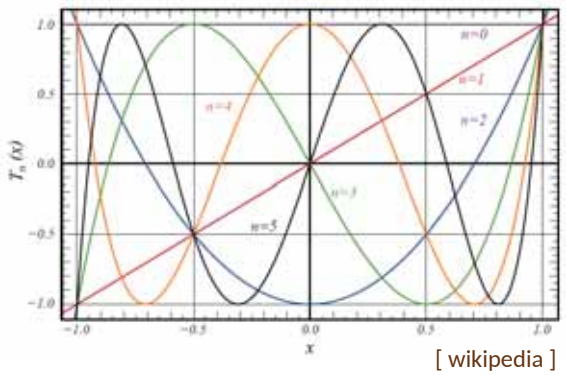
$$T_0(x) = 1 \qquad T_1(x) = x \qquad \Big\} \quad T_{n+1}(x) = 2\,x\,T_n(x) - T_{n-1}(x)$$

$$T_2(x) = 2\,x^2 - 1$$

$$T_3(x) = 4\,x^3 - 3\,x$$

$$T_4(x) = 8\,x^4 - 8\,x^2 + 1$$

$$T_5(x) = 16\,x^5 - 20\,x^3 + 5\,x$$



[wikipedia]

- synthesis — characteristics of the input reflection coefficient (magnitude):

„maximally flat characteristics“

$$|\Gamma_{in}|^2 = \frac{c_f^2 \cos^{2N}\left(\frac{\pi}{2}\frac{f}{f_m}\right)}{1 + c_f^2 \cos^{2N}\left(\frac{\pi}{2}\frac{f}{f_m}\right)}$$

„Tschebyscheff characteristics“

$$|\Gamma_{in}|^2 = \frac{c_t^2 \left[T_N\left(s \cos\frac{\pi}{2}\frac{f}{f_m}\right)\right]^2}{1 + c_t^2 \left[T_N\left(s \cos\frac{\pi}{2}\frac{f}{f_m}\right)\right]^2}$$

$$c_f^2 = \frac{\Gamma_0^2}{1-\Gamma_0^2} \quad \Gamma_0 = \frac{R-1}{R+1} \quad R = \frac{Z_{load}}{Z_0} \quad c_t^2 = \frac{\Gamma_{max}^2}{1-\Gamma_{max}^2} \quad s = \cosh\left(\frac{1}{N}\,\text{ar cosh}\left(\frac{R-1}{\sqrt{R}}\,\frac{\sqrt{1-\Gamma_{max}^2}}{2\,\Gamma_{max}}\right)\right)$$

- practical design:

„maximally flat characteristics“

R impedance ratio

n number of sections

Z_i ... from the tables

„Tschebyscheff characteristics“

w bandwidth

R impedance ratio

n number of sections

max VSWR

Z_i

} from the tables

... tabulated values are normalized to $Z_0 = 1$

... the impedance distribution in multi-section transformers is symmetric:

... n=2 : $Z_1 = Z_{1,\text{Tabelle}} \cdot Z_0 \quad , \quad Z_2 = R\, Z_0 / Z_{1,\text{Tabelle}}$

... n=3 : $Z_1 = Z_{1,\text{Tabelle}} \cdot Z_0 \quad , \quad Z_2 = Z_0 \cdot \sqrt{R} \quad , \quad Z_3 = R\, Z_0 / Z_{1,\text{Tabelle}}$

... n=4 : $Z_1 = Z_{1,\text{Tabelle}} \cdot Z_0$, $Z_2 = Z_{2,\text{Tabelle}} \cdot Z_0$, $Z_3 = R\, Z_0 / Z_{2,\text{Tabelle}}$, $Z_4 = R\, Z_0 / Z_{1,\text{Tabelle}}$

„Tschebyscheff characteristics“

[L. Young, „Tables for cascaded homogeneous quarter-wave transformers, IRE Transactions MTT, 1959, S. 233-237; correction: IRE Trans. MTT, 1960, S. 243-244]

MAXIMUM VSWR FOR 2-SECTION QUARTER-WAVE TRANSFORMERS

Impedance Ratio, R	Bandwidth, W								
	0.10	0.20	0.30	0.40	0.50	0.60	0.80	1.00	1.20
1.25	1.00	1.00	1.01	1.01	1.02	1.03	1.05	1.08	1.11
1.50	1.00	1.01	1.01	1.02	1.03	1.05	1.09	1.15	1.22
1.75	1.00	1.01	1.02	1.03	1.05	1.07	1.13	1.21	1.32
2.00	1.00	1.01	1.02	1.04	1.06	1.08	1.16	1.27	1.41
2.50	1.00	1.01	1.03	1.05	1.08	1.12	1.22	1.37	1.58
3.00	1.00	1.01	1.03	1.06	1.10	1.14	1.27	1.47	1.74
4.00	1.00	1.02	1.04	1.08	1.13	1.19	1.37	1.64	2.04
5.00	1.01	1.02	1.05	1.09	1.15	1.23	1.45	1.80	2.33
6.00	1.01	1.03	1.06	1.11	1.17	1.26	1.53	1.95	2.60
8.00	1.01	1.03	1.07	1.13	1.22	1.33	1.67	2.23	3.13
10.00	1.01	1.04	1.08	1.15	1.25	1.38	1.80	2.50	3.64

MAXIMUM VSWR FOR 4-SECTION QUARTER-WAVE TRANSFORMERS

Impedance Ratio, R	Bandwidth, W								
	0.10	0.20	0.30	0.40	0.50	0.60	0.80	1.00	1.20
1.25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.03
1.50	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.02	1.06
1.75	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.03	1.08
2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.04	1.10
2.50	1.00	1.00	1.00	1.00	1.00	1.01	1.02	1.06	1.14
3.00	1.00	1.00	1.00	1.00	1.00	1.01	1.03	1.07	1.17
4.00	1.00	1.00	1.00	1.00	1.00	1.01	1.03	1.09	1.22
5.00	1.00	1.00	1.00	1.00	1.01	1.01	1.04	1.11	1.27
6.00	1.00	1.00	1.00	1.00	1.01	1.01	1.05	1.13	1.31
8.00	1.00	1.00	1.00	1.00	1.01	1.02	1.06	1.16	1.39
10.00	1.00	1.00	1.00	1.00	1.01	1.02	1.07	1.18	1.46

... these values for example on slide 8.10

„Tschebyscheff characteristics“

[L. Young, „Tables for cascaded homogeneous quarter-wave transformers, IRE Transactions MTT, 1959, S. 233-237; correction: IRE Trans. MTT, 1960, S. 243-244]

w = 10%

R	n=2	n=3	n=4	
	Z_1	Z_1	Z_1	Z_2
1.00	1.00000	1.00000	1.00000	1.00000
1.25	1.05755	1.02843	1.01414	1.07232
1.50	1.10703	1.05227	1.02586	1.13530
1.75	1.15067	1.07290	1.03591	1.19146
2.00	1.18986	1.09113	1.04473	1.24239
2.50	1.25835	1.12238	1.05972	1.33252
3.00	1.31725	1.14868	1.07223	1.41113
4.00	1.41585	1.19171	1.09250	1.54503
5.00	1.49741	1.22645	1.10873	1.65794
6.00	1.56755	1.25578	1.12235	1.75657
8.00	1.68501	1.30391	1.14455	1.92490
10.00	1.78219	1.34289	1.16242	2.06710

w = 30%

R	n=2	n=3	n=4	
	Z_1	Z_1	Z_1	Z_2
1.00	1.00000	1.00000	1.00000	1.00000
1.25	1.05903	1.02952	1.01486	1.07306
1.50	1.10985	1.05431	1.02719	1.13673
1.75	1.15474	1.07577	1.03777	1.19354
2.00	1.19511	1.09476	1.04706	1.24508
2.50	1.26581	1.12736	1.06287	1.33636
3.00	1.32676	1.15486	1.07607	1.41603
4.00	1.42915	1.19992	1.09752	1.55190
5.00	1.51420	1.23642	1.11472	1.66660
6.00	1.58761	1.26731	1.12917	1.76689
8.00	1.71119	1.31815	1.15279	1.93828
10.00	1.81407	1.35949	1.17184	2.08328

w = 50%

R	n=2	n=3	n=4	
	Z_1	Z_1	Z_1	Z_2
1.00	1.00000	1.00000	1.00000	1.00000
1.25	1.06205	1.03184	1.01643	1.07456
1.50	1.11564	1.05865	1.03009	1.13961
1.75	1.16311	1.08192	1.04182	1.19773
2.00	1.20593	1.10254	1.05214	1.25052
2.50	1.28121	1.13805	1.06973	1.34412
3.00	1.34643	1.16811	1.08448	1.42596
4.00	1.45672	1.21763	1.10849	1.56581
5.00	1.54908	1.25796	1.12783	1.68414
6.00	1.62941	1.29227	1.14415	1.78783
8.00	1.76591	1.34913	1.17092	1.96543
10.00	1.88089	1.39572	1.19264	2.11613

w = 120%

R	n=2	n=3	n=4	
	Z_1	Z_1	Z_1	Z_2
1.00	1.00000	1.00000	1.00000	1.00000
1.25	1.08650	1.05636	1.03560	1.08683
1.50	1.16292	1.10495	1.06576	1.16342
1.75	1.23199	1.14805	1.09214	1.23248
2.00	1.29545	1.18702	1.11571	1.29572
2.50	1.40979	1.25594	1.15681	1.40907
3.00	1.51179	1.31621	1.19218	1.50943
4.00	1.69074	1.41972	1.25182	1.68360
5.00	1.84701	1.50824	1.30184	1.83358
6.00	1.98768	1.58676	1.34555	1.96694
8.00	2.23693	1.72383	1.42054	2.19954
10.00	2.45663	1.84304	1.48458	2.40096

… these values for the example on slide 8.10

„maximally flat characteristics“

[L. Young, „Tables for cascaded homogeneous quarter-wave transformers, IRE Transactions MTT, 1959, S. 233-237; correction: IRE Trans. MTT, 1960, S. 243-244]

R	n=2	n=3	n=4	
	Z_1	Z_1	Z_1	Z_2
1.00	1.00000	1.00000	1.00000	1.00000
1.25	1.05737	1.02829	1.01405	1.07223
1.50	1.10668	1.05202	1.02570	1.13512
1.75	1.15016	1.07255	1.03568	1.19120
2.00	1.18921	1.09068	1.04444	1.24206
2.50	1.25743	1.12177	1.05933	1.33204
3.00	1.31607	1.14793	1.07176	1.41051
4.00	1.41421	1.19071	1.09190	1.54417
5.00	1.49535	1.22524	1.10801	1.65686
6.00	1.56508	1.25439	1.12153	1.75529
8.00	1.68179	1.30219	1.14356	1.92323
10.00	1.77828	1.34089	1.16129	2.06509

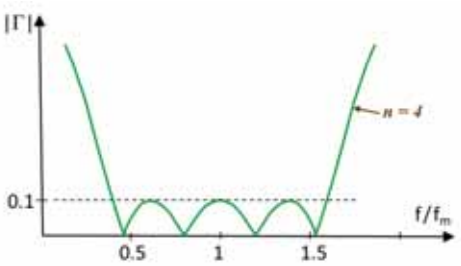
- example: Tschebyscheff, n = 4 , $Z_0 = 50\Omega$, $Z_{load} = 200\Omega$, $|\Gamma|_{max} = -20dB$

→ R = 4, VSWR = 1.22 $VSWR = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+0.1}{1-0.1} = 1.22$, $|\Gamma| = -0.1 = -20\,dB$

→ w = 1.20 (slide **8.8**) ($f_{min} = 0.4$, $f_{max} = 1.6$)

→ $Z_{1,table} = 1.252$, $Z_{2,table} = 1.684$ (slide **8.9**)

→ $Z_1 = 62.6\Omega$, $Z_2 = 84.2\Omega$,
$Z_3 = 118.8\Omega$, $Z_4 = 159.7\Omega$

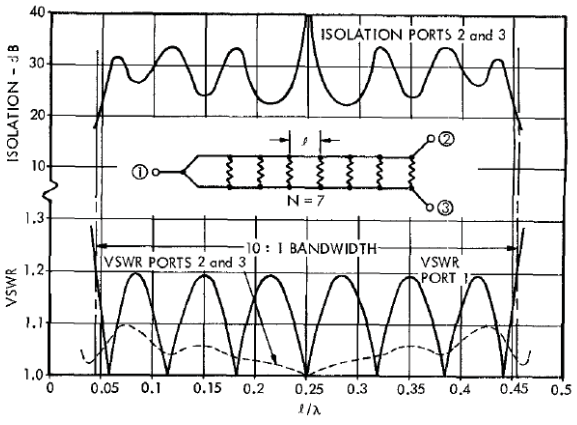


- applications:

. broadband power divider:

standard-Wilkinson (1 section):
bandwidth 1.4:1 at VSWR<1.2

7-section-divider:
bandwidth 10:1 at VSWR<1.2

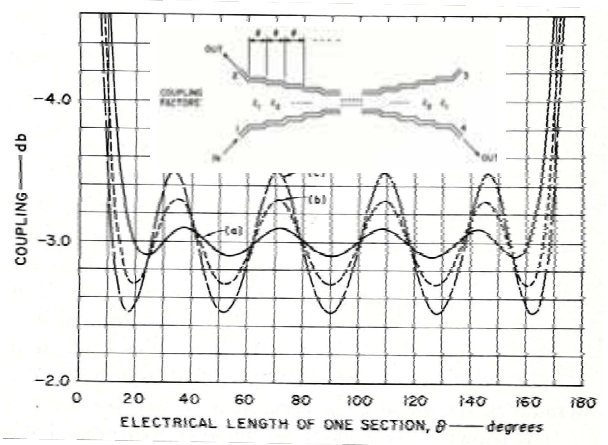


[S.B. Cohn, „A class of broadband three-port TEM-mode hybrids," IEEE Transactions MTT, 1968, S. 110-116]

. broadband coupler:

9-section-coupler with
bandwidth 11:1

3 different 9-section-couplers: smaller ripple corresponds to smaller bandwidth and vice versa

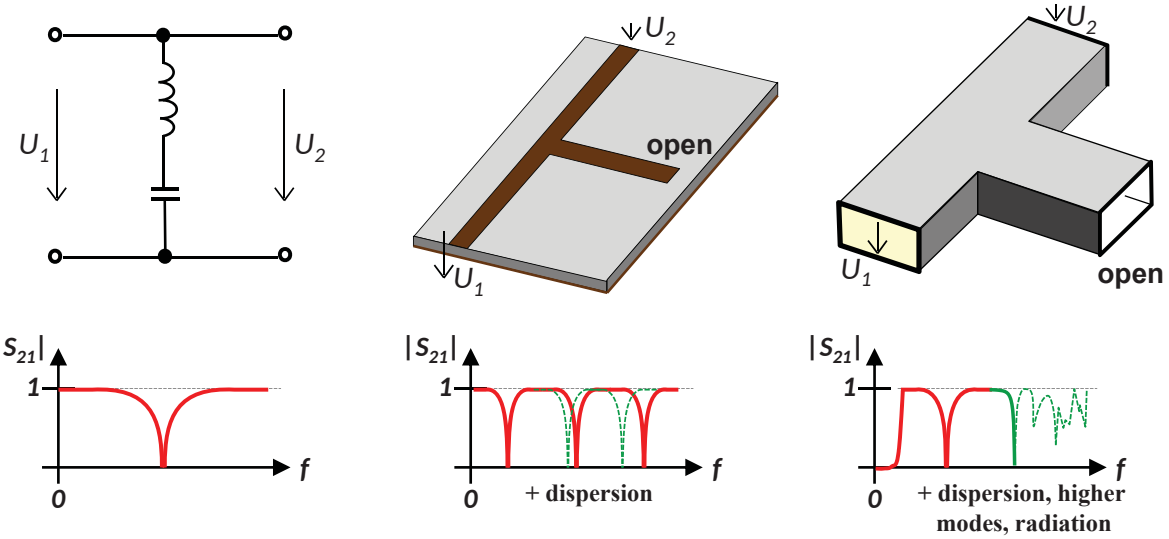


[E.G.Cristal, L. Young, „Theory and tables of optimum symmetric TEM-mode coupled-transmission-line directional couplers," IEEE Transactions MTT, 1965, S. 544-558]

Radio frequency / microwave filters:

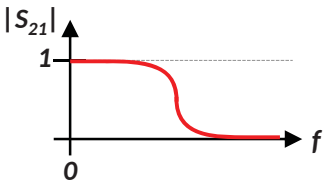
- RF filters are typically based on „distributed" (not: lumped) components

i.e.: dimensions not small compared to wavelength

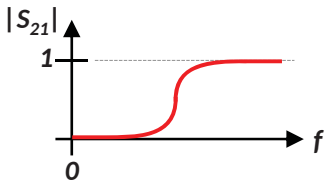


Categories :

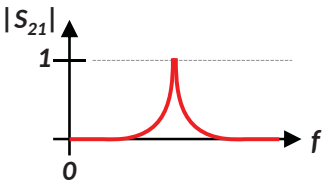
low-pass



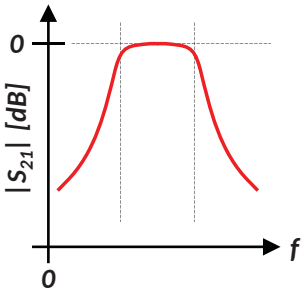
high-pass



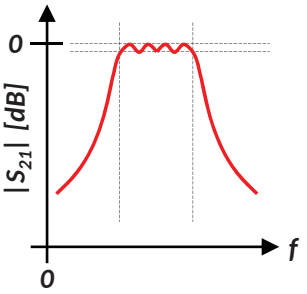
band-pass



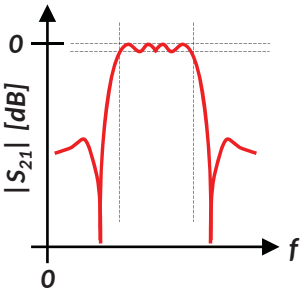
flat / Butterworth



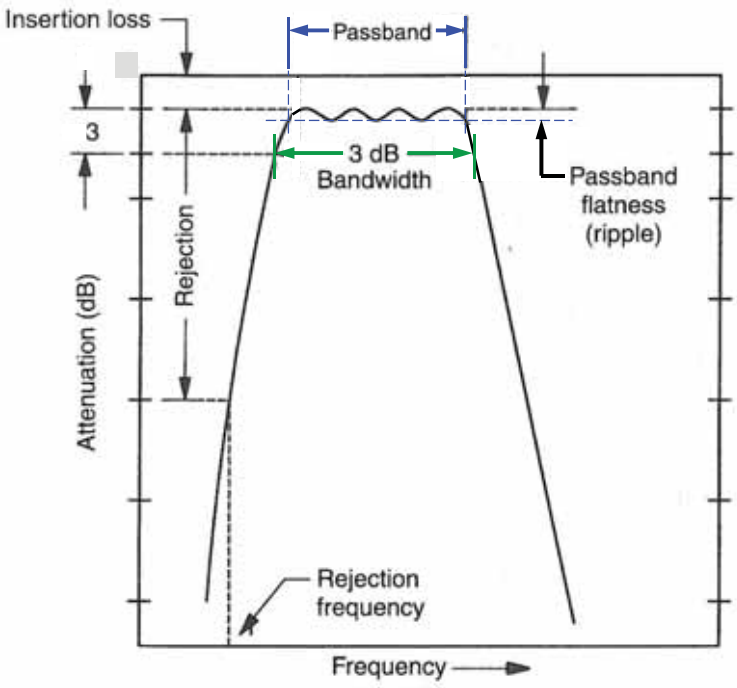
equal ripple / Tschebyscheff



with attenuation pole / elliptic / Cauer

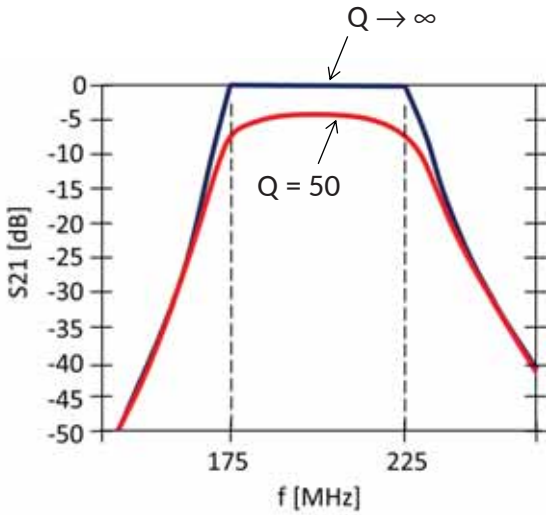


Terms :



Quality factor / Q-factor :

- dissipative loss („heat“) in the resonators increases the insertion loss in the passband and limits the maximum insertion loss at the attenuation poles (filter slopes are not much affected by loss)



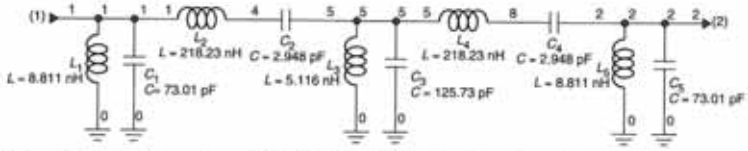
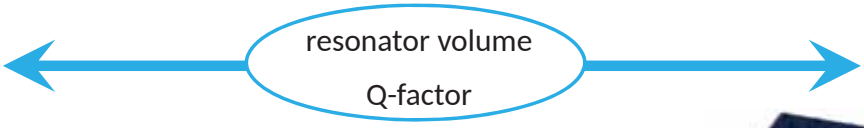


Figure 9.10-2 Five-section 0.1-dB-ripple Chebyshev bandpass filter for 175 to 225 MHz.

[J.F. White, *High frequency techniques*, IEEE Press / Wiley, 2004, S. 365]

- resonator loss / Q-factor is determined by:
  - . current density on lines (design, frequency)
  - . metal electric conductivity (material, temperature)
  - . displacement current density (design, frequency)
  - . dielectric loss, tanδ (frequency, temperature)

high Q-factor requires large resonator volume (see next slide)





planar microstrip filters



ceramic block filters



metal coax cavity filters

- exceptions: superconducting, optical, acoustic-wave filters

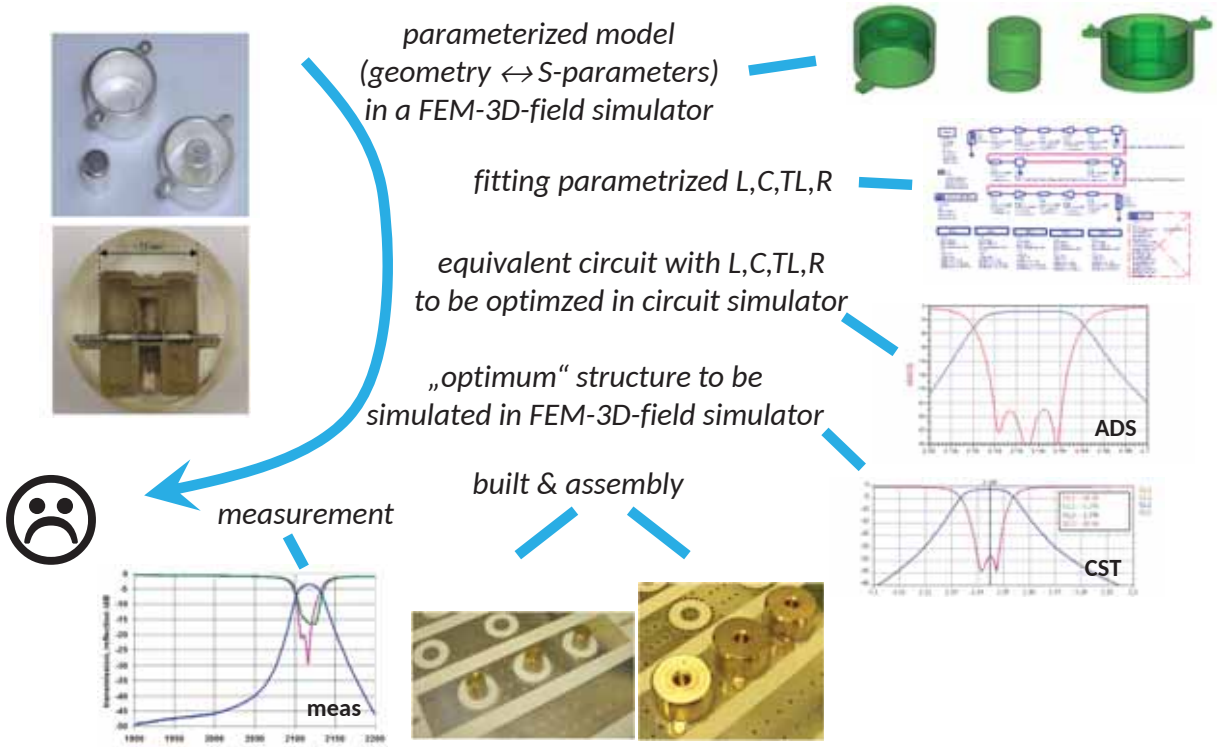


[mini-circuits]

Designing filters by computer optimization:

- a filter based on L,C,R is calculated very fast
- optimization (*random*, *gradient*) often gives good results fast for filters of low order
- higher-order filters and filters with attenuation poles require a good starting point for optimization
- a „good starting point“ can come from (simplified, e.g., neglecting loss) classical filter synthesis
- realizing L,C by „microwave means“ (transmission lines, coupled lines, 3D structures and resonators) requires long calculation time (3D field simulation) for a single analysis – optimizers will become much too slow. Then, a parameterized equivalent circuit of the 3D structure, using L,C,R,TL, with geometrical parameters linked to electrical parameters, must be found. This parameterized equivalent circuit can be optimized very fast.
- the higher the frequency, the less accurate are models based on L,C,R,TL

- example: 3D coax resonator for PCB mount; f ~ 2.1 GHz



Filter synthesis (one possibility of many):

1) synthesis of a normalized lowpass (loss-less) … can be built with lumped L,C

2) transformation to a normalized highpass or bandpass … can be built with lumped L,C

3) transformation to a transmission-line circuit „Richards transformation“

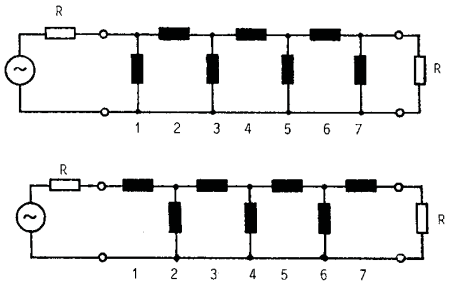
4) transformation to a circuit with separated reactances „Kuroda equivalence“ … can be etched as planar circuit

1) low-pass ladder network

numbering from source to load;

lowpass → series-L, shunt-C

2 variants



normalization

system impedance $Z_0 \rightarrow R = 1\Omega$

edge frequency (@ 3 dB) $\omega_c \rightarrow \Omega = 1$ rad/s

n = 7 … number of reactances

… here ω_c@-3dB; but other definitions possible, e.g., ω_c@IL="ripple" for Tschebyscheff filters

tabulated are values of g_i (i = 1…n), e.g.:

g Values for Butterworth Low-Pass Prototype Filter

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
2	1.4142	1.4142						
3	1.0000	2.0000	1.0000					
4	0.7654	1.8478	1.8478	0.7654				
5	0.6180	1.6180	2.0000	1.6180	0.6180			
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176		
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	
8	0.3902	1.1111	1.6629	1.9616	1.9616	1.6629	1.1111	0.3902

… these values for the example on slide 8.23

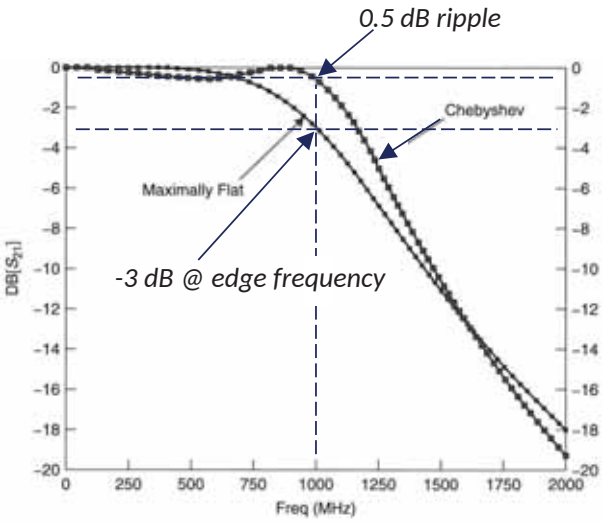
Tschebyscheff:

additional parameter: *ripple*

makes sense: tabulated values define bandwidth at ripple-insertion-loss (not at -3dB!)

bei even N the load impedance is not one!

… therefore, typically for Tschebyscheff only odd N



0.1 dB ripple / S11,max = -16.4 dB

N	g1	g2	g3	g4	g5	g6	g7
3	1,0316	1,1474	1,0316				
5	1,1468	1,3712	1,9750	1,3712	1,1468		
7	1,1812	1,4228	2,0967	1,5734	2,0967	1,4228	1,1812

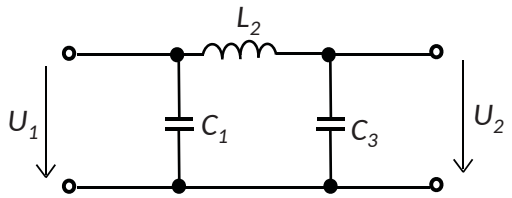
… these values for the example slide 8.31

re-normalization:

$$C = \frac{1}{Z_0}\,\frac{g_i}{\omega_c}$$

$$L = Z_0\,\frac{g_i}{\omega_c}$$

example:



Butterworth / maximal flat, N = 3
system impedance Z_0 50Ω
edge frequency / 3dB-insertion at 1 GHz

→ $g_1 = 1$, $g_2 = 2$, $g_3 = 1$ (s. slide 8.21)

$$C_1 = C_3 = \frac{1}{Z_0}\,\frac{g_i}{\omega_c} = \frac{1}{50\,\Omega}\,\frac{1}{2\pi \cdot 1\,\text{GHz}} = 3.18\,\text{pF}$$

$$L_2 = Z_0\,\frac{g_i}{\omega_c} = 50\,\Omega \cdot \frac{2}{2\pi \cdot 1\,\text{GHz}} = 15.92\,\text{nH}$$

2) high-pass

- starting with the normalized lowpass,

  exchange C ↔ L

  invert $g_i := 1/g_i$

  re-normalization as with the lowpass

3) band-pass

- starting with the normalized lowpass,

  definitions: f_1 … lower 3dB band limit

  f_2 … upper 3dB band limit

  $f_0 = \sqrt{f_1 f_2}$ … band mid frequency (geometric mean)

  $b_w = (f_2 - f_1)/f_0$ … fractional bandwidth

substitution:

series-L of normalized lowpass } replaced by { *series* -LC- resonator

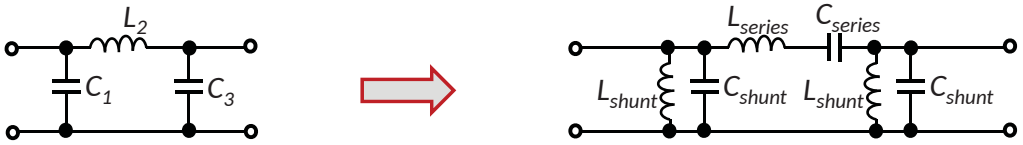
$$L_{\text{series}}^{\text{norm}} = \frac{g_2}{b_w}$$

$$C_{\text{series}}^{\text{norm}} = \frac{1}{L_{\text{series}}^{\text{norm}}}$$

shunt-C of normalized lowpass } replaced by { *parallel* -LC- resonator

$$L_{\text{shunt}}^{\text{norm}} = \frac{1}{C_{\text{shunt}}^{\text{norm}}}$$

$$C_{\text{shunt}}^{\text{norm}} = \frac{g_{1,3}}{b_w}$$



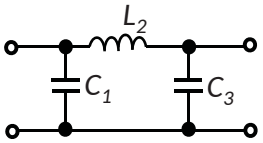
re-normalization:

using band mid frequency f_0 : $L_i = Z_0 \frac{L_i^{\text{norm}}}{2\pi f_0}$, $C_i = \frac{1}{Z_0} \frac{C_i^{\text{norm}}}{2\pi f_0}$

Example: low-pass: Butterworth / maximally flat,
N = 3, Z_0 = 50Ω, fc3dB = 1 GHz

→ $g_1 = 1$, $g_2 = 2$, $g_3 = 1$

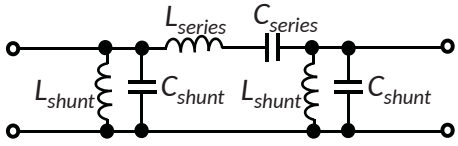
→ $C_1 = C_3 = 3.2$ pF ; $L_2 = 15.9$ nH



band-pass: 3dB bandwidth 0.8 GHz – 1.2 GHz,
f_0 = 0.980 GHz, bw = 0.408

geometric mean

normalized:



→ $L_{\text{series}}^{\text{norm}} = \frac{g_2}{b_w} = \frac{2}{0.408} = 4.90$

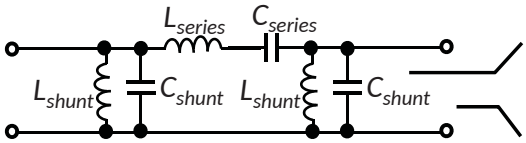
$C_{\text{series}}^{\text{norm}} = \frac{1}{L_{\text{series}}^{\text{norm}}} = 0.204$

$C_{\text{shunt}}^{\text{norm}} = \frac{g_{1,3}}{b_w} = \frac{1}{0.408} = 2.45$

$L_{\text{shunt}}^{\text{norm}} = \frac{1}{C_{\text{shunt}}^{\text{norm}}} = 0.408$

re-normalization: ...

Example: re-normalization: $f_0 = 0.980$ GHz



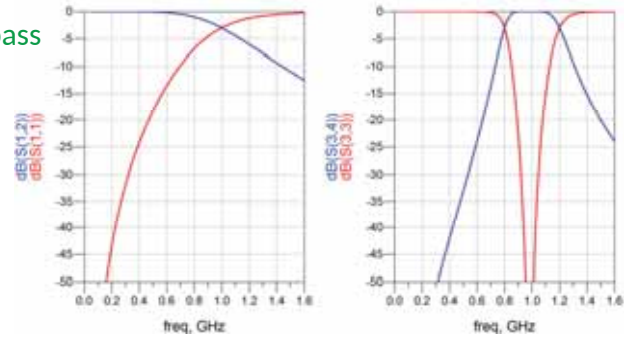
$$L_{series} = Z_0 \frac{L_{series}^{norm}}{2\pi f_0} = \frac{50\,\Omega \cdot 4.90}{2\pi \cdot 0.980\,\text{GHz}} = 39.8\,\text{nH}$$

$$C_{series} = \frac{1}{Z_0} \frac{C_{series}^{norm}}{2\pi f_0} = \frac{0.204}{50\,\Omega \cdot 2\pi \cdot 0.980\,\text{GHz}} = 0.66\,\text{pF}$$

$$L_{shunt} = Z_0 \frac{L_{shunt}^{norm}}{2\pi f_0} = \frac{50\,\Omega \cdot 0.408}{2\pi \cdot 0.980\,\text{GHz}} = 3.31\,\text{nH}$$

$$C_{shunt} = \frac{1}{Z_0} \frac{C_{shunt}^{norm}}{2\pi f_0} = \frac{2.45}{50\,\Omega \cdot 2\pi \cdot 0.980\,\text{GHz}} = 7.96\,\text{pF}$$

low-pass band-pass



ADS simulation

4) Richards-Transformation

... an inductance L will be replaced by a transmission line (length less than λ/4) with short-circuited end. Two degrees of freedom: line length and line impedance.

- transmission-line with shorted end:

$$Z_{sc}(z = -d) = jZ_0 \tan\frac{2\pi d}{\lambda}$$

- length 90 degrees at frequency f_R :

$$Z_{\text{shorted_line}}(f) = jZ_0 \tan\left(\frac{\pi}{2}\frac{f}{f_R}\right)$$

- mapping f $\leftrightarrow$ Ω (additional degree of freedom is f_R (where the line length is 90°))

$$\Omega = \tan\left(\frac{\pi}{2}\frac{f}{f_R}\right)$$

- equivalent reactance of inductance and line with shorted end:

$$j\,\omega\,L \equiv j\,\Omega Z_0$$

- similarly, a line with open end can replace a capacitor:

$$Z_{oc}(z = -d) = \frac{-j\,Z_0}{\tan\frac{2\pi d}{\lambda}}$$

$$Y_{\text{open_line}}(f) = jY_0 \tan\left(\frac{\pi}{2}\frac{f}{f_R}\right) = j\Omega Y_0 \equiv j\omega C$$

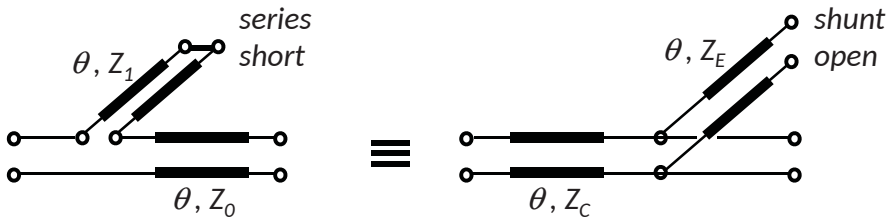
5) Kuroda identities

... if the actual circuit technology can not realize the specific reactances (lines), then the Kuroda identity changes to an identical circuit of different topology, which may be realizable

$$Z_C = n\,Z_0$$

$$Z_E = \frac{n}{n-1} Z_0$$

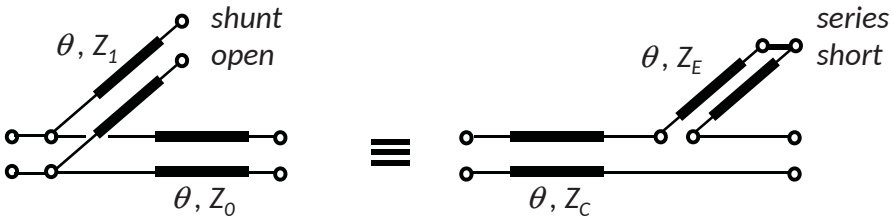
$$n = 1 + \frac{Z_1}{Z_0}$$



$$Z_C = \frac{Z_0}{n}$$

$$Z_E = \frac{n-1}{n} Z_0$$

$$n = 1 + \frac{Z_0}{Z_1}$$



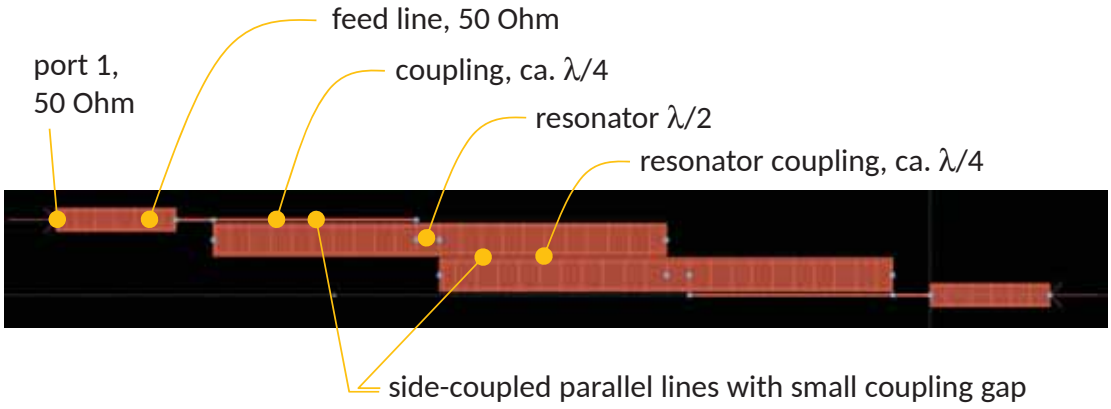
... above are Kuroda identities of first order; there exist also those of second order

Band-pass filter: half-wavelength resonator-filter

The ***direct-coupled stub filters*** (shown before), when used as band-pass, are preferred for medium-to-large passband bandwidth.

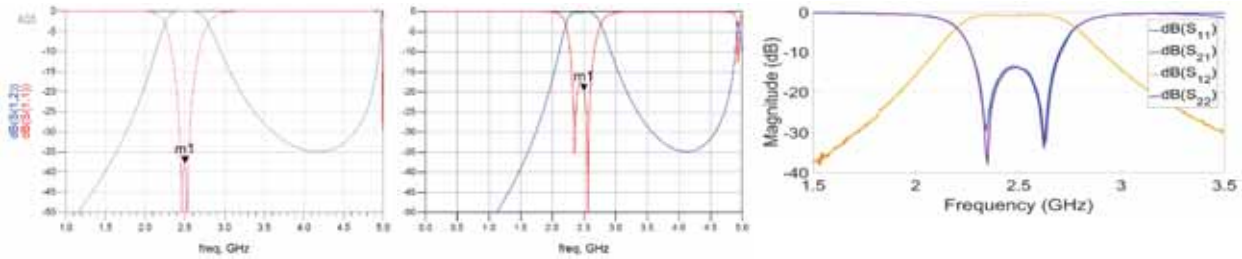
Band-pass filters for narrow-to-medium bandwidth can be realized preferably as ***coupled-resonator filter***.

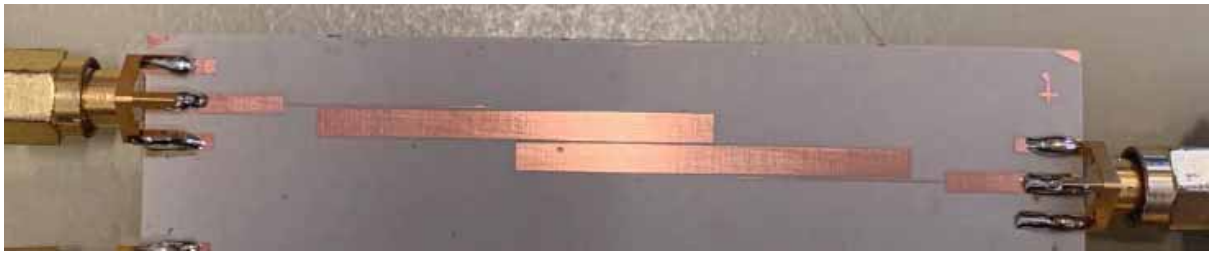
Example of a two-resonator- ***coupled-resonator filter*** (microstrip layout):



Example of a two-resonator- ***coupled-resonator filter*** (microstrip layout):

circuit simulation *layout simulation* *measurement*





[Bretzke, Satheeskanth, Lab Course 2024]

Design of planar direct-coupled stub filters – examples:

1) low-pass Tschebyscheff 0.1dB ripple, N=3, Z_0=50Ω, f_c=1GHz
   … realisation in *microstrip* with 3 open stub lines

2) discussion of a shorted stub line in *microstrip*

3) low-pass Butterworth, N=5, Z_0=50Ω, f_c=1.4GHz

4) bandwidth increase by *stub line matching*

A look into an old measurement device reveals: filters everywhere, including *stub filters*:

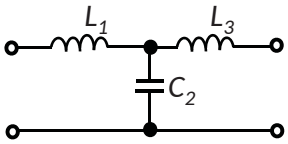


- example 1:

„freely chosen" as there is an additional degree of freedom

N=3 Tschebyscheff 0.1dB ripple, Z_0=50, fc = 1 GHz, f_R = 1.4 GHz

→ g1=g3=1.0316, g2=1.1474 (slide 8.22) → L1=L3=8.21nH, C2=3.65pF



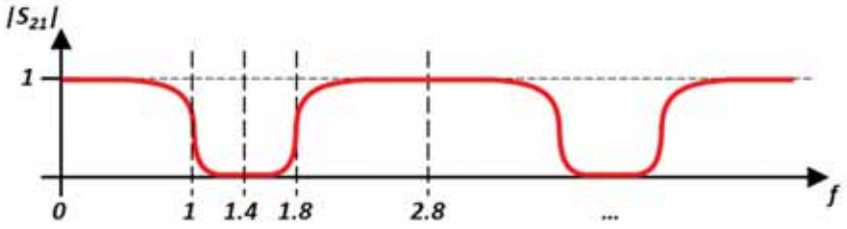
$$C = \frac{1}{Z_0}\frac{g_i}{\omega_c} \qquad L = Z_0\frac{g_i}{\omega_c}$$

$$g_2 Y_0 = \omega_c C \equiv \Omega_c\, Y_{0,\text{open_line}} \qquad g_1 Z_0 = \omega_c L \equiv \Omega_c\, Z_{0,\text{shorted_line}}$$

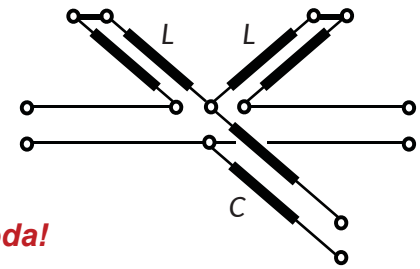
$$\Omega_c = \tan\left(\frac{\pi}{2}\frac{f = 1GHz}{f_R = 1.4GHz}\right) = 2.076$$

$$Y_{0,\text{open_line}} = \frac{g_2 Y_0}{\Omega_c} = (90.5\,\Omega)^{-1} \qquad Z_{0,\text{shorted_line}} = \frac{g_1 Z_0}{\Omega_c} = 24.8\,\Omega$$

→ the choice of f_R determines the frequency-periodicity of the filter:

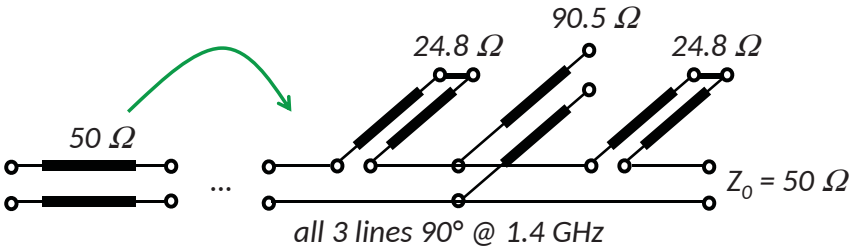


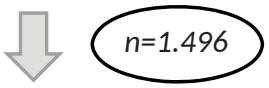
→ Can this be realized as a transmission line circuit?

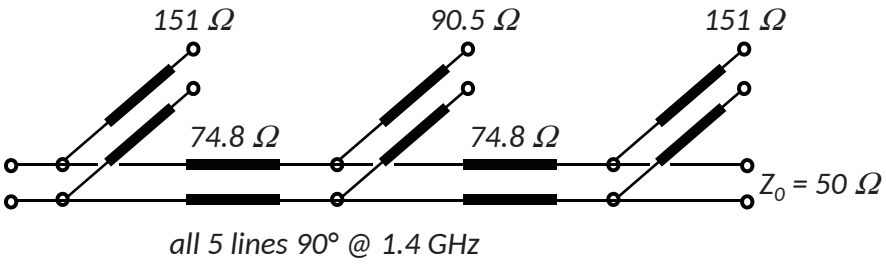


… not in microstrip! therefore: Kuroda!

example 1, cont´d:

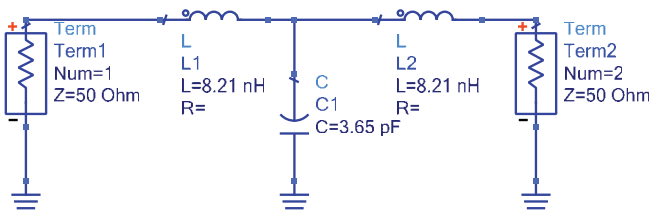


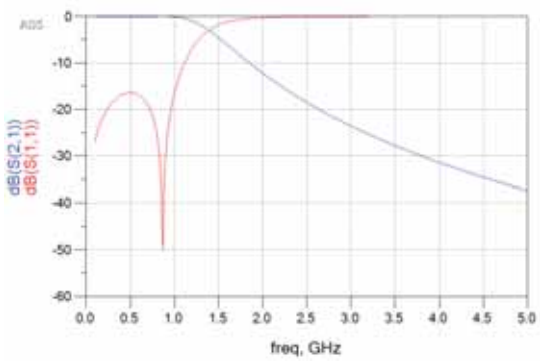


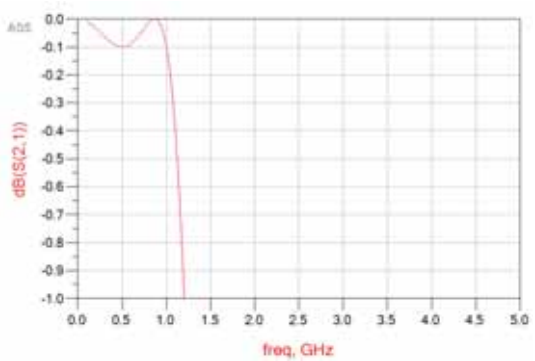


example 1, cont´d:

1) LC lowpass simulation (values of components: see slide 8.31):

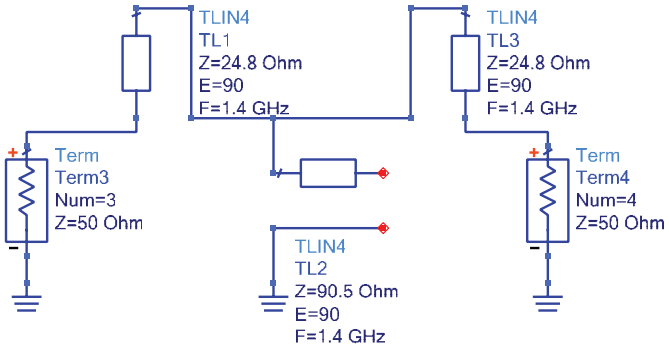


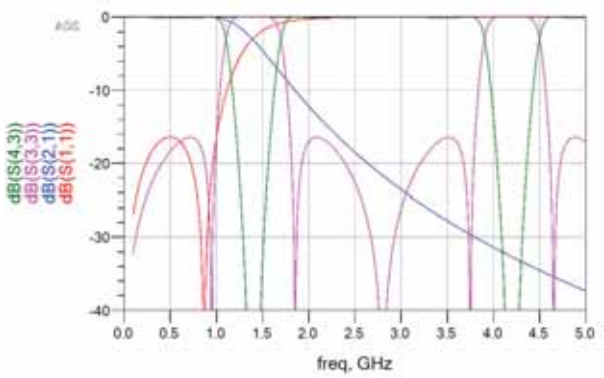


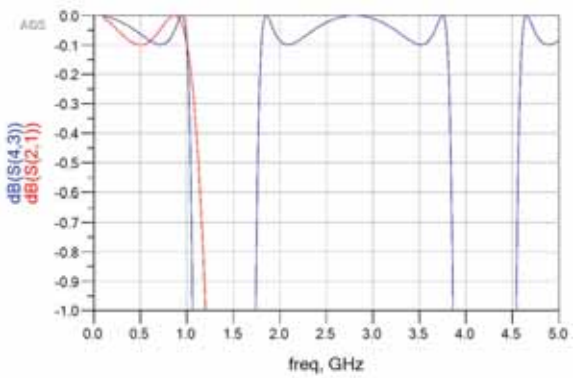


example 1, cont´d:

2) substitution of reactances by ideal transmission lines (see slide 8.31):

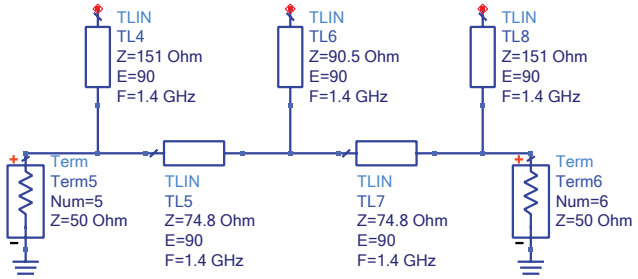


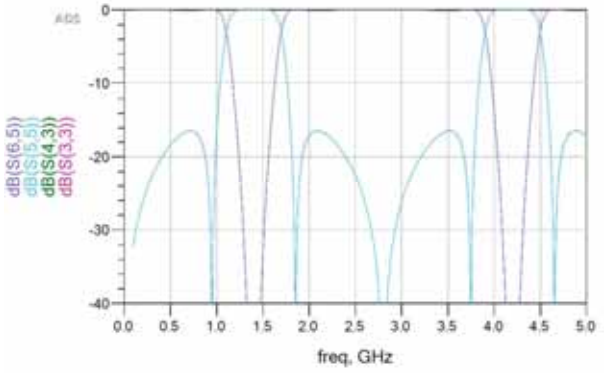


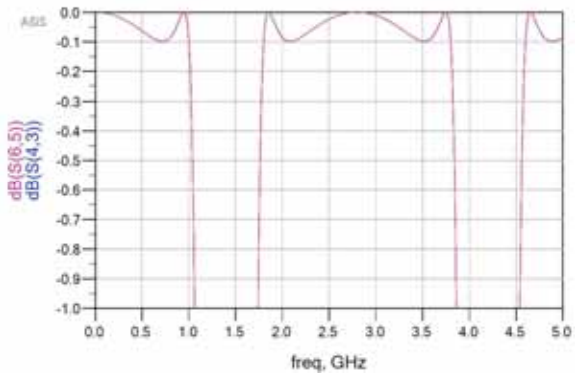


example 1, cont´d:

3) no change after applying Kuroda identity (see slide 8.33):



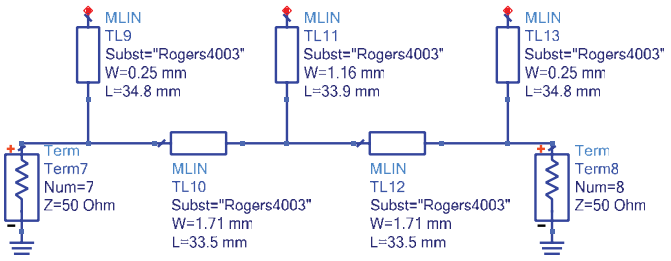


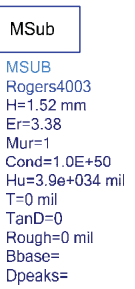


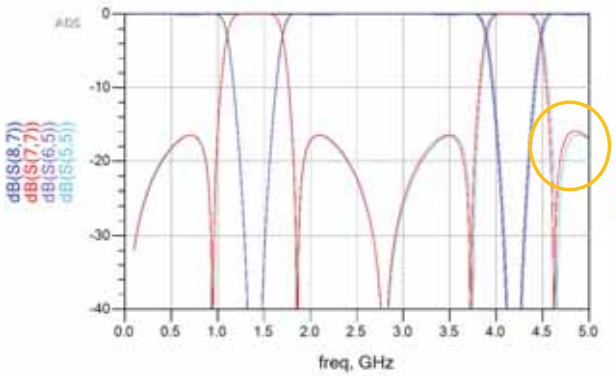
example 1, cont´d:

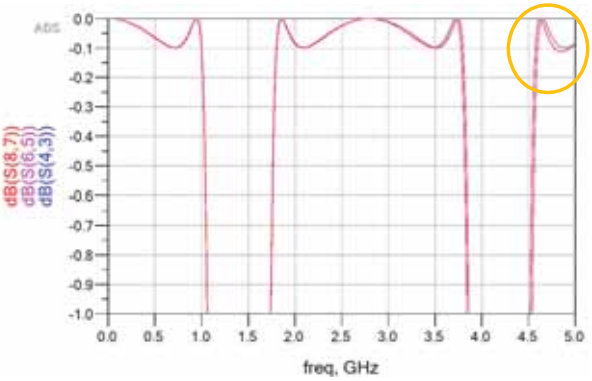
4) little change by substitution of ideal lines by microstrip lines (only some dispersion at higher frequency):

50 Ohm: W=3.52mm , 90 Grad @ 1.4 GHz : L = 32.6 mm
74.8 Ohm: W=1.71mm , 90 Grad @ 1.4 GHz : L = 33.5 mm
90.5 Ohm: W=1.16mm , 90 Grad @ 1.4 GHz : L = 33.9 mm
151 Ohm: W=0.25mm , 90 Grad @ 1.4 GHz : L = 34.8 mm



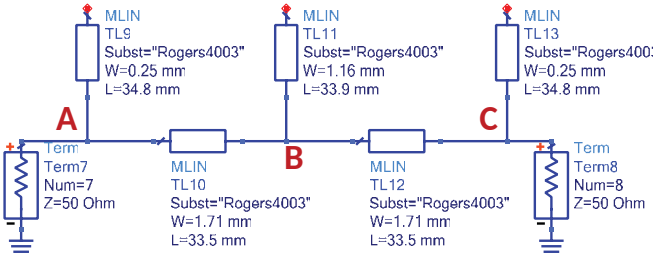


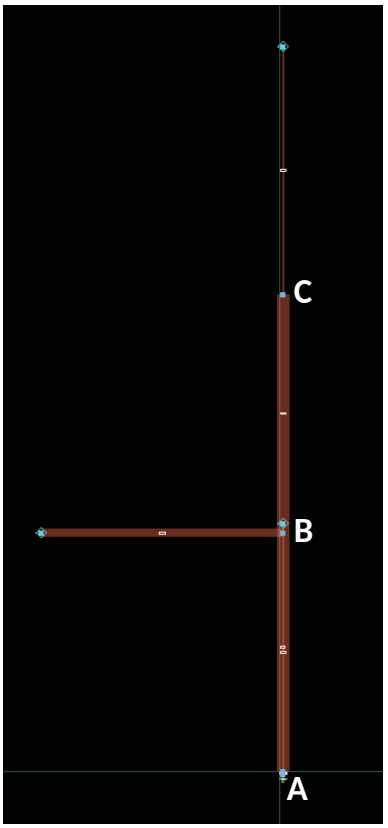




example 1, cont´d:

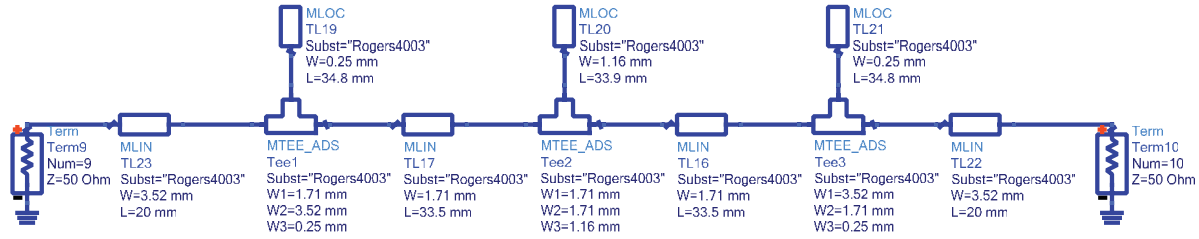
5) but: „real“ layout impossible!

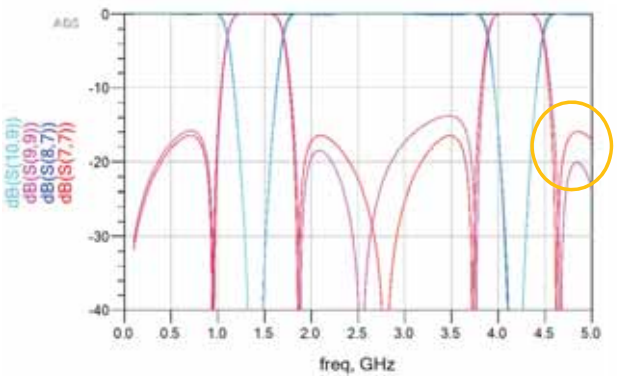


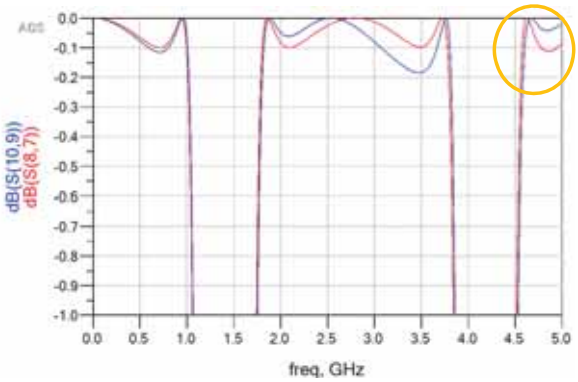


example 1, cont´d:

6) realistic microstrip circuit with elements „OPEN“ and „TEE“:

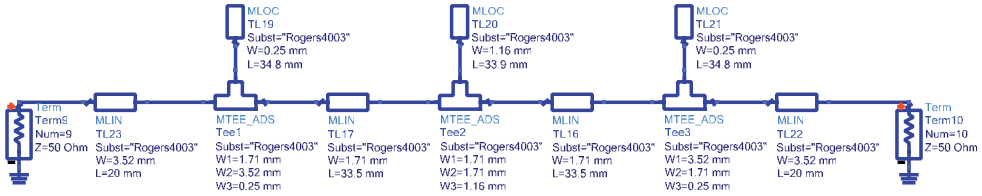


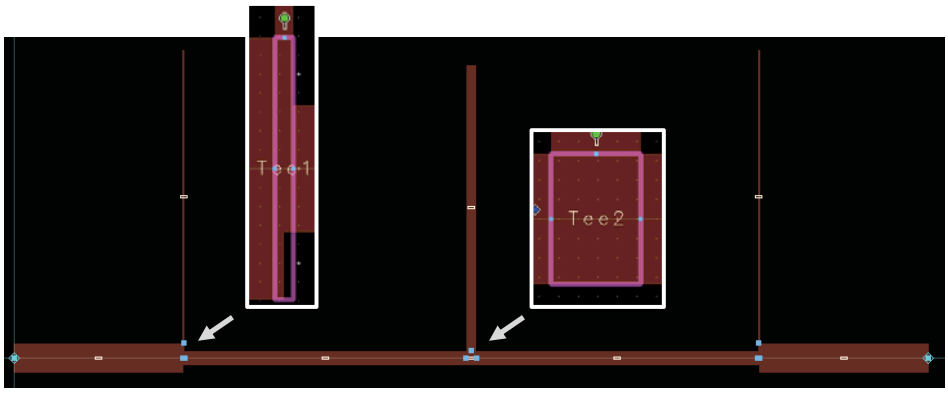




example 1, cont´d:

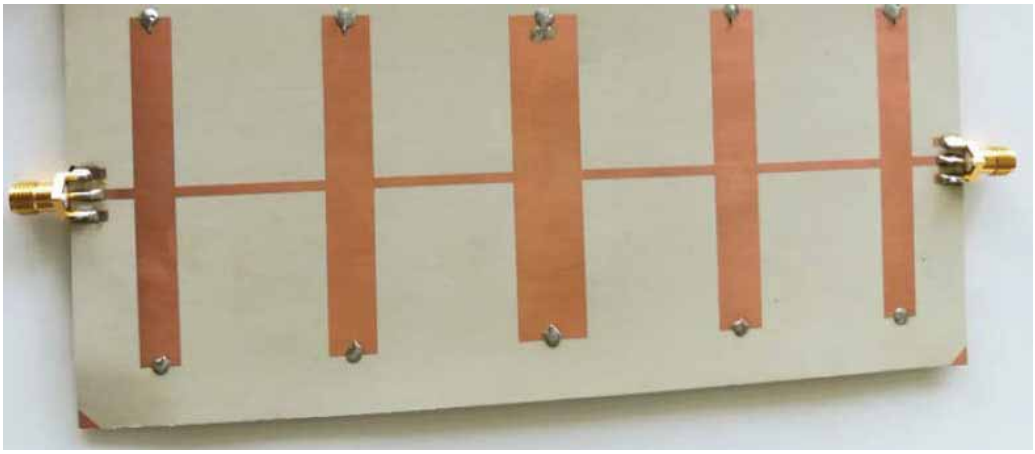
7) now the „real“ layout, but: „OPEN“ and „TEE“ change the circuit!





Example 2:

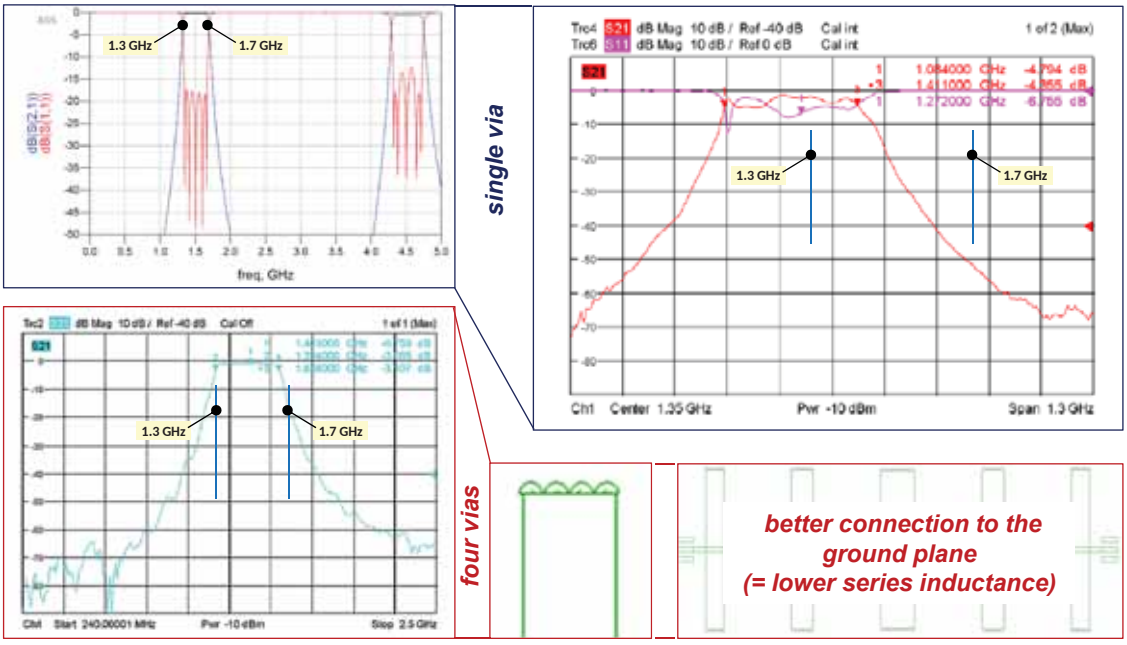
bandpass filter (1.3 GHz – 1.7 GHz) with 5 shorted stub lines:



- doubled (=parallel) stubs because the needed stub impedance is very low

Example 2, cont´d:

- a short circuit with only one via at the end of the stub line: very bad!



Example 3:

- Filter N=5 Butterworth, Z_0=50Ω, f_c=1.4GHz, f_R=2GHz

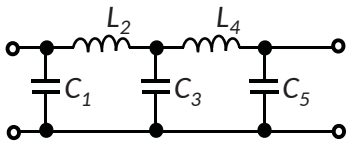


TABLE 9.3-1 g Values for Butterworth Low-Pass Prototype Filter

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
2	1.4142	1.4142	1					
3	1.0000	2.0000	1.0000	1				
4	0.7654	1.8478	1.8478	0.7654	1			
5	0.6180	1.6180	2.0000	1.6180	0.6180	1		

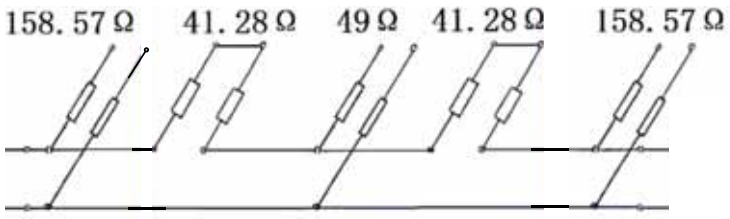
$g_1 = g_5 = 0.618$

$g_2 = g_4 = 1.618$

$g_3 = 2$

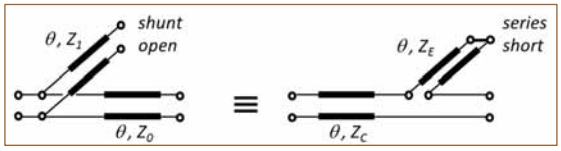
Example 3, cont´d:

- Calculate L,C using g. Then, Richards-transformation (Z_0=50Ω):

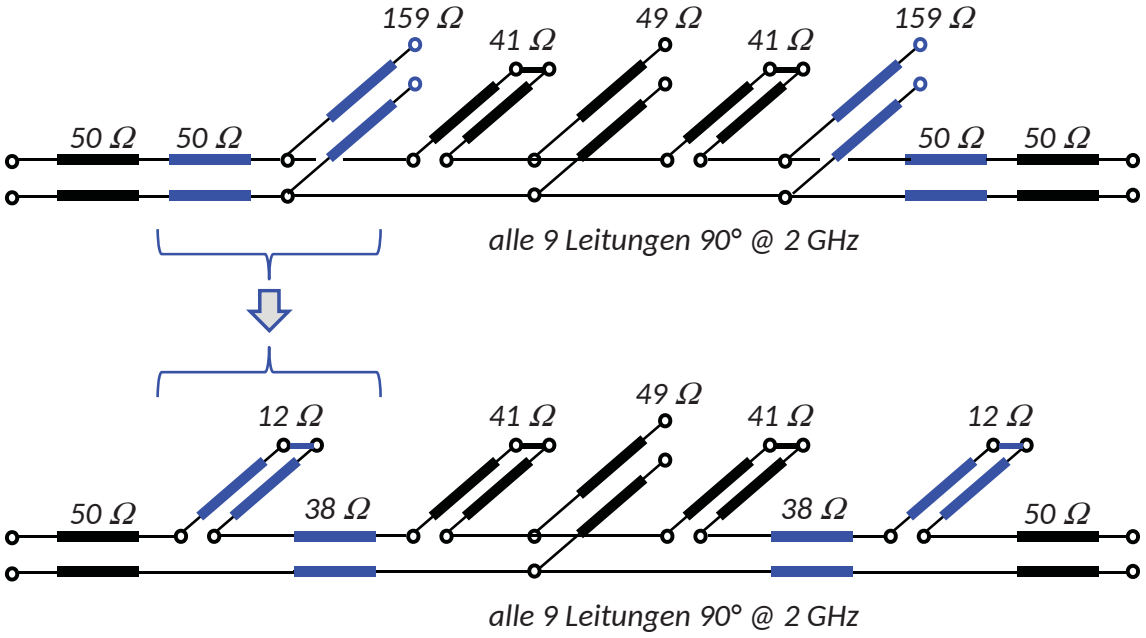


- at the input and outputs, line sections with 50Ω of arbitrary length can be added!

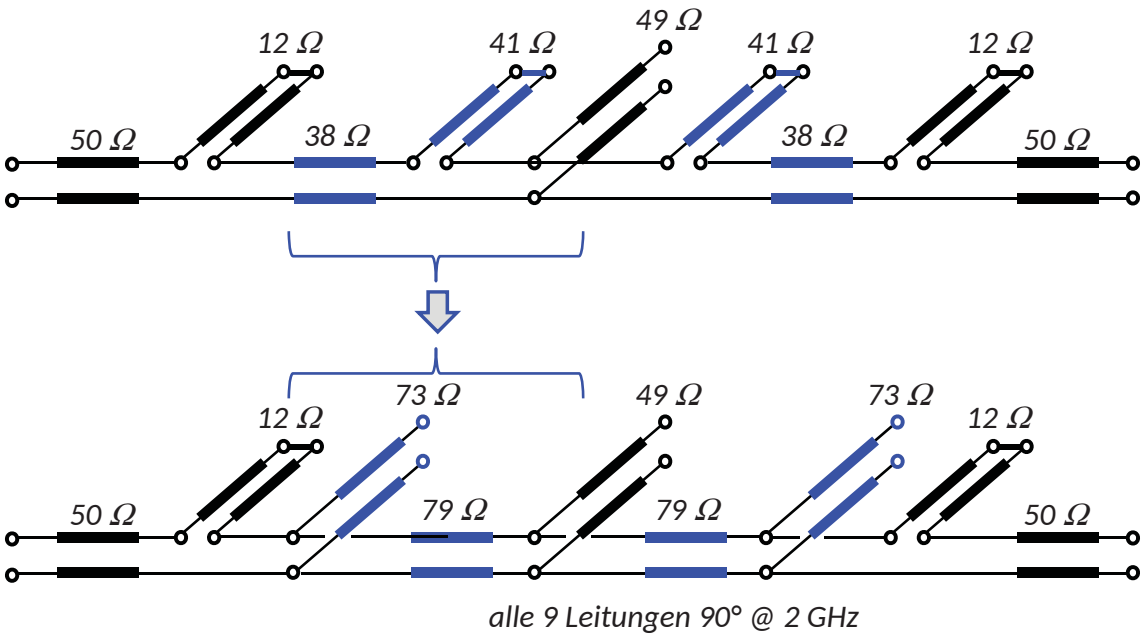
- Kuroda:
  replace the „outer“
  open shunt stubs
  by
  shorted series stubs:



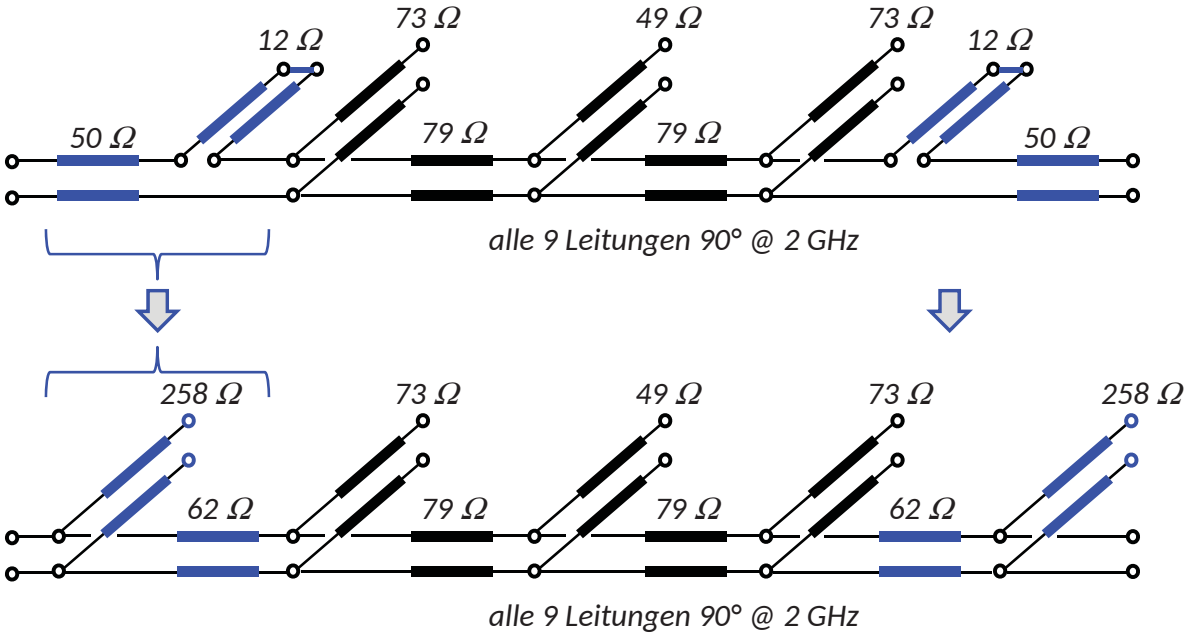
- first application of Kuroda (right & left sides – everything symmetrical):



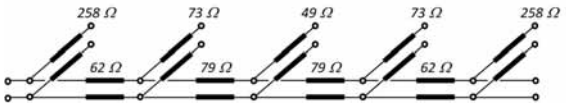
- second application of Kuroda (right & left sides – everything symmetrical):



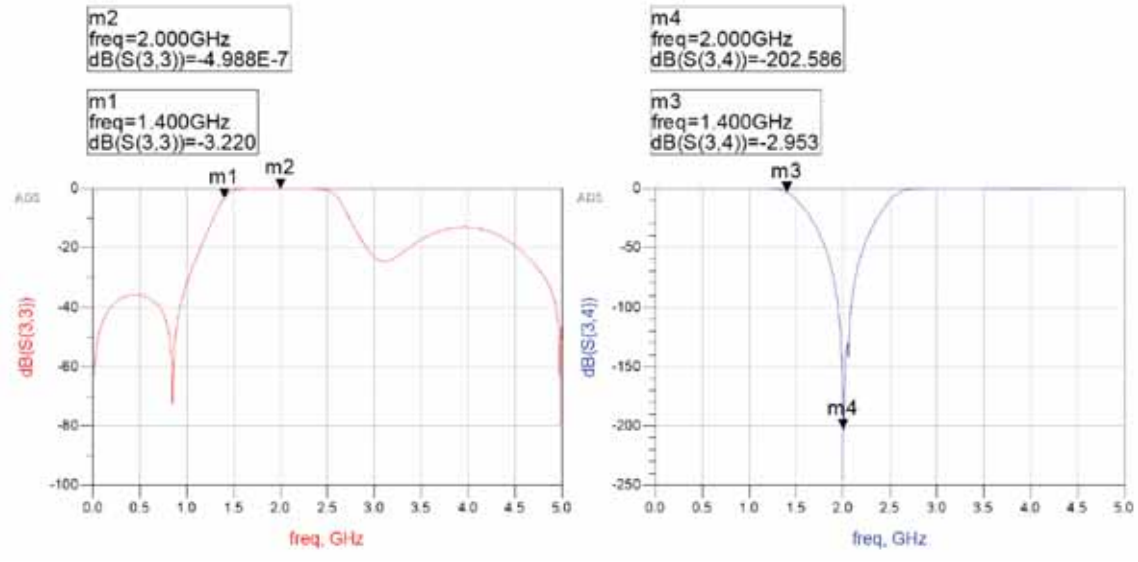
- third application of Kuroda (right & left sides – everything symmetrical):



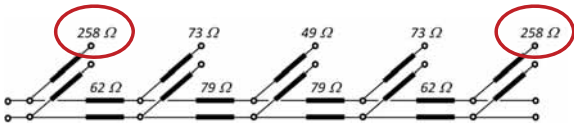
Example 3, cont´d:



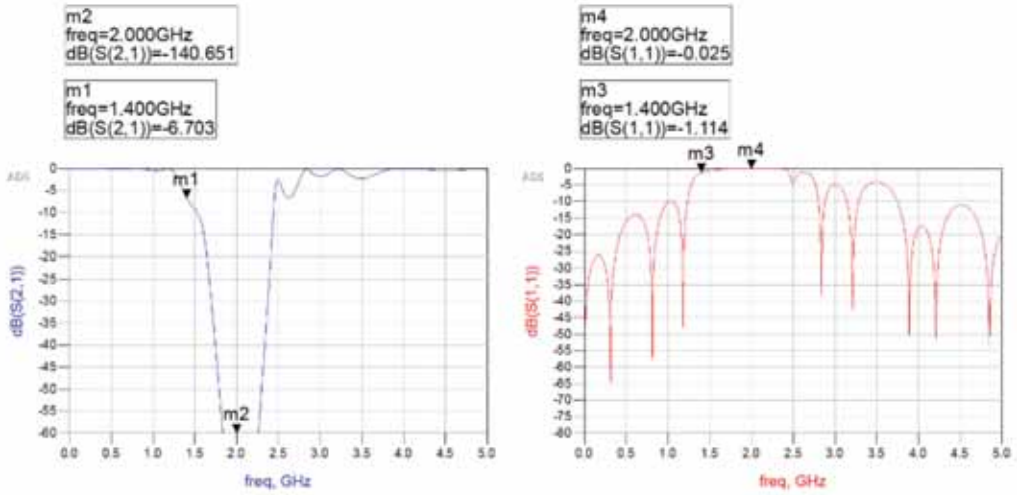
- The simulation model in *microstrip technology* will be complemented by TEE elements and OPEN effects and re-optimized:



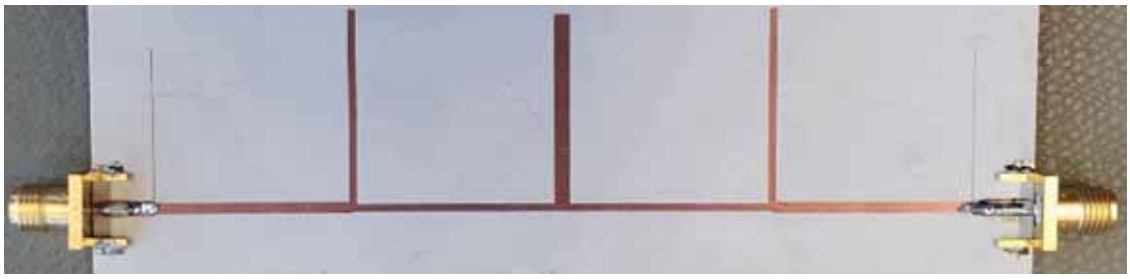
Example 3, cont´d:

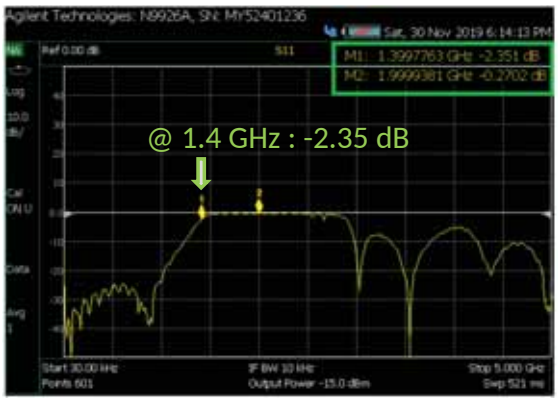


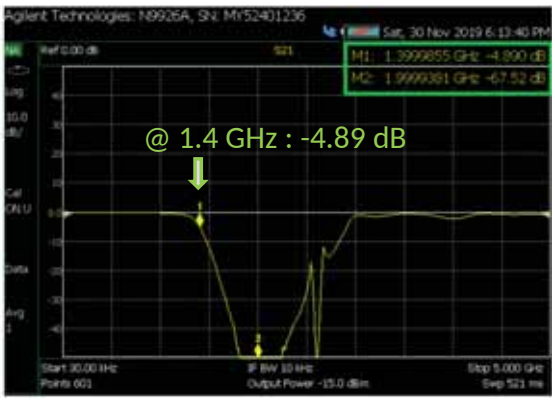
- The stub line with Z=258Ω cannot be realized in the chosen technology (why: it is way too narrow). Instead, the narrowest-possible line with (by the technology) of width 0.15mm is chosen (Z=147Ω). The filter properties deteriorate:



Example 3, cont´d:

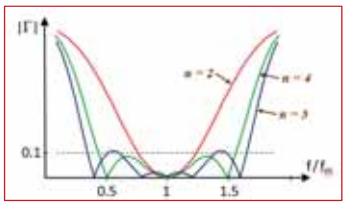






Example 4:

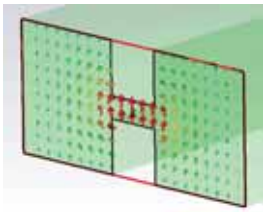
- impedance-bandwidth of a single-step impedance transformers can be increased by an additional resonance (half-wavelength resonator)

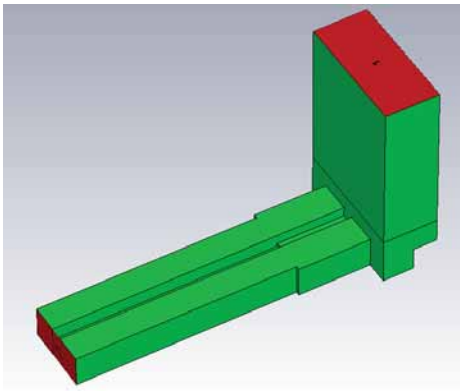


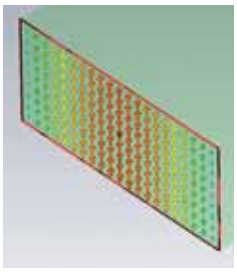
- repetition: limited bandwidth when using only one resonance → additional resonance needed:

- shown below is the transition from WR90 rectangular waveguide to a *double ridge* waveguide at 9.4 GHz:

double ridge waveguide:
13 mm width





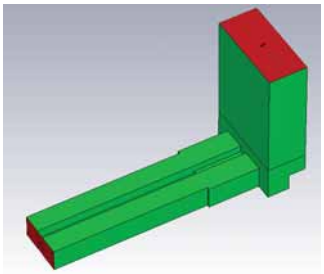


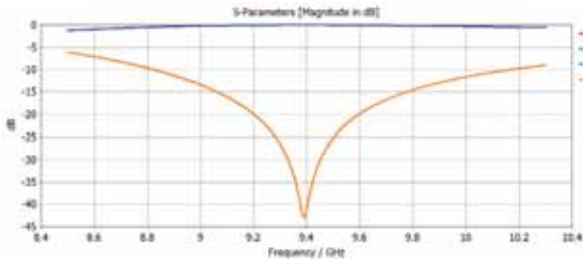
WR90: 22.86 mm width

Example 4, cont´d:

with λ/4 trafo:

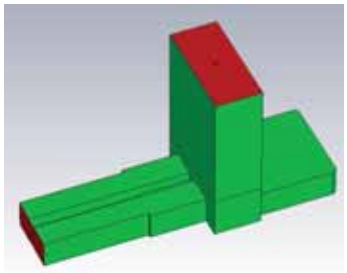
-20 dB bandwidth 4.3%
-30 dB bandwidth 1.3%

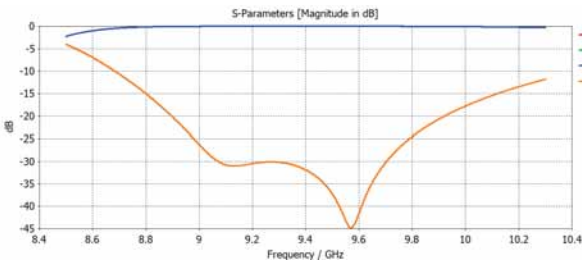




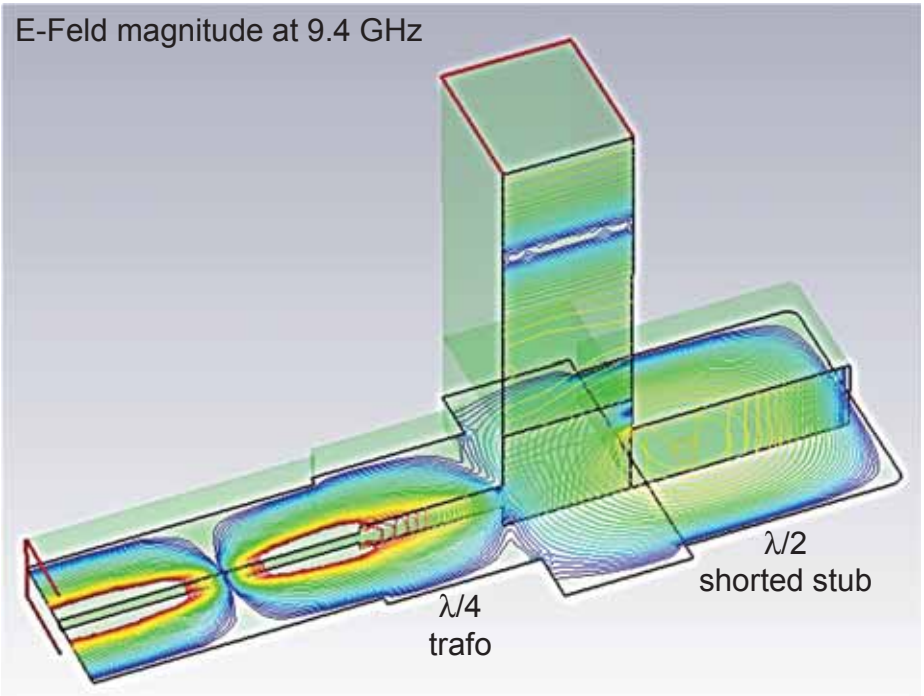
with λ/4 trafo and λ/2 shorted stub:

-20 dB bandwidth 11.5%
-30 dB bandwidth 6.9%





Example 4, cont´d:



The End